Underwrite for Collection: Five Connected Research Articles

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This research edition connects randomized credit-offer experiments, sequential collection decisions, reject inference, Treasury/ALM control and bank-controlled agentic deployment. The five articles retain their separate decision units and carry a common evidence and governance discipline.

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Navigating the equations

Displayed equations are numbered by article: (2.1) belongs to Article 2. Appendix numbers carry an A: (A2.1) belongs to its mathematical appendix. An aligned derivation has one number for the complete displayed block. Inline mathematical symbols remain unnumbered.

Equation references and page numbers are clickable. The register at the end of this edition lists every displayed block by topic and derivation step. Section labels such as A2.5 identify appendix sections; equation labels appear in parentheses.

Follow this idea	Connected equations
Cash and value	(1.2), (A1.3), (A4.11)
Policy evaluation	(1.10), (1.11), (A1.26)
Survival and competing risks	(A1.35), (A2.13), (A3.37)
Hierarchies and hyperpriors	(A2.8), (A2.9), (A3.20)
Pólya–Gamma computation	(A2.28), (A3.30), (A5.33)
Credit-limit decisions	(2.35), (2.38), (A5.25)
Selection and sensitivity	(A3.7), (A3.16)
Bellman and POMDP control	(2.1), (A2.6), (A4.26), (A4.28)
Reviewer uncertainty	(5.1), (5.2), (5.3)

Testing Credit Offers for Collectability

From randomized trials to contextual bandits in sustainable MSME lending

Underwrite for Collection · LinkedIn research article · 8 October 2026

Abstract

Credit pricing decisions influence who takes up a loan, how much cash the business must generate, and whether repayment can remain sustainable after disbursement. This makes experimentation in credit different from optimizing clicks or short-term conversions. A randomized controlled trial (RCT) can estimate the effect of specified offers for an eligible population; a contextual bandit can adapt allocation among those offers as evidence accumulates. This article develops a cautious progression from fixed randomization to response-adaptive allocation, explains context, actions, reward and policy, and sets out the assumptions behind Linear UCB, logistic Thompson sampling, inverse-propensity scoring and doubly robust evaluation. It also addresses delayed repayment outcomes, adverse selection, fairness, selective labels and the legal controls relevant to Kenyan credit. The practical aim is to learn which permitted offer supports durable repayment, not to maximize nominal yield in isolation.

Research questions

1. Among eligible Kenyan MSME applicants, what is the effect of a cash-flow-aligned credit offer on take-up, durable repayment, net realized cash and borrower outcomes compared with a defined standard offer?
2. When does response-adaptive contextual allocation improve offer learning enough to justify its loss of fixed-allocation simplicity, given delayed outcomes, changing cohorts and limited overlap?
3. What experimental, data-protection, fairness and human-review safeguards are needed before contextual bandits can inform credit offers or later collection actions in the *Underwrite for Collection* lifecycle?

The question starts with a repayment pathway

The *Underwrite for Collection* work began with a practical concern: a credit decision is not complete when funds are disbursed. The loan must fit the borrower's actual cash generation and the lender must be able to monitor whether that repayment source remains available. For an operating MSME, cash is shaped by sales, working capital, suppliers, seasonality, tax, owner withdrawals and existing obligations. For a ring-fenced receivables stream, repayment depends on eligible invoices, obligor performance, dilution, assignment, account control, servicing continuity, collection timing and the waterfall. A price that appears profitable on a spreadsheet can be uncollectable if its instalment profile conflicts with cash conversion or if the underlying receivable is disputed. Credit experimentation should therefore evaluate the offer as a whole and follow the cash journey after the decision. Financing ring-fenced receivables is the general proposition used throughout this series. In the worked examples that use a special-purpose vehicle (SPV), the vehicle is the chosen structure for holding the receivables and allocating their collections. The underwriting evidence follows the receivables, collection controls and contractual payment priorities of that structure.

An RCT and a contextual bandit are related, but they answer questions differently. A prespecified RCT assigns among allowed offers according to a fixed design so that effects can be estimated with a clear comparison. The allocation need not be uniform: it can be stratified by risk or product, or use unequal probabilities established in the protocol. A contextual bandit also uses randomized assignment, but it updates allocation as outcomes arrive, often assigning more frequently to actions that appear to perform well for particular contexts. That adaptation can improve learning efficiency or participant outcomes under some designs, yet it changes the data-generating process and complicates inference. The decision to adapt should follow

a demonstrated need, a credible reward signal and an analysis plan capable of accounting for changing probabilities [1], [2].

In credit, the action is not an abstract "arm." It is a specified offer: principal or limit, annual percentage rate and fees, term, instalment frequency, grace period, collateral or guarantee conditions, eligibility requirements and servicing conditions. If a loan is fixed-rate, the price is ordinarily set at origination under the contract; the lender cannot treat the rate as a free monthly control variable after disbursement. A revolving facility or renewal may have permitted repricing mechanics, but those must be stated in the contract and policy. Collection actions such as reminders, support referrals or an agreed payment-date adjustment are a separate post-disbursement action set. The bandit methods can inform both settings, but the experimental population, rights, rewards and authority differ. Clear action definitions prevent a pricing experiment from silently becoming a collection intervention.

Four elements of a contextual bandit

Equation (1.2) uses the discounted cash convention derived in equations (A1.1), (A1.3), (A1.4). The institution's internal funding charge is developed in equation (A4.11).

At decision time t, the contextual bandit observes a feature vector X_t, selects action A_t from an approved set, and later observes a reward R_t. A policy $\pi(a \mid x)$ maps context to an action probability. In a credit offer setting, X_t can include only information available and permitted at that time: verified income or business cash-flow features, debt obligations, product type, season, application channel and economic conditions. It should also capture data quality and uncertainty so the policy can decline to automate or route a case for review. The action set might include a small number of independently approved schedules and rates, not arbitrary price values. The observed reward arrives after the offer, take-up, disbursement and subsequent repayment path, so it is neither immediate nor always fully observed.

The reward should reflect the decision objective and the time period. For a complete one-period loan, with financing expense recorded explicitly, a realised cash contribution can be written:

$$\begin{aligned} R = {} & \text{cash received} - \text{initial advance} \\ & - \text{funding cost} - \text{servicing cost} \end{aligned} \tag{1.1}$$

Cash received includes actual principal, interest, fees and recoveries. A missed receipt has already reduced that amount, so the same shortfall is counted once. A real credit offer generates dated cash flows, and an asset-level present-value convention can instead handle funding through its declared discount curve:

$$V(a, x) = \mathbb{E}\left[-P_0(a) + \sum_{t=1}^{T} d_t \{ Q_t(a) - \mathsf{OPEX}_t(a) \} \,\middle|\, X = x \right] \tag{1.2}$$

Here P_0 is the initial advance, Q_t is actual borrower and recovery cash, OPEX_t is attributable operating cost, and d_t is the chosen discount factor. This convention keeps an additional funding charge out of the same present-value calculation. A contractual-receipts formulation deducts the discounted cash shortfall instead; the appendix derives the reconciliation between the two presentations. The calculation should account for take-up, prepayment, delinquency, cure, recovery timing, funding and capital conventions, with the horizon extended to include the relevant recovery cash or an explicitly valued terminal position. It should report affordability, complaints and other borrower outcomes beside financial value. A scalar reward may support optimization, but its weights must be explicit; legal rights and safety limits remain constraints outside that scalar.

This formulation explains why a simple "interest earned × probability of repayment minus principal × probability of default" can mislead. Interest is earned over time and depends on payment timing and balance;

principal repayment is not a loss event; recovery can occur after default; prepayment changes exposure; and operating, funding and capital costs matter. A model may estimate probability of default, while the offer policy also needs take-up, utilization, loss given default, exposure at default and cash-flow timing. Expected credit loss under IFRS 9 is a probability-weighted discounted cash shortfall and should not be confused with a pricing reward or economic capital [3]. The research protocol should preserve outcome components, then document any approved aggregation rule so reviewers can see whether a nominal margin gain was achieved through higher loss, lower take-up, borrower strain or selection of a different cohort.

What the experiment estimates

Before assigning offers, the research team should define its estimand: for example, the effect of offering a cash-flow-aligned repayment schedule rather than a monthly schedule among applicants who pass the same eligibility and affordability screen. The treatment is the offer, not merely the rate printed in a model. The target population should be described by product, market, period and entry criteria. A primary outcome might be verified net cash recovered by month 12, sustained repayment status at month 12, or net contribution per eligible applicant; secondary outcomes could include take-up, delinquency timing, cure, prepayment, restructuring, complaints and hardship referrals. The denominator must match the question. Profit per funded loan is not profit per eligible applicant, and a lower default rate among originated borrowers can arise because fewer applicants accepted the offer.

An intention-to-treat analysis compares outcomes by randomized assignment, whether or not the applicant accepted, completed documentation or drew the facility. It estimates the effect of making the offer available under the observed take-up process. An analysis restricted to those who accepted may reintroduce selection, because acceptance can depend on price, urgency, outside options and unobserved financial conditions. If the research question concerns the effect of actually receiving credit, a randomized offer might be used as an instrumental variable only under additional assumptions; that estimate typically describes applicants whose take-up changed because of the offer, not everyone. Assignment, offer delivery, acceptance, origination and repayment should therefore be recorded and reported separately [4].

The protocol should state the randomization unit and method, control offer, treatment offers, eligibility rules, allocation probabilities, analysis windows, sample-size assumptions, noncompliance plan, censoring rules, safety monitoring and stop conditions. Stratification by product, season, region or risk band can improve balance if chosen before looking at outcomes. Cluster assignment may be necessary where loan officers handle several applicants and might carry treatment behavior across cases, or where a group guarantee creates spillovers among members. Repeated testing should account for multiple comparisons, interim monitoring and planned stopping. Randomization does not remove data errors, non-adherence, policy changes or economic shocks; it creates a basis for comparison when design and execution remain faithful enough to the protocol.

The flow below shows the order of evidence and authority. The eligibility gate comes before assignment, and the randomized comparison remains visible even if allocation later adapts. A decision to update the offer probabilities depends on outcome maturity, support, customer outcomes and independent review. If those checks fail, the experiment stays fixed, pauses or returns to the approved policy; the algorithm does not loosen eligibility or invent a new price. Recording each transition lets analysts distinguish an offer effect from take-up, origination and later payment. It also gives borrowers and reviewers a traceable route from the original decision to the outcome used in subsequent analysis.

Moving from fixed RCT allocation to response-adaptive allocation

Suppose the current policy assigns among two or three offers with known probabilities. A contextual bandit estimates expected reward

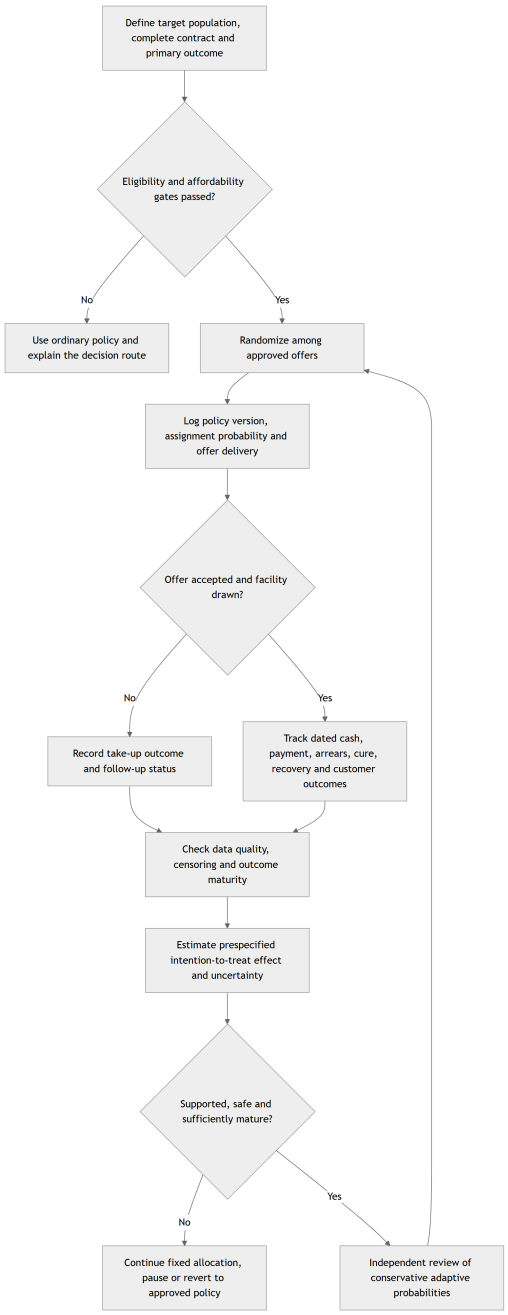


Figure 1: Offer experiment and governed learning

$$\mu(x, a) = \mathbb{E}[R \mid X = x, A = a] \tag{1.3}$$

and selects actions as evidence develops. In response-adaptive randomization, probabilities change over time according to accrued outcomes, sometimes with a minimum allocation retained for each approved action. A simple mixture can be written

$$\pi_t(a \mid x) = (1 - \varepsilon_t)\hat{\pi}_t(a \mid x) + \varepsilon_t q(a \mid x), \tag{1.4}$$

where $\hat{\pi}_t$ is the current estimated best action, q preserves controlled exploration, and ε_t follows a predeclared schedule. This is a design family, not a universal formula. Credit outcomes mature slowly, so a policy that updates on incomplete early cases can amplify noise, react to seasonal composition or chase transient take-up rather than durable repayment [5].

Linear upper-confidence-bound methods assume that expected reward can be adequately approximated as

$$\mu(x, a) = x^{\mathsf{T}}\theta_a \tag{1.5}$$

or by a shared linear action-feature model. The algorithm chooses an action using a predicted mean plus an uncertainty bonus, which encourages learning where the model is uncertain. The assumption can be useful as a transparent baseline, but the relation between business cash context and risk-adjusted value may be nonlinear, and the uncertainty bonus is not itself a credit safety guarantee. A contextual Thompson sampler instead draws plausible model parameters from a posterior and selects the action that appears best in that draw. For binary outcomes, logistic or hierarchical Bayesian models can represent default probabilities, but reward may also depend on take-up, loss severity and cash-flow timing. Both algorithms require a well-defined action set, calibrated models and governance; neither justifies "exploring" an offer that fails affordability or product rules.

Response adaptation can reduce the number of applicants assigned to actions that appear inferior, but it can also reduce statistical precision for those actions. If one arm's allocation probability approaches zero, later policy comparisons have weak support and inverse-probability weights become unstable. A fixed randomized benchmark, a minimum exploration floor among eligible offers, or a batched adaptive design can help preserve evidence. Analysts need to account for adaptive data collection in confidence intervals and policy-value estimates rather than applying a fixed-sample formula without adjustment. A batched approach waits for a cohort or outcome window before updating; this can fit credit better than reacting to every early payment. The experiment should specify when an outcome is mature enough to influence allocation and what monitoring will detect drift or a change in cohort mix [6].

Delayed outcomes need careful handling

The competing-risk probability below connects to the incidence derivation in equation (A1.35) and the shared hazard likelihood in equation (A2.13).

Credit reward is delayed because repayment, default, cure and recovery unfold over a loan's life. First-payment default, early delinquency, missed instalments, utilization and payment promises can be informative leading indicators, but they are proxies for later outcomes, not replacements for them. A proxy can be useful if its relationship to the final objective is tested across products, cohorts and economic periods, and if the policy records which outcomes are interim versus mature. Optimizing a proxy can create Goodhart's law: staff may improve the measured early event while lifetime loss, affordability or durable cure deteriorates. Every interim dashboard should therefore show outcome maturity and avoid relabelling a 30-day metric as a lifetime-risk result.

Survival analysis helps represent event timing and incomplete follow-up. A Cox model can estimate the hazard shown below, while the survival probability follows by integrating that hazard under the model. If prepayment, cure, restructuring or default can occur as competing events, a single-event model that treats every alternative exit as non-informative censoring may answer the wrong question. A multi-state or competing-risk model can distinguish transitions. The Cox model estimates event risk under its assumptions; it does not automatically provide an unbiased causal comparison between offers if action assignment is confounded or adaptive probabilities are ignored. Time-to-default is one input to offer value and monitoring, not a substitute for experiment design [7].

$$\lambda_i(t \mid X_i, a) = \lambda_0(t) \exp\big(X_i^\top \beta^H + g^H(a, X_i) + u^H_{s(i)} + v^H_{r(i)} + w^H_{c(i)}\big) \tag{1.6}$$

$$S_i(t \mid X_i, a) = \exp\left[-\int_0^t \lambda_i(u \mid X_i, a)\, du\right] \tag{1.7}$$

The hierarchical specification preserves sector, region and origination-vintage effects in both the hazard model and any fixed-horizon logistic model used for offer evaluation. Model label H denotes the hazard parameters; L denotes separate logistic parameters. For each model, sector, region and vintage effects are zero-centred Gaussian conditional on their respective positive scales, and each scale receives a calibrated half-Student-t prior with three degrees of freedom. Non-centred standard-normal variables multiplied by those scales provide the computational representation. Article 2 spells out the complete hierarchy and prior sensitivity. The offer effect g^H and its logistic counterpart have distinct parameters and their causal interpretation comes from the experiment. Baseline hazards, fixed-horizon probabilities and competing-event incidence retain their own outcome definitions, so borrowing information across groups preserves what each model actually estimates.

For a competing-risk analysis, define T as the time to the first terminal event and J as its cause. Default, voluntary prepayment and contractual payoff can be distinct terminal causes when the question concerns which exit happens first. Cure and restructuring are often better represented as intermediate states in a multi-state process, because a cured or modified loan can later default, repay or enter another state. The default cumulative-incidence function for a specified offer and applicant context is:

$$F^D(\tau \mid x, a) = \Pr(T \le \tau, J = D \mid X = x, A = a) \tag{1.8}$$

The cause-specific hazard for event type *k* is:

$$\lambda_k(t \mid x, a) = \lim_{\Delta t \to 0} \frac{1}{\Delta t} \times \Pr\big(t \le T < t + \Delta t, J = k \mid T \ge t, X = x, A = a\big) \tag{1.9}$$

The cumulative incidence depends on the hazards of all competing causes. Treating prepayment as ordinary independent censoring in a Kaplan–Meier calculation can overstate absolute default incidence because prepaid loans can no longer default under the original contract. Aalen–Johansen estimates transition probabilities and cumulative incidence in a multi-state setting; Fine–Gray models the subdistribution hazard linked to cumulative incidence. These methods answer different descriptive or associational questions. Neither alone identifies a causal offer effect: that still depends on randomization or defensible adjustment, action-probability logs, support, and a clearly specified population and contract [8], [9].

Interim proxy outcomes should be handled in a prespecified hierarchy. The protocol can designate matured net cash or loss as the primary endpoint; first-payment default or 30-day delinquency as early monitoring signals; and contact, take-up or promise-to-pay as process outcomes. A proxy may support a cautious interim stop for harm, but using it to declare a winner requires evidence linking it to the final endpoint. If early outcomes are used in a Bayesian model to inform allocation, the model should include their uncertainty and avoid double-counting the same evidence at maturity. Matured cohorts must later test whether a policy selected on early signals retained the expected benefit. A bandit should be allowed to pause or revert to standard policy when data quality, outcomes or fairness monitoring fall outside predeclared bounds.

Selection, take-up and reject inference

Historical loan performance contains selective labels: repayment is generally observed for applicants who received a loan under a previous process. Even within a randomized pricing experiment, a trial among already-approved applicants estimates effects within that approval region; it does not reveal how previously rejected applicants would repay. The potential outcome must be tied to a defined offer a and horizon τ, such as $D_i^a(\tau)$. A model trained on funded loans can predict under assumptions, but a synthetic label sampled from a posterior is not an observed repayment record. The last section of the supplied source raises this issue directly. The companion article, "Who Gets a Repayment Label?", develops the identification problem, support conditions, sensitivity analysis and controlled offer designs in more detail [10], [11].

The first experiment should usually be a prespecified, fixed-allocation pilot rather than immediate online adaptation. It can establish eligible regions, outcome latency, delivery rates, take-up, data quality and preliminary treatment effects. If a lender later considers controlled expansion near an approval boundary, the protocol must define the population and offer, maintain affordability and legal safeguards, and clarify whether assignment is to an offer or an encouragement. Randomized offers generate evidence only for those screened into that trial and for the specified contract. They do not justify generalizing to applicants who fail legal, identity, affordability or minimum risk criteria. An exploration probability is never a substitute for independent approval of the action and population.

Off-policy evaluation and the limits of propensities

When a trial or bandit has logged the probability of each action, inverse-propensity scoring can estimate the value of a target policy π from data collected under a logging policy π_0. A simplified estimator is

$$\widehat{V}_{\mathsf{IPS}}(\pi) = \frac{1}{n} \sum_i \frac{\pi(A_i \mid X_i)}{\pi_0(A_i \mid X_i)} R_i. \tag{1.10}$$

This corrects action proportions only under assumptions that include accurate propensities, consistency, sufficient overlap and appropriate outcome observation. If a policy was deterministic, an action never taken has no empirical support. Tiny propensities produce large and unstable weights; clipping may reduce variance but introduces bias that should be reported. The probability must reflect the actual conditional assignment after eligibility, affordability, legal, operational and human-review gates have determined which actions were available.

A doubly robust estimator combines an outcome model $m(x, a)$ with propensity weighting. One contextual form is

$$\widehat{V}_{\mathrm{DR}}(\pi) = \frac{1}{n}\sum_i \left[\sum_a \pi(a \mid X_i) m(X_i, a) + \frac{\pi(A_i \mid X_i)}{\pi_0(A_i \mid X_i)} (R_i - m(X_i, A_i)) \right]. \tag{1.11}$$

Under standard conditions, it can be consistent if either the propensity model or outcome model is correctly specified, but it cannot repair unmeasured confounding, poor overlap, misspecified outcomes and policies outside the data support all at once [1], [12]. Evaluation reports should include weight distributions, effective sample size, support maps, uncertainty intervals, sensitivity to clipping and a direct comparison with the randomized benchmark. "Doubly robust" names a property under assumptions; it is not a certificate that the result is robust to every way the data may be wrong.

Fairness and legal safeguards

Fairness constraints require a defined decision and outcome. Equal opportunity usually compares true-positive rates across groups; equalized odds compares both true-positive and false-positive rates conditional on the outcome [13]. They do not mean equal accuracy, and no single metric resolves access, price, error burden, affordability, treatment quality and recovery outcomes at once. In credit, protected attributes may be legally restricted or unavailable for routine pricing, while still raising a need for carefully governed evaluation. Proxy variables can arise from geography, business type, account behavior or device data. A model card should state which fairness diagnostics were used, what labels they depend on, where rejected applicants create blind spots, and what action follows if disparities are found. Feature exclusion or adversarial training may be evaluated, but neither alone guarantees fair treatment.

The Data Protection Act, 2019 sets duties for lawful, fair and transparent processing, purpose limitation, data minimisation, accuracy, retention, data-subject rights and security. Section 31 requires a data-protection impact assessment where processing is likely to result in high risk to rights and freedoms; section 35 addresses certain decisions based solely on automated processing that produce legal effects or significantly affect a person [14]. The Data Protection (General) Regulations, 2021 identify high-risk processing requiring a DPIA in regulation 49 and address its conduct, prior consultation, consideration and audit in regulations 50–53. Regulation 22 adds safeguards for automated individual decision-making, including notice, meaningful information about logic, error prevention, appropriate statistical procedures, bias reduction and human intervention [15]. A human reviewer must have enough information, authority and time to assess and change a recommendation. Consent is not the only possible lawful basis, and a model's research utility does not itself establish a lawful purpose or permission to repurpose transaction data.

The legal perimeter depends on institution and action. The Digital Credit Providers Regulations, 2022 include provisions on responsible lending, complaints and debt-collection conduct for providers within their scope; regulation 20, for example, prohibits specified abusive collection practices [16]. The CBK framework and applicable banking, SACCO, MFI, consumer-protection and credit-reference rules must be mapped to the entity, product and decision. A pricing offer at origination, a credit limit change, a bureau report and a collection contact are different acts with different rules. The experiment should therefore record controller/processor roles, purpose and lawful basis, notice, data sources, retention, cross-border transfers, automated decision implications and complaint routes. Complete any required DPIA before processing and get Kenyan legal review for the actual institution rather than importing a generic "fair-lending law" claim from another jurisdiction.

What Fannie Mae and Freddie Mac contribute to the method

Fannie Mae and Freddie Mac publish loan-performance datasets and documentation for defined US mortgage populations. Freddie Mac's Single-Family Loan-Level Dataset includes origination, monthly performance and actual-loss information for specified acquired loans; Fannie Mae's Single-Family Loan Performance Data describes selected fixed-rate mortgage cohorts and quarterly performance updates [17], [18]. These resources can inform data architecture: stable loan identifiers, origination snapshots, reporting dates, delinquency transitions, modifications, prepayment, liquidation, loss fields, release cut-offs, field definitions and versioned corrections. They are useful examples of how a large portfolio can be made analytically reusable through documentation and time-indexed performance records. The methodological lesson is to preserve the relationship between the offer at origination and the monthly repayment path rather than training on an undated "good/bad" outcome alone.

The US datasets are not Kenyan MSME loan data and do not provide the missing outcomes of rejected Kenyan applicants. Their mortgages are selected by acquisition and program rules, with different legal, product, security, economic and servicing environments. They cannot estimate the causal effect of a Kenyan credit offer or collection practice. The local research question requires Kenyan records with suitable permissions, definitions for business and receivables exposures, account and cash-flow provenance, and governance for natural-person data contained in business files. External datasets can validate data-pipeline concepts, cohort logic and event definitions; local portfolio evidence must establish local performance. The distinction matters for reproducibility and for avoiding claims that a method tested on US mortgages transfers directly to Kenyan micro and small enterprises.

A worked example with explicit assumptions

Suppose a lender has two fully approved offers for a seasonal trader who passes the same affordability and product rules: **Offer A** has a KES 100,000 principal, a 12-month monthly schedule, and total contractual cost C_A; **Offer B** has the same principal and a seasonal schedule aligned to verified receipts, with total contractual cost C_B. The numerical amounts below are illustrative only; they are not proposed rates or evidence about a Kenyan portfolio. Within an eligible research population, a fixed 1:1 randomized pilot compares the offers. It records offer delivery, take-up, instalment timing, business receipts, owner support, arrears, cures, recovery, costs and complaints. It does not randomize credit to applicants who fail minimum eligibility or use coercive collection to accelerate cash.

Assume 200 eligible applicants are randomized, 100 per offer; 72 in A and 75 in B accept and draw. At twelve months, 58 A-loans and 64 B-loans are current or fully repaid, while mature loss and recovery are still being reconciled. The raw funded-loan rates are:

$$\frac{58}{72} = 80.6\% \tag{1.12}$$

$$\frac{64}{75} = 85.3\% \tag{1.13}$$

The difference calculated from the unrounded fractions is 4.8 percentage points (4.7778 before rounding). These denominators include only those who accepted, so the raw comparison is not the primary causal estimate. The intention-to-treat comparison uses all 100 assigned to each offer and a prespecified outcome such as verified net cash per eligible applicant by month 12. It reports uncertainty, follow-up completeness, payment timing, loss severity and customer outcomes. The observed counts do not establish significance or a recommendation without a powered design and complete data.

Now suppose an adaptive policy proposes assigning the seasonal schedule more often because early repayment looks better. Before changing allocation, the lender verifies that all observed cases have had comparable follow-up, that the apparent improvement is not driven by lower take-up of a particular risk segment, and that interim delinquency predicts mature loss for this product. It then uses a response-adaptive rule with a minimum probability floor for both offers, randomized benchmark cases and predeclared pause conditions. It records $\pi_t(a \mid x)$ after all action gates and preserves a holdout or valid evaluation stream. This step can reduce allocation to an inferior schedule while continuing to learn, but the choice remains governed by affordability, fair access, loss budgets, legal review and outcome maturity. If the evidence remains noisy, the responsible result may be to retain the fixed pilot rather than adapt.

A practical implementation sequence

The first stage establishes the baseline: define the product and offer catalog, map historical policy and exceptions, reconcile application and loan identifiers, create timestamped cohort data, define outcome maturity and build a support map. The second stage runs a prespecified fixed-allocation experiment inside the approved eligibility set. The third stage analyzes causal effects and heterogeneity with appropriate uncertainty, while separating offer assignment, take-up, loan origination and later repayment. The fourth stage can test contextual allocation in a batched, conservative design with explicit minimum probabilities, benchmark assignment and action gates. Only after outcome quality and adaptive inference are credible should an online bandit be considered. Each stage produces a reviewable artifact: protocol, data dictionary, model card, code version, trial report and approval record.

The ongoing control framework should assign owners for model risk, product policy, affordability, data protection, collections, complaints, finance and independent validation. It should track calibration, offer acceptance, allocation probabilities, portfolio mix, cohort vintage, early and matured outcomes, delinquency migration, cure, recovery, loss severity, overrides, complaints, hardship referrals, data corrections and group-level outcomes. A policy change should require evidence that the new version improves the declared objective without breaching hard constraints. Drift, low overlap, poor data quality, adverse fairness patterns or unexpectedly high losses should trigger a pause or rollback. The system should preserve the prior approved policy and reproducible logs so the institution can explain what information was considered and why a particular offer was made.

Conclusion

Contextual bandits can extend credit-pricing experiments by learning which approved offer performs better for which observed context. Their value depends on a careful RCT foundation: contract-specific actions, a meaningful outcome, complete assignment logs, adequate overlap, mature repayment data and a valid comparison. Adaptation is not inherently superior to fixed randomization; it trades allocation efficiency against inferential clarity and can react badly to delayed or proxy outcomes. Linear UCB and Bayesian Thompson sampling are candidate methods under different model assumptions, while IPS and doubly robust estimators can evaluate policies only within supported, well-logged data. The offer itself must be affordable and lawful before any algorithm chooses among alternatives.

For *Underwrite for Collection*, the central outcome is sustainable, evidenced repayment. The price, size and timing of credit shape future collections, but they are not identical to post-disbursement servicing decisions. A business financed as a going concern requires evidence about operating cash; a ring-fenced receivables stream requires evidence about eligible invoices, legal transfer, obligor payment and account control. This first article establishes how a controlled offer study can measure permitted choices. The second moves from that origination decision to repeated servicing actions and evolving borrower states; the third examines the missing outcomes created by historical rejection; the fourth carries the control idea to the institution's balance sheet. Each stage changes the decision unit and evidence required, while preserving the same discipline: define the action, observe outcomes, state assumptions and keep governance in the design.

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Publication note. All worked numbers are hypothetical. This article proposes a research design and model-governance pathway; it does not report a completed Kenyan RCT, approved price policy or production bandit. Confirm the current legal framework and institution-specific scope before publication.

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Article 1 mathematical appendix: offers, cash outcomes and experimental learning

This appendix derives the mathematical relationships used in Article 1. All monetary examples are hypothetical. Each derivation states the population and information conditions needed for its interpretation. Probabilities are dimensionless; currency amounts are in Kenyan shillings (KES); the time unit is stated for each example. Equations concern an approved eligible population and a defined contract and servicing regime. A statistical comparison becomes a causal offer comparison when consistency, the stated assignment conditions and the relevant interference assumptions hold.

A1.1 Notation and the unit of analysis

Symbol	Definition
$i = 1, \dots, n$	Eligible applicant or prespecified randomisation unit
X_i	Information available before that unit's assignment
H_{i-1}	Information available to the allocation rule before assignment i, including outcomes already received
$A_i \in \mathcal{A}(X_i)$	Assigned approved offer; $\mathcal{A}(X_i)$ is the eligible action set
$e_i(a \mid X_i, H_{i-1})$	Actual conditional assignment probability after eligibility gates
$\pi(a \mid x)$	Target policy whose expected outcome is evaluated
$U_i(a)$	Indicator that applicant i would accept and draw offer a
$Y_i(a)$	Prespecified applicant-level outcome under offer a
R_i	Observed scalar reward, with declared accounting and observation conventions
t and τ	Time since origination and the endpoint horizon
d_t	Discount factor from time t to the valuation date
T_i, J_i, C_i	Event time, event cause, and right-censoring time
$\mu(x, a)$	Conditional expected reward for action a
θ, β	Regression coefficient vectors, defined in the relevant model

The same letter is not used for take-up and survival time here: U denotes take-up and T denotes event time. The article's T as a cash-flow horizon is written τ below. All probabilities and expectations are conditional on membership in the eligible study population unless another population is explicitly specified.

A1.2 Construct the reward from a dated cash ledger

Let P_0 be the advance at origination, Q_t actual borrower and recovery receipts, O_t attributable operating and servicing payments, and B_t external funding cash flow when it is explicitly included in the chosen financing convention. A realised asset-level discounted contribution, valued using a declared discount curve and with funding handled through that curve, is

$$W = -P_0 + \sum_{t=1}^{\tau} d_t(Q_t - O_t). \tag{A1.1}$$

The advance and subsequent principal receipts are both represented. A missed principal payment reduces Q_t; subtracting that same principal again as a realised loss would count the shortfall twice. If the analysis instead uses an accounting margin, its revenue and loss components must be defined so they reconcile to this cash ledger. For example, interest and fee income less funding expense, servicing expense and net credit charge is an accounting presentation with a separate principal roll-forward. The article's compact reward expression is meaningful only after making that choice explicit.

To derive a cash-shortfall presentation, let Q_t^{ctr} be the contractual receipts and define the present value of their shortfall by

$$L_{\text{PV}} = \sum_{t=1}^{\tau} d_t \left(Q_t^{\text{ctr}} - Q_t\right). \tag{A1.2}$$

Substitution gives the identity

$$\begin{aligned} W &= -P_0 + \sum_t d_t (Q_t^{\text{ctr}} - O_t) - L_{\text{PV}} \\ &= -P_0 + \sum_t d_t (Q_t - O_t). \end{aligned} \tag{A1.3}$$

Consequently, the article's expected-value form can be implemented consistently as

$$V(a,x) = E\left[-P_0(a) + \sum_t d_t\{Q_t^{\text{ctr}}(a) - O_t(a)\} - L_{\text{PV}}(a) \mid X = x\right]. \tag{A1.4}$$

If CF_t already denotes actual net cash including the initial advance and recovery shortfalls, a further deduction of L_{PV} is unnecessary. If funding cash flows B_t are modelled explicitly, include their signs and dates and use a valuation convention consistent with that financing perspective. An FTP charge and an external funding cost require an explicit reconciliation; they cannot both be deducted for the same economic funding requirement without explanation. The IFRS 9 effective-interest-rate discount convention for an impairment allowance has its own purpose and is retained separately from the pricing discount curve.

Numerical example. Advance KES 100,000 today. Suppose twelve end-of-month receipts are KES 9,500, monthly servicing cost is KES 100, and the monthly discount rate is 1%. With full payment,

$$\begin{aligned} d_t &= (1.01)^{-t}, \\ W &= -100{,}000 + 9{,}400 \sum_{t=1}^{12} (1.01)^{-t} \\ &= -100{,}000 + 9{,}400 \frac{1 - (1.01)^{-12}}{0.01} \\ &= 5{,}797.73 \text{ KES}. \end{aligned} \tag{A1.5}$$

If the final KES 9,500 receipt is missing and no later recovery is expected within the valuation model, its present-value shortfall is $9{,}500/(1.01)^{12}$. Reduce W by that amount once. An eventual recovery is entered at its expected or realised recovery date, with its collection costs, rather than treated as immediate cash at default.

The value per eligible applicant also includes take-up. If C_a^{offer} is the cost of delivering the offer and W_a is contribution conditional on origination,

$$E[Y(a) \mid X = x] = -C_a^{\text{offer}}(x) + \Pr\{U(a) = 1 \mid x\}\, E[W_a \mid U(a) = 1, x]. \tag{A1.6}$$

This is the law of total expectation, with a zero originated-loan contribution for non-takers and separately recorded offer costs. Other applicant outcomes, such as complaints or business distress, need their own definitions for non-takers. The separation explains why profit per originated loan and contribution per applicant can rank offers differently.

A1.3 Random assignment, intention to treat and take-up

The credit experiment separates assigned offers and subsequent behaviour using the experimental-design logic in [4].

For two offers a and b, define the eligible-population intention-to-treat estimand

$$\Delta_{\text{ITT}} = E[Y(a) - Y(b)]. \tag{A1.7}$$

Assume consistency, no relevant interference between randomisation units, and assignment independent of potential outcomes within the randomisation strata. Then

$$\begin{aligned} E[Y \mid A = a] &= E[Y(a) \mid A = a] = E[Y(a)], \\ E[Y \mid A = b] &= E[Y(b) \mid A = b] = E[Y(b)]. \end{aligned} \tag{A1.8}$$

The difference of the two assigned-group means therefore estimates Δ_{ITT}. With stratification, calculate stratum-specific comparisons and aggregate using the target stratum proportions; an unweighted average across differently sized strata generally targets a different population.

Conditioning on acceptance changes the comparison:

$$E[Y \mid A = a, U = 1] = E[Y(a) \mid U(a) = 1]. \tag{A1.9}$$

The corresponding mean under b conditions on $U(b) = 1$. The two accepting groups can have different risk and business characteristics. Randomisation of A does not make these two post-assignment subgroups identical. Record acceptance as an outcome of the offer and retain every assigned applicant in the ITT denominator.

Reproduce the article's counts. In each arm 100 applicants are assigned. Take-up is 72% and 75%; current-or-repaid status among the funded loans is

$$\frac{58}{72} = 80.5556\%, \qquad \frac{64}{75} = 85.3333\%, \tag{A1.10}$$

giving a funded-subgroup difference of 4.7778 percentage points. Define a distinct applicant-level endpoint $J(a) = 1$ if an applicant both originates and is current or fully repaid at twelve months. Its assigned-group estimates are 58% and 64%, giving an ITT difference of 6 percentage points for that combined endpoint. This endpoint is fully defined for non-takers, whose $J = 0$ denotes no origination; it does not label them as defaulters. A simple independent-binomial standard error for the difference is

$$\widehat{\mathsf{SE}} = \sqrt{\frac{0.64(0.36)}{100} + \frac{0.58(0.42)}{100}} = 0.06885. \tag{A1.11}$$

The normal-approximation 95% interval is approximately $0.06 \pm 1.96(0.06885)$, or −7.49 to 19.49 percentage points. It is a rough fixed-design illustration; the realised cash endpoint requires its actual cash observations and its own uncertainty analysis.

For a finite population of $N = n_a + n_b$ randomly assigned units, the randomisation variance of the difference in means is

$$\mathsf{Var}(\widehat{\Delta}) = \frac{S_a^2}{n_a} + \frac{S_b^2}{n_b} - \frac{S_{a-b}^2}{N}, \tag{A1.12}$$

where S_a^2, S_b^2 are finite-population potential-outcome variances and S_{a-b}^2 is the variance of individual effects. The last quantity is unobserved because only one potential outcome is seen for each unit. Dropping its non-negative term yields the usual conservative variance estimator. Cluster assignment requires cluster-level inference; treating members of a common guarantee group as independent can understate uncertainty.

A first planning approximation for equal-sized arms comparing binary endpoints with common planning rate p_0 and detectable difference δ is

$$n_{\text{per arm}} \approx \frac{2(z_{1-\alpha/2} + z_{1-\beta})^2 p_0(1-p_0)}{\delta^2}. \tag{A1.13}$$

Here α is the two-sided TypeI error target and $1-\beta$ is power. At 5% significance,80% power, $p_0 = 0.5$ and $\delta = 0.10$, using unrounded normal quantiles, the approximation is 392.44, rounded up to 393 per arm. This is a planning calculation under independent complete follow-up. Unequal allocation, censoring, multiplicity, cluster correlation and sequential stopping require the corresponding design calculation. For equal clusters of size m and exchangeable intracluster correlation ρ, the variance inflation $1 + (m-1)\rho$ follows by summing the m variances and $m(m-1)$ covariances of the cluster total.

When an offer is an instrument. Let Z be a random encouragement to take a fixed contract, $U(z)$ take-up, and $Y(u)$ an outcome defined under receipt and non-receipt. Under random encouragement, exclusion of a direct Z effect, monotonicity $U(1) \geq U(0)$, a non-zero first stage and the relevant consistency/interference conditions,

$$\frac{E[Y \mid Z=1] - E[Y \mid Z=0]}{E[U \mid Z=1] - E[U \mid Z=0]} = E[Y(1) - Y(0) \mid U(1) > U(0)]. \tag{A1.14}$$

To see why, partition applicants into never-takers, always-takers and compliers. Exclusion makes the first two groups' contribution to the assignment effect zero. The numerator is the complier proportion times their average receipt effect; the denominator is the complier proportion. Price and repayment-schedule changes can directly change outcomes even among people who take up both offers, so they require a different causal model unless this exclusion condition is defensible.

Source connection: Karlan and Zinman, "Observing Unobservables," motivates separating randomised contract information, take-up and repayment. The algebra above states the additional conditions for each estimand explicitly.

A1.4 Adaptive probabilities and the linear-UCB calculation

The linear confidence-bound construction follows [5]; inference under adaptive assignment follows [6].

Let $\hat{\pi}_i$ and q_i each be probability distributions on the approved set $\mathcal{A}(X_i)$. For a registered exploration weight $0 \leq \epsilon_i \leq 1$,

$$\pi_i(a \mid X_i, H_{i-1}) = (1-\epsilon_i)\hat{\pi}_i(a \mid X_i, H_{i-1}) + \epsilon_i q_i(a \mid X_i, H_{i-1}). \tag{A1.15}$$

Non-negativity follows from the component distributions; summing over a gives $(1-\epsilon_i)+\epsilon_i = 1$. If $q_i(a) \geq q_{\text{min}} > 0$ and $\epsilon_i \geq \epsilon_{\text{min}} > 0$ for an eligible action, then $\pi_i(a) \geq \epsilon_{\text{min}} q_{\text{min}}$. With two actions, uniform q, $\epsilon = 0.2$ and a current best-action point mass, the probabilities become 0.9 and 0.1. The floor applies only after hard eligibility restrictions. The realised final assignment probability must also account for any actual allocation-stage override mechanism; a deterministic human change cannot simply retain the pre-override randomisation probability.

For a linear reward model, use an action-context feature vector $\phi_i(a) \in \mathbb{R}^d$ and assume

$$R_i = \phi_i(A_i)^\top \theta + \varepsilon_i, \qquad E[\varepsilon_i \mid X_i, A_i, H_{i-1}] = 0. \tag{A1.16}$$

For ridge parameter $\lambda > 0$, minimise the accumulated squared errors plus $\lambda\|\theta\|^2$. Differentiating and equating to zero gives

$$\begin{aligned} B_i &= \lambda I + \sum_{j<i} \phi_j(A_j)\phi_j(A_j)^\top, \\ b_i &= \sum_{j<i} \phi_j(A_j) R_j, \\ \hat{\theta}_i &= B_i^{-1} b_i. \end{aligned} \tag{A1.17}$$

If a confidence ellipsoid is $\|\theta - \hat{\theta}_i\|_{B_i} \leq c_i$, the Cauchy–Schwarz inequality gives

$$\phi^\top \theta \leq \phi^\top \hat{\theta}_i + c_i \sqrt{\phi^\top B_i^{-1} \phi}. \tag{A1.18}$$

The LinUCB choice maximises that upper bound over permitted actions. Its claimed confidence coverage requires an appropriate c_i and assumptions such as bounded features and conditionally sub-Gaussian reward noise. A generic tuning constant has an exploration interpretation but no automatic safety or coverage guarantee. For an orthogonal two-dimensional example, $B = \text{diag}(4,1)$, $b = (8,1)^\top$, and $c = 1$ give $\hat{\theta} = (2,1)^\top$. The scores for unit-vector actions are 2.5 and 2.0. Greater uncertainty raises the second action's bonus from 0.5 to 1, while the policy constraints still determine whether either action is available.

Logistic Thompson sampling. For a binary event, set $p_\theta(x,a) = \{1 + \exp[-\phi(x,a)^\top\theta]\}^{-1}$ and update the specified posterior. Draw $\tilde{\theta}_i$ from that posterior and choose

$$A_i \in \arg\max_{a \in \mathcal{A}(X_i)} \widetilde{Q}_i(X_i, a). \tag{A1.19}$$

Here $\widetilde{Q}_i$ combines the sampled event probability with the documented take-up, exposure, severity, timing and cost model. If the event is default, maximising $p_{\tilde{\theta}}$ itself would maximise default probability. The allocation probability of Thompson sampling is the posterior probability that an action wins this value comparison,

subject to tie rules and eligibility; it is different from the sampled default probability. Hierarchical coefficient draws and Pólya–Gamma computation are derived in Article 3's appendix. Their sector, region and vintage effects use the same evidence and time-window definitions.

Source connection: Li and colleagues, "A Contextual-Bandit Approach to Personalized News Article Recommendation," develops the linear-UCB approach. Durand and colleagues, "Contextual Bandits for Adaptive Treatment Allocation," discusses controlled adaptive allocation. Hadad and colleagues, "Confidence Intervals for Policy Evaluation in Adaptive Experiments," addresses uncertainty under adaptive sampling.

A1.5 Derive inverse-propensity policy evaluation

The evaluation setting and overlap requirements follow [1], [12].

For a fixed target policy π evaluated on the study's context distribution, define

$$V(\pi) = E_X\left[\sum_a \pi(a \mid X)\mu(X,a)\right]. \tag{A1.20}$$

Assume the observed reward agrees with its assigned potential reward, the action assignment is conditionally exchangeable given recorded context and history, observed outcomes satisfy the declared measurement rule, and $e_i(a) > 0$ whenever $\pi(a) > 0$. Conditional on X_i, H_{i-1},

$$\begin{aligned} E\left[\frac{\pi(A_i \mid X_i)}{e_i(A_i \mid X_i, H_{i-1})} R_i \,\middle|\, X_i, H_{i-1}\right] &= \sum_a e_i(a)\frac{\pi(a \mid X_i)}{e_i(a)}\mu(X_i,a) \\ &= \sum_a \pi(a \mid X_i)\mu(X_i,a). \end{aligned} \tag{A1.21}$$

Averaging and applying iterated expectation proves the IPS identity

$$\widehat{V}_{\mathsf{IPS}}(\pi) = \frac{1}{n}\sum_i \frac{\pi(A_i \mid X_i)}{e_i(A_i \mid X_i, H_{i-1})} R_i. \tag{A1.22}$$

The proof permits predictable adaptive logging probabilities, but precision calculations must account for that adaptive process. A target policy selected using the same evaluated rewards also needs sample splitting or an analysis that addresses policy selection. When context or reward distributions change with calendar time, the estimator targets the declared mixture of those contexts and times; transport to a different period needs additional assumptions.

For weights w_i, the common concentration diagnostic is

$$n_{\mathsf{eff}} = \frac{(\sum_i w_i)^2}{\sum_i w_i^2}. \tag{A1.23}$$

It measures weight concentration, not a universal replacement for the actual sampling variance. Clipping w_i to $\min(w_i, c)$ changes the conditional expectation in the proof and generally introduces bias. The self-normalised estimator $\sum_i w_i R_i / \sum_i w_i$ is a ratio estimator; it can be consistent under appropriate conditions but does not inherit exact finite-sample unbiasedness from the IPS sum.

A1.6 Derive the doubly robust estimator and its remainder

The augmented estimator follows the doubly robust policy-evaluation construction in [1].

Let $\widehat{m}(x, a)$ be an outcome regression and $\widehat{e}_i(a)$ a working logging-probability model, fitted without using the evaluated observation's outcome. Define

$$\psi_i = \sum_a \pi(a \mid X_i)\widehat{m}(X_i, a) + \frac{\pi(A_i \mid X_i)}{\widehat{e}_i(A_i)}\{R_i - \widehat{m}(X_i, A_i)\}. \tag{A1.24}$$

Taking conditional expectation under the true e_i yields

$$E[\psi_i \mid X_i, H_{i-1}] = \sum_a \pi(a) \left[\widehat{m}_a + \frac{e_i(a)}{\widehat{e}_i(a)}(\mu_a - \widehat{m}_a)\right]. \tag{A1.25}$$

Subtract the desired conditional value to obtain

$$E[\psi_i \mid X_i, H_{i-1}] - \sum_a \pi(a)\mu_a = \sum_a \pi(a) \left(\frac{e_i(a)}{\widehat{e}_i(a)} - 1\right)(\mu_a - \widehat{m}_a). \tag{A1.26}$$

The remainder vanishes if the logging model is correct or if the outcome regression is correct. This is the double-robustness property. Estimation error, same-sample overfitting, weak overlap and adaptive dependence still affect consistency and uncertainty; cross-fitting and appropriate sequential inference address specific parts of those problems.

Reproducible four-record example. Observe actions (a, b, a, b) and rewards $(8, 4, 6, 2)$ under equal 0.5 logging probabilities. Evaluate a policy with probabilities $(0.75, 0.25)$. The weights are $(1.5, 0.5, 1.5, 0.5)$, giving

$$\widehat{V}_{\mathsf{IPS}} = (12 + 2 + 9 + 1)/4 = 6, \qquad n_{\mathsf{eff}} = 4^2/5 = 3.2. \tag{A1.27}$$

With working regressions $\widehat{m}_a = 6$ and $\widehat{m}_b = 4$, the direct-policy prediction is 5.5 for every record. Weighted residuals are $(3, 0, 0, -1)$, so

$$\widehat{V}_{\mathsf{DR}} = 5.5 + (3 + 0 + 0 - 1)/4 = 6. \tag{A1.28}$$

This small calculation checks arithmetic; it supplies no meaningful empirical precision estimate or policy-performance evidence.

Source connection: Dudík, Langford and Li, "Doubly Robust Policy Evaluation and Learning," and Wang, Agarwal and Dudík, "Optimal and Adaptive Off-Policy Evaluation in Contextual Bandits," establish the policy-evaluation framework. The expectation proof here shows exactly which assumptions the article invokes.

A1.7 Delayed outcomes and censoring

Let $M_i = 1$ indicate that the complete target-horizon reward is observed by the analysis date, and let $g_i = \mathsf{Pr}(M_i = 1 \mid X_i, A_i, H_{i-1})$. If outcome observation is conditionally independent of the completed reward given these variables and $g_i > 0$, then

$$E\left[\frac{M_i R_i}{g_i}\middle| X_i, A_i, H_{i-1}\right] = E[R_i \mid X_i, A_i, H_{i-1}]. \tag{A1.29}$$

Combining this observation weight with the action weight gives

$$\widehat{V} = \frac{1}{n}\sum_i \frac{\pi(A_i \mid X_i)}{e_i(A_i \mid X_i, H_{i-1})}\frac{M_i R_i}{g_i}. \tag{A1.30}$$

A 0.8 observation probability produces a 1.25 observation weight. If recent loans have zero probability of reaching the twelve-month endpoint by the analysis date, this identity cannot recover their realised twelve-month rewards; use mature cohorts or a clearly labelled forecasting model. If observation depends on unmeasured financial distress, the conditional observation assumption also fails. Early delinquency and final loss belong in a joint, prespecified model if both are used; treating them as independent repeated evidence about the same loan overstates information.

For event data, observe $\widetilde{T}_i = \min(T_i, C_i)$ and $\delta_i = 1\{T_i \le C_i\}$. Under independent right censoring conditional on the model covariates, the event-time likelihood contribution, omitting a separate censoring factor, is

$$L_i = \lambda_i(\widetilde{T}_i)^{\delta_i} \exp\{-\Lambda_i(\widetilde{T}_i)\}, \qquad \Lambda_i(t) = \int_0^t \lambda_i(u)\,du. \tag{A1.31}$$

An observed event contributes its density $\lambda_i S_i$; a censored record contributes survival S_i. Censoring does not mean a zero eventual loss. Administrative censoring, informative missingness and competing termination are different mechanisms and receive different treatment.

A1.8 Hazard, survival and competing-risk incidence

The hazard specification follows [7], with competing-risk and transition distinctions supported by [8], [9].

The event hazard satisfies $\lambda(t) = f(t)/S(t)$, where $S(t) = \Pr(T > t)$ and $f(t) = -S'(t)$. Rearranging gives

$$\frac{d\log S(t)}{dt} = -\lambda(t), \qquad S(t) = \exp\left\{-\int_0^t \lambda(u)\,du\right\}. \tag{A1.32}$$

For a Cox specification $\lambda(t \mid x, a) = \lambda_0(t)\exp\{x^\top\beta + g(a, x)\}$ with time-constant predictor,

$$S(t \mid x, a) = \exp\{-\Lambda_0(t)\exp[x^\top\beta + g(a, x)]\}. \tag{A1.33}$$

The hazard ratio compares instantaneous rates among those still at risk. It is not generally a probability ratio or a causal effect. Time-varying coefficients or proportional-hazards failures require an appropriately extended model.

With mutually exclusive terminal causes $k = 1, \dots, K$, define cause-specific hazard $\lambda_k(t)$ among loans still free of every terminal event. Their sum determines overall survival:

$$S(t) = \exp\left\{ -\int_0^t \sum_k \lambda_k(u)\, du \right\}. \tag{A1.34}$$

The probability of surviving to just before u and then exiting by cause k in du is $S(u-)\lambda_k(u)du$. Integration derives the cumulative-incidence function:

$$F_k(\tau) = \Pr(T \le \tau, J = k) = \int_0^\tau S(u-)\lambda_k(u)\, du, \qquad S(\tau) + \sum_k F_k(\tau) = 1. \tag{A1.35}$$

For constant annual hazards $\lambda_D = 0.04$ for default and $\lambda_P = 0.16$ for prepayment, the one-year default incidence is

$$F_D(1) = \frac{0.04}{0.20}(1 - e^{-0.20}) = 3.6254\%. \tag{A1.36}$$

The single-event expression $1 - e^{-0.04} = 3.9211\%$ describes a different calculation that removes the prepayment hazard. Treating prepayment as a risk-set exit is appropriate when estimating a cause-specific default hazard, but converting that hazard to absolute default incidence requires all competing hazards. This distinction explains the article's warning about treating prepayment as ordinary independent censoring in an absolute-risk calculation.

At event times t_j, let n_j be the at-risk count, d_{kj} cause-k events and $d_j = \sum_k d_{kj}$. The competing-risk product-limit construction is

$$\widehat{S}(t) = \prod_{t_j \le t} (1 - d_j/n_j), \qquad \widehat{F}_k(t) = \sum_{t_j \le t} \widehat{S}(t_j-)\frac{d_{kj}}{n_j}. \tag{A1.37}$$

Independent censoring conditional on the relevant strata and appropriately measured event causes are required. Cure and restructuring may instead be intermediate states. If $Q(t)$ contains transition intensities $q_{rs}(t)$, with $q_{rr} = -\sum_{s \ne r} q_{rs}$, the transition matrix solves

$$\frac{\partial P(s,t)}{\partial t} = P(s,t)Q(t), \qquad P(s,s) = I. \tag{A1.38}$$

The Aalen–Johansen estimator replaces intensity increments by observed transitions per risk set and forms the ordered matrix product

$$\widehat{P}(s,t) = \prod_{s < u \le t} (I + d\widehat{A}(u)). \tag{A1.39}$$

For transition interpretation under the stated model, the state description must carry enough history for the assumed Markov structure, or the analysis must explicitly extend it. The Fine–Gray subdistribution hazard instead satisfies

$$\tilde{\lambda}_D(t) = \frac{F_D'(t)}{1 - F_D(t)}, \qquad F_D(t) = 1 - \exp\left\{ -\int_0^t \tilde{\lambda}_D(u)\, du \right\}. \tag{A1.40}$$

Its denominator differs from all-cause event-free survival. Cause-specific Cox and Fine–Gray regression therefore attach coefficients to different quantities. Randomised or otherwise identified offer comparisons remain necessary for causal interpretation of differences between offers.

Source connection: Cox, "Regression Models and Life-Tables"; Aalen and Johansen, "An Empirical Transition Matrix for Non-Homogeneous Markov Chains Based on Censored Observations"; and Fine and Gray, "A Proportional Hazards Model for the Subdistribution of a Competing Risk."

A1.9 Fairness measures and uncertainty shown to reviewers

The conditional fairness rates follow [13].

Let G denote an evaluation group, $Y = 1$ the declared positive outcome and $\widehat{Y} = 1$ the declared positive decision. Then

$$\mathsf{TPR}_g = \mathsf{Pr}(\widehat{Y} = 1 \mid Y = 1, G = g), \qquad \mathsf{FPR}_g = \mathsf{Pr}(\widehat{Y} = 1 \mid Y = 0, G = g). \tag{A1.41}$$

Equal opportunity equates the TPR across groups; equalized odds equates both TPR and FPR. Accuracy also depends on the group's outcome prevalence p_g:

$$\mathsf{Accuracy}_g = p_g\,\mathsf{TPR}_g + (1 - p_g)(1 - \mathsf{FPR}_g). \tag{A1.42}$$

Therefore equalized odds does not generally imply equal accuracy. The positive label must be explicit: default detection and approval of sustainable repayment use different label conventions. With selective labels, these empirical rates initially describe the observed-outcome population. Weighted or model-based population rates require the corresponding selection assumptions and support.

A posterior credible interval for an identified conditional mean effect and a predictive interval for a future realised outcome answer different questions. For model parameters ϑ, future outcome Y_{new} and data D, the law of total variance gives

$$\mathsf{Var}(Y_{\mathsf{new}} \mid D) = E[\mathsf{Var}(Y_{\mathsf{new}} \mid \vartheta, D) \mid D] + \mathsf{Var}(E[Y_{\mathsf{new}} \mid \vartheta, D] \mid D). \tag{A1.43}$$

The first term is future outcome variability conditional on the model; the second is posterior parameter uncertainty. For human review, retain expected contribution, the effect or risk credible interval, the predictive cash range and any approved threshold-exceedance probability, each labelled with its horizon and assumptions. These quantities become causal evidence when the underlying comparison is causally identified and the evaluation addresses adaptive assignment, take-up, missingness and support.

Source connection: Hardt, Price and Srebro, "Equality of Opportunity in Supervised Learning," defines the group-rate distinctions. The uncertainty decomposition is an application of iterated conditional expectation and variance.

Further methodological reading

Additional sources retained from the article bibliography: [19].

From a One-Step Bandit to Sequential Credit Control

How to learn which collection action helps, for whom, and when

Underwrite for Collection · Article 2 of 5 · LinkedIn research article · 8 October 2026

Abstract

Credit models often estimate whether a borrower may miss a payment, but a lender must also decide what to do next, what evidence supports that action, and how the choice changes the borrower's later options. This article develops a measured path from a randomized collection pilot and contextual bandit to sequential decision-making, while treating affordability, lawful authority, privacy, fair treatment and human review as operating conditions. It explains why reject inference is a separate identification problem, how survival and transition models can describe evolving repayment, and when a partially observed Markov decision process is justified. A small hypothetical example illustrates delayed cash, cost and customer outcomes without presenting synthetic numbers as observed evidence. The final section maps the method to Kenyan data-protection law and to the practical lifecycle architecture developed in *Underwrite for Collection*.

This is the second article in a five-article sequence. Article 1 established the experimental unit as a complete credit offer and showed how fixed randomization can precede conservative contextual allocation. Here the decision unit changes: after disbursement, the lender may face repeated choices as payments, disputes, hardship, collateral and servicing evidence arrive. Article 3 isolates the selective-label problem that limits what historical approval data can tell us; Article 4 applies sequential control to bank-level funding, hedging and balance-sheet choices. The purpose of this progression is to keep the methods matched to their decisions: one-off offer allocation, repeated borrower servicing, missing repayment counterfactuals and institution-wide Treasury control require related but distinct evidence and governance.

Research questions

1. Among legally and operationally eligible Kenyan MSME credit accounts, what is the causal effect of a support-oriented collection action on verified net cash recovered and sustained repayment, relative to standard servicing?
2. Under what data, overlap, selection and governance assumptions can contextual-bandit or sequential off-policy evaluation estimate the value of alternative collection policies, and how sensitive are the estimates to reject inference and unobserved caseworker information?
3. When do repeated actions, borrower-state transitions and partially observed cash-flow conditions justify moving from randomized pilots and contextual bandits to model-predictive control, constrained MDPs or POMDPs, while preserving affordability, privacy, fairness and meaningful human review?

The question begins with the borrower's cash journey

The starting point for this work was practical: lenders can approve a facility that appears affordable at origination and still fail to support sustainable repayment across the months that follow. An enterprise may trade through several bank accounts, mobile-money wallets, payment aggregators and suppliers; a short cash dip may be seasonal, while a longer decline may signal that the business model or repayment schedule has become strained. A collection team then faces a decision under uncertainty: make a reminder, ask for updated information, offer a payment-date adjustment where contract and policy allow it, route the case to a trained officer, or take no additional action. The research question is therefore not simply "Who will default?" It is "Which permissible action, for this borrower and this observed state, improves durable repayment and overall outcomes?"

That framing follows the complete credit lifecycle, connecting the borrower's operating activity, financing arrangement, servicing decisions and eventual cash recovery. The evidence chain begins with the enterprise

and its purpose, follows the actual financing structure and cash source, records payments and servicing events, monitors changes after disbursement, and measures realised cure, recovery and loss. A business loan depends on the going concern: operating cash, working capital, obligations, owner involvement and continued trading. A facility backed by a ring-fenced receivables stream depends on eligible receivables, assignment or transfer effectiveness, obligor performance, dilution, account control, collection timing and a functioning waterfall. Those propositions need different state variables, interventions and recovery plans, even where the same lender holds both exposures.

The distinction matters once intervention is modelled over time. A business cash-flow dip may make a brief, respectful contact useful if it helps verify a temporary timing mismatch; a recurring shortfall may call for a human review of affordability, restructuring eligibility or business viability. In a ring-fenced receivables facility, the same apparent payment delay could instead arise from a disputed invoice, concentration in one payer, set-off, dilution, servicer interruption or a failure to transfer receipts to the controlled account. Treating both as a generic "high-risk borrower" state hides the mechanism that the action is meant to change. A credible decision system represents the source of repayment and the evidence about it, then constrains the action set to the actual product, contract, legal role and borrower circumstances.

What the one-step contextual bandit can and cannot answer

A contextual bandit is a useful first experimental design when a team observes a context x_t, selects one action a_t, and measures a defined reward r_t over a fixed follow-up window. The context might include days past due, recent payment regularity, verified cash-flow indicators, prior contact outcomes and the product's servicing rules. The action could be one of a small set of approved reminders or an invitation to speak with a trained officer. The method estimates which action tends to work better for which observed context, while recording the probability with which the action was assigned. It is "one-step" because the principal question is the effect of today's choice over its declared outcome window; it does not, by itself, learn an optimal series of actions across a changing borrower state.

The minimum experimental record is more precise than a model score. It includes the eligible population; the decision timestamp; the information available at that moment; the actions the case could actually receive; the policy and model versions; all safety and capacity gates; the randomization mechanism; the realized assignment probability; what was offered, delivered, received and accepted; any human override and reason; and outcomes with a prespecified maturity date. If a legal or affordability gate removes two actions, the logged propensity must be conditional on the remaining allowed actions, not the probability before the gate. This is essential for both transparent experimentation and later off-policy evaluation: a logged probability that does not describe the action actually available cannot correct the selection process.

The reward should reflect the research objective rather than a convenient system event. "Contact delivered" is not repayment; "promise to pay" is not cash collected; "account credited" may be a related-party transfer rather than trading income; and a short-term cure may reverse before the next billing cycle. A stronger primary outcome can combine verified net cash recovered by a specified date with repayment durability, while separately reporting the borrower's cost, complaint, hardship, repeat-delinquency and escalation outcomes. One can predefine a vector of outcomes and a decision rule rather than immediately compress every consequence into one scalar. When an organization does use a scalar objective, it should publish how weights were chosen, preserve legally protected constraints outside the reward, and test whether an apparent gain comes from shifting cost or burden to the borrower.

A randomized pilot is the right benchmark before adaptive assignment. Within a narrow, approved eligibility region, a team can randomize among low-risk, support-oriented actions or compare an action with the current standard process. It should not randomize access to a legally required notice, a complaint route, a hardship safeguard, or a treatment that policy says must be provided. Nor should it offer unaffordable credit or withhold service merely to create labels. Random allocation can make groups comparable in expectation, but small samples, non-compliance, spillovers between staff, differential contact success and delayed out-

comes still require careful analysis. A pilot should therefore state its estimand, unit of assignment, follow-up horizon, stopping rules and operational guardrails before the first case is enrolled.

Sequential control changes the mathematical problem

Sequential decision-making becomes relevant when today's action can change later states, future choices or the information available to the lender. A borrower who receives a reminder may pay, request assistance, dispute the debt, become temporarily unreachable, or experience no change. A case officer's choice can also alter future data: a conversation may reveal that funds are seasonal, a receivable is disputed or the owner has suffered a shock. A finite-horizon Markov decision process represents a decision date t, state s_t, allowed action set $A(s_t)$, transition law $P(s_{t+1} \mid s_t, a_t)$ and stage outcome r_t. The value function is

$$V_t(s) = \max_{a \in A(s)} \mathbb{E}[r_t(s, a, S_{t+1}) + \gamma V_{t+1}(S_{t+1})]\,, \tag{2.1}$$

with a declared horizon and discount convention. This equation describes a framework, not proof that the lender has the data or authority to implement it.

The action set is part of the model specification, not a footnote. It depends on the product agreement, the stage of arrears, known hardship, the collection channel, permitted contact times, complaints, active disputes, vulnerability, data quality and the person or team with authority to act. A hard gate can remove an action when a rule or policy makes it unavailable; a soft penalty can merely make an unlawful or harmful action appear unattractive under some reward weights. For this reason, a legally required notice, a borrower's right to object or a binding affordability restriction belongs in the eligible-state and action logic. The optimizer should choose only among actions that pass those gates, while staff remain able to pause automation and route a case for review.

The Markov assumption says that the represented state contains the information needed to predict the next transition for the chosen purpose. It is a modelling assumption to test, not a guarantee provided by machine learning. If material history is omitted—such as a prior promise-to-pay, a complaint, a payment reversal, an active dispute or a recent restructuring—the transition model may treat two different situations as identical. If the process is partly hidden, a partially observed Markov decision process (POMDP) can represent a latent condition z_t, an observation o_t, an action a_t, state transitions and an observation likelihood. The agent's information is then a belief distribution

$$b_t(z) = \Pr(z_t = z \mid o_{1:t}, a_{1:t-1}). \tag{2.2}$$

A belief state is useful only where its hidden state and observation model are operationally meaningful and validated.

Irregular payment and contact intervals can make a fixed daily or monthly grid misleading. A semi-Markov formulation can represent the elapsed time until a transition, so that "paid 14 days after contact" and "paid 14 days after a revised due date" are not confused. Survival analysis can model time to default, cure, payment, prepayment or another event, while competing-risk methods distinguish which event occurs first. A Cox model such as

$$\lambda_i(t) = \lambda_0(t) \exp\!\left(x_i^{\top}\beta^H + u^H_{s(i)} + v^H_{r(i)} + w^H_{c(i)}\right) \tag{2.3}$$

expresses a relative event hazard; it does not itself choose the next intervention. A hierarchical logistic model such as

$$\begin{aligned}\text{logit}(p_{i,\tau}) = \alpha^L + x_i^\top \beta^L \\ + u^L_{s(i)} + v^L_{r(i)} + w^L_{c(i)}\end{aligned} \tag{2.4}$$

predicts a defined binary outcome horizon. These models can inform a transition model, but repeated scoring alone does not constitute reinforcement learning.

Put the risk models inside the decision problem

Equations (A2.8), (A2.9) specify the full logistic hierarchy and hyperpriors; equation (A2.13) specifies the separate survival likelihood. The selection analysis in Article 3 applies the same named levels in equations (A3.19), (A3.20).

A hierarchical Bayesian logistic model estimates fixed-horizon default probability, while a hierarchical Cox proportional-hazards model describes time to default. These models answer complementary questions. A fixed-horizon model can be written

$$D_i(\tau) \sim \text{Bernoulli}(p_{i,\tau}) \tag{2.5}$$

with

$$\begin{aligned}\text{logit}(p_{i,\tau}) = \alpha^L + X_i^\top \beta^L \\ + u^L_{s(i)} + v^L_{r(i)} + w^L_{c(i)}\end{aligned}; \tag{2.6}$$

it estimates default probability by a specified horizon. A Cox model writes

$$\lambda_i(t \mid X_i) = \lambda_0(t) \exp\!\big(X_i^\top \beta^H + u^H_{s(i)} + v^H_{r(i)} + w^H_{c(i)}\big), \tag{2.7}$$

with survival

$$S_i(t) = \exp\left(-\int_0^t \lambda_i(u)\, \mathrm{d}u\right) \tag{2.8}$$

under the model. Hierarchical effects partially pool sparse groups, but they do not erase model uncertainty or create evidence in unsupported regions. The decision system should report the group, vintage, contract and data support behind each estimate, and use posterior intervals rather than treating a point prediction as a known risk.

A time-to-default estimate does not by itself require a sequential controller. If a fixed-term loan's price and repayment schedule are set at origination, the offer decision is made at that point; later survival information helps monitoring and portfolio forecasts, but it does not automatically authorize monthly repricing. Sequential control is relevant when the institution makes repeated, permitted decisions—such as a collection support action, a revolving-line decision under its contract, or a renewal offer—and when each choice can change later states or observations. A useful state can include the original contract, payment and contact history, verified business cash, product-specific repayment source, macro conditions, current arrears, prior restructuring and data confidence. The action list must then separate origination terms from post-disbursement servicing actions, each with its own legal and operational authority.

The transition model should represent the events that matter to the policy. An account can remain current, miss a payment, cure, restructure, default, prepay, refinance or enter recovery; a ring-fenced receivables facility may instead experience payer delay, invoice dispute, dilution, servicing interruption or cash-control failure. These states are not interchangeable. A multi-state model can estimate transitions among them, while cause-specific hazards can describe event timing. The controller's stage outcome can include incremental verified cash, intervention cost and a declared measure of repayment durability, but a legally required notice, affordability limit, complaint pause or prohibited practice should be enforced as a hard gate. A human reviewer can override a recommendation when new evidence or context changes the case, and the override reason should be logged so that the next evaluation can distinguish policy from execution.

The hierarchy has the same named levels in both models: sector s, region r and origination vintage c. Model label L identifies the fixed-horizon logistic model; H identifies the event-hazard model. Each model has its own coefficients, group effects and scale parameters, fitted to its declared outcome and likelihood. For every model m, the following non-centred representation supplies sector, region and vintage effects and their half-Student-t scale priors. The second argument of Normal denotes variance; B^m is a positive-definite coefficient covariance. The logistic intercept receives its own prior. For a fully Bayesian piecewise-constant baseline hazard, interval ℓ receives a positive Gamma prior with declared shape a_ℓ and rate b_ℓ. Cause-specific hazards repeat this structure with a separate event label. Any sharing between outcomes requires an explicitly specified joint model and validation.

$$u_s^m = \sigma_u^m z_{us}^m, \qquad z_{us}^m \sim N(0,1) \tag{2.9}$$

$$v_r^m = \sigma_v^m z_{vr}^m, \qquad z_{vr}^m \sim N(0,1) \tag{2.10}$$

$$w_c^m = \sigma_w^m z_{wc}^m, \qquad z_{wc}^m \sim N(0,1) \tag{2.11}$$

$$\begin{gathered}\sigma_k^m \sim \mathsf{HalfStudentT}(3, s_{0,k}^m), \\ k \in \{u,v,w\}, \quad m \in \{L,H\}\end{gathered} \tag{2.12}$$

$$\alpha^L \sim N(0, s_\alpha^2), \qquad \beta^m \sim N(0, B^m) \tag{2.13}$$

$$\lambda_{0,\ell} \sim \mathsf{Gamma}(\mathsf{shape} = a_\ell, \mathsf{rate} = b_\ell) \tag{2.14}$$

An IFRS 9 transition model is an accounting component, not the whole controller

Markov transition matrices are often used to describe migration among credit states, including current, Stage 2, Stage 3, cure and write-off. Such a matrix can support lifetime cash-flow projections, but IFRS 9 stages are accounting classifications governed by the standard's significant-increase-in-credit-risk and credit-impaired requirements. An MDP used to choose an action is a different object: it combines states, actions, transitions, rewards and a decision objective. A transition model can inform the MDP, while the accounting engine separately applies IFRS 9 measurement and staging requirements. Stage 3 is not automatically an absorbing state, and cure or improvement requires policy-consistent reassessment. A model should not define Stage 2 merely as a "bad state" for reward optimization or minimize provisions by discouraging recognition of deterioration [1].

The ECL calculation must remain distinct from the bandit reward and capital measures. IFRS 9 expected credit loss is a probability-weighted present value of cash shortfalls over the applicable horizon using reasonable

and supportable information; 12-month ECL is the portion of lifetime cash shortfalls associated with default events possible in the next twelve months, not simply cash shortfalls occurring during those twelve months [1]. The simplified approximation

$$\text{EL} \approx \text{PD} \times \text{LGD} \times \text{EAD} \tag{2.15}$$

can explain one expected-loss component under simplifying assumptions, but it is not a complete ECL model or economic-capital measure. If an intervention causes earlier payment, its effect on timing, recovery and expected cash shortfall should be assessed under the accounting method. The policy reward can then include separately defined operational costs or incremental net cash, without embedding accounting recognition or provisions as a proxy for borrower welfare.

Fairness diagnostics need precise targets

Credit fairness analysis should match the decision being evaluated. For a binary classification, equal opportunity compares true-positive rates across groups; equalized odds compares true-positive and false-positive rates conditional on the outcome; disparate-impact ratios compare selection rates under a chosen definition. These measures have different meanings, can conflict and depend on the label, threshold, base rates and population. They do not mean "equal accuracy," and none alone proves legal compliance or fair collection. Collection outcomes are especially difficult because the lender observes repayment after its own offer and servicing decisions, and because prior treatment can influence payment. A diagnostic should state which group, outcome, time horizon and action are being compared, how uncertainty and small samples are handled, whether the protected attribute is lawfully available for this analysis, and what accountable human process follows a concerning result [2].

Fairness constraints should be tested alongside—not instead of—hard eligibility and conduct rules. A constrained policy may be represented as

$$\max_{\pi} \mathbb{E}_{\pi}\left[\sum_{t} \gamma^{t} r_{t}\right] \tag{2.16}$$

subject to

$$\mathbb{E}_{\pi}\left[\sum_{t} c_{k,t}\right] \leq B_{k} \tag{2.17}$$

for each defined risk or burden k, plus action-level eligibility restrictions. Yet statistical parity constraints can be inappropriate if the outcome or exposure is misspecified, while group-specific thresholds can introduce other policy choices. Historical labels may be selective and may encode prior access patterns; group evaluation therefore requires a support analysis and sensitivity to unobserved outcomes. Feature removal is not enough because proxies may remain. A governance report should separate descriptive disparity, causal treatment effect, policy constraint and legal assessment rather than presenting one metric as a verdict.

Economic capital and RAROC require separate, declared conventions

Economic capital and risk-adjusted return on capital can inform the financial assessment of an action when their measurement conventions are explicit. A portfolio loss variable L can be simulated from borrower exposures, default timing, loss severity, recovery and dependence. Under an institution-defined horizon and confidence level, one illustrative economic-capital convention is

$$\mathsf{EC}_a = \mathsf{VaR}_a(L) - \mathbb{E}[L]; \tag{2.18}$$

another is $\mathsf{ES}_a(L) - \mathbb{E}[L]$. The institution must name its measure, horizon, confidence, diversification treatment and allocation method. It should also validate calibration and stress behavior. Economic capital is not the same as regulatory capital, and there is no universal confidence level or risk measure that every lender must use for pricing. The reported value is a model estimate under specified assumptions, not a cash reserve or a direct amount that can simply be added to a customer's rate.

An illustrative RAROC may be written

$$\mathsf{RAROC} = \frac{\begin{array}{c}\text{risk-adjusted income} - \text{expected loss} - \text{funding cost} \\ -\text{operating cost} - \text{tax adjustments}\end{array}}{\text{economic capital}}, \tag{2.19}$$

but institutions define numerator, denominator, horizon and allocation differently. A pricing or collection action may change income, expected loss, take-up, balance duration, recovery, funding and required capital simultaneously. The controller should not optimize headline RAROC without retaining borrower outcomes, absolute loss limits, concentration constraints and affordability. A portfolio can show high RAROC on a thin or volatile denominator while carrying unacceptable exposure; a product with lower transaction-level return might support diversification or strategic access. Use RAROC as a governed decision lens with sensitivity analysis and ALCO or credit authority approval, rather than a universal reward or hurdle silently embedded in code.

For a worked hypothetical calculation, suppose a policy action is estimated to add KES 1,100 in verified net cash per eligible account over the chosen horizon, adds KES 250 in staff and operating cost, and uses KES 5,000 of allocated economic capital. The incremental cash after that direct cost is KES 850.

$$\frac{\text{KES } 850}{\text{KES } 5{,}000} = 17\% \tag{2.20}$$

is only an illustrative incremental cash-to-capital ratio; it is not automatically the institution's RAROC because it omits income attribution, funding, expected loss, tax, capital allocation and the institution's approved horizon. If a risk committee uses a 15% hurdle, the scenario does not prove that the policy passes it. The example is useful precisely because it reveals which terms still need a controlled, reconciled definition before decision-makers compare an intervention with alternatives.

Default dependence: Clayton can be a challenger, not a contagion story

Portfolio dependence affects tail loss and concentration, but a copula is a dependence model, not a mechanism that automatically explains contagion. A Clayton copula with parameter

$$\theta > 0 \tag{2.21}$$

has lower-tail dependence

$$\lambda_L = 2^{-1/\theta} \tag{2.22}$$

and no upper-tail dependence in its standard bivariate form; its Kendall rank correlation is

$$\tau = \frac{\theta}{\theta + 2}. \tag{2.23}$$

It can represent stronger dependence in the lower tail for variables oriented so that the relevant adverse outcome lies there. It does not by itself create time-varying contagion, identify a macroeconomic shock or imply that default probabilities jump at the instant another borrower defaults. A Gaussian copula with finite correlation has zero asymptotic tail dependence, but it still represents dependence at finite quantiles. Model selection should compare the hypothesized mechanism, portfolio evidence, tail diagnostics, stress scenarios and alternatives such as t-copulas or factor structures [3].

The capital framework must preserve two tracks. Regulatory capital uses the standards and local rules applicable to the institution and exposure class. Basel's IRB risk-weight functions are based on an asymptotic single-risk-factor framework with prescribed parameters and eligibility rules; that does not mean every regulatory capital calculation is a fitted Gaussian copula, nor that an internal Clayton model can replace the regulatory calculation [4]. An internal economic-capital model can use a richer dependence structure if it is justified, validated and governed, but results should be labelled internal and should not be reported as regulatory capital. A difference between internal and regulatory estimates is a risk-management signal for analysis, not proof of regulatory arbitrage or evidence that the bandit should automatically increase prices.

Pólya–Gamma computation supports posterior estimation

The Gaussian conditional posterior is derived in equation (A2.28) and expanded for selective labels in equation (A3.30). Its service-facing numerical form appears in equations (A5.33), (A5.34).

For hierarchical logistic regression, Pólya–Gamma augmentation gives a convenient conditional-Gaussian representation. With binary $y_i \in \{0, 1\}$, linear predictor

$$\eta_i = x_i^{\mathsf{T}} \beta, \tag{2.24}$$

and

$$\kappa_i = y_i - \frac{1}{2}, \tag{2.25}$$

introduce

$$\omega_i \sim \mathsf{PG}(1, \eta_i). \tag{2.26}$$

Given ω, the likelihood contribution is quadratic in β; with a Gaussian prior, the conditional posterior is multivariate normal, with covariance of the form:

$$\Sigma^* = \left(X^{\mathsf{T}} \Omega X + \Sigma_0^{-1}\right)^{-1} \tag{2.27}$$

and a corresponding mean determined by $X^{\mathsf{T}} \kappa$ and the prior mean [5]. Hierarchical priors add group-level parameters and may require additional sampling steps. The method can make posterior draws practical for logistic PD or selection components, subject to convergence checks, effective sample size, prior sensitivity and computational diagnostics. "Exact" refers to the augmentation identity and conditional sampler under the specified model; it does not mean that a model is exact about reality or that labels are identified.

Those posterior draws can support uncertainty-aware policy evaluation, such as estimating a range of expected loss or the probability that one allowed action outperforms another. They should not be interpreted as a license for the algorithm to explore every action the posterior considers plausible. The experiment still needs a safe action set, sufficient support, minimum allocation rules, a control group, outcome maturity and human governance. Posterior uncertainty can be understated by model misspecification, unobserved selection, changing macro conditions or incomplete histories. A robust implementation checks chain mixing and convergence, posterior predictive performance, calibration across vintages and sensitivity to priors and feature construction. Where the model is too uncertain for a consequential decision, the system should be able to defer to a pre-approved rule or trained reviewer rather than convert uncertainty into a high-stakes experiment.

In the compact Pólya–Gamma equations, the design matrix is expanded to include the intercept, approved feature bases, sector indicators, region indicators and vintage indicators. Its coefficient vector stacks the corresponding logistic coefficients and group effects; the prior covariance carries their current hierarchy scales. The logistic Thompson-sampling notation later in the article uses the same expansion when the hierarchical model supplies its probabilities. The hazard model preserves those named group levels in its own likelihood and posterior computation. This convention makes a short matrix expression reproducible: the registered design specification states every column, constraint, group mapping and coefficient block, while the mathematical appendix gives the likelihoods and conditioning steps used to fit them.

Half-Student-t priors connect partial pooling to decision uncertainty

A practical collection model must learn how much sectors, regions, institutions and origination cohorts differ while retaining useful estimates for groups with limited observations. This is where a half-Student-t prior with three degrees of freedom enters the hierarchy. For a group effect u_j on the log-odds scale, write u_j as a positive scale σ multiplied by a standard-normal variable z_j. The prior on σ governs the plausible amount of between-group variation and consequently the strength of partial pooling. A smaller inferred scale draws group effects closer together; a larger scale permits stronger differences supported by the observations. The selected prior scale s_0 is calibrated through prior-predictive default probabilities and compared with plausible alternatives, following the sensitivity discipline for hierarchical variance parameters [6].

$$u_j = \sigma z_j, \qquad z_j \sim \mathcal{N}(0,1) \tag{2.28}$$

$$\sigma \sim \text{HalfStudent-t}(\nu = 3, \text{scale} = s_0) \tag{2.29}$$

The Gaussian scale-mixture representation makes the construction concrete. Draw a positive precision λ from a Gamma distribution and draw a signed auxiliary quantity q from a normal distribution whose variance depends on λ; the absolute value of q has the required half-t distribution [7]. In these equations, Gamma uses shape and rate, so three Student-t degrees of freedom correspond to Gamma shape 3/2 and rate 3/2. The symbol s_0 denotes the prior scale, and the second argument of Normal below denotes variance. This construction gives a precise meaning to the heavy-tailed scale prior. In the collection hierarchy, σ describes variation across groups, while u_j remains Gaussian conditional on σ. A t-copula, discussed earlier, has a separate job describing dependence between outcomes.

$$\lambda \sim \text{Gamma}(\text{shape} = 3/2, \text{rate} = 3/2) \tag{2.30}$$

$$q \mid \lambda \sim \mathcal{N}(0, s_0^2/\lambda), \qquad \sigma = |q| \tag{2.31}$$

These scale variables and the Pólya–Gamma variables fit into different steps of a blocked inference procedure. Given the current hierarchy scales and the Pólya–Gamma variables, the regression and Gaussian group-effect coefficients have the normal conditional distribution described above [5]. The hierarchy scales, auxiliary variables and any correlation parameters receive their own validated updates. Thus the useful computational result is conditional Gaussian inference within a larger posterior calculation. The chosen parameterisation still needs convergence, effective-sample-size and sensitivity checks. Operationally, the resulting draws carry uncertainty about sparse groups into predicted default, cure or other separately specified transition probabilities. Those distributions can then inform limit-review decisions and comparison of permitted collection actions. The model record should preserve which prior, data window and calibration version generated the distributions used for each decision [8].

The practical reason for this construction is to make calibrated uncertainty computationally feasible at the frequency the credit process needs. An offline fitting service can reuse the conditional-Gaussian coefficient blocks, update the necessary latent variables and scales, and publish an approved set of posterior draws or summaries for scoring. Selection of the half-t scale and parameterisation balances plausible group variation, prior-predictive behaviour, numerical stability and computation time. Validation compares calibration, tail probabilities, effective sample size, convergence and decision stability with a reference fit. The serving system then uses the versioned artefact within its approved data and policy window; a scheduled or event-triggered refresh follows change control and validation. Combined with an identified treatment comparison, these calculations give a causal credit decision process usable uncertainty about outcomes and action value, while preserving the assumptions under which each estimate was produced.

Bayesian decision boundaries turn predictions into limit-review evidence

The expected-value boundary in equation (2.35) is derived in Appendix A2.12. The distinct risk-cap, incremental-benefit and predictive-cash probabilities used by a reviewer appear in equations (5.1), (5.2), (5.3).

A credit-limit adjustment is a decision about prospective exposure, future cash generation and the consequences of each permitted action. Consider a proposed extension over a declared horizon. Let p be its default probability, G the net gain conditional on repayment, L the net loss conditional on default, and O an explicitly valued opportunity cost of declining a creditworthy case. All three monetary quantities use the same horizon, currency and cash-flow convention. The comparison below holds the payoffs fixed and averages over the posterior distribution of p given evidence D. Set O to zero when the payoff comparison already captures the relevant foregone income, so the same benefit is counted once. Under this simple two-outcome, linear-payoff model, the break-even rule depends on the posterior mean $\bar{p}$; uncertainty receives a separate policy test [9].

$$\bar{p} = \mathbb{E}[p \mid D] \tag{2.32}$$

$$\mathbb{E}[V(\text{extend}) \mid D] = (1-\bar{p})G - \bar{p}L \tag{2.33}$$

$$\mathbb{E}[V(\text{decline}) \mid D] = -(1-\bar{p})O \tag{2.34}$$

$$p^* = \frac{G+O}{G+O+L} \tag{2.35}$$

For a hypothetical extension, suppose net gain on repayment is KES 1,200, net loss on default is KES 18,800, and the comparison uses O=0. The break-even posterior mean default probability is 6%. At a posterior mean

of 4%, the extension has expected value KES 400 under the stated assumptions. This calculation makes the economics visible: approved income and costs determine G, while prospective utilisation, recoveries, recovery timing and collection costs determine L. A revolving facility therefore needs a stress-drawdown assumption; its current drawn balance alone is insufficient to specify the loss. Larger limits, different repayment schedules or slower recovery change the comparison. The next decision step checks verified repayment capacity, aggregate obligations, concentration and the institution's approved exposure limits before selecting an action.

$$p^* = \frac{1{,}200}{1{,}200 + 18{,}800} = 6\% \tag{2.36}$$

$$\mathbb{E}[V(\text{extend}) \mid D] = 0.96 \times 1{,}200 - 0.04 \times 18{,}800 = \text{KES } 400 \tag{2.37}$$

A posterior exceedance rule adds a direct uncertainty check. Let p_cap be an approved product risk cap and ε the largest acceptable posterior probability of exceeding it. Calculate the exceedance probability from the model's posterior draws, recording both policy values and the reason for their selection. When that probability exceeds ε, the decision may route to verification, a smaller permitted offer, retention of the current limit or an authorised reviewer, according to the contract and credit policy. Two cases with the same posterior mean can receive different review routes because their uncertainty differs. Sparse or conflicting records should prompt a proportionate response to the evidence gap. Any consequential limit change also follows the information, explanation, correction and human-review arrangements developed in the article's Kenyan legal framework [10], [11].

$$\begin{gathered} \Pr(p > p_{\text{cap}} \mid D) > \varepsilon \\ \Longrightarrow \text{apply the approved uncertainty-review route} \end{gathered} \tag{2.38}$$

Meaningful human review needs a compact evidence packet that keeps several uncertainty measures visible. A credible interval for the default-probability parameter describes uncertainty about risk over the stated horizon, and the posterior cap-exceedance probability shows how that risk relates to the approved limit. Under an identified experimental or causal design, a separate credible interval can describe the conditional average incremental effect of an intervention on net cash or sustained cure. A posterior predictive cash interval also includes variation in the future realised outcome. The reviewer should see the posterior distribution of the difference between permitted actions, together with model versions, data age, affordability and support in comparable cases. Overlap between separate action intervals is interpreted through that comparison and the decision costs. For sparse groups, the packet explains pooling and prior sensitivity. Authorised options include requesting verification, maintaining the current limit, choosing a supported alternative or referring the case, with the decision and rationale recorded.

The connection to sequential collection is the effect a limit decision can have on the borrower's subsequent state. An extension may finance a verified operating cycle and support repayment; it can also increase utilisation, future instalments and exposure during financial stress. The controller therefore evaluates action-specific outcomes, costs and transitions using evidence that supports those actions. A default model trained under historical limits supplies an estimate for its documented conditions; causal evidence is needed to estimate how changing the limit changes repayment. Within a permitted sequential policy, posterior draws can enter the belief state or transition model, and the payoff comparison can inform action values. Affordability, aggregate exposure, lawful processing, contract terms and approval authority define which actions are available at each review date.

Reject inference leads to article 3

These methods raise a further question: how can a lender assess repayment when an application was previously rejected? This is a natural link to the companion article, "Who Gets a Repayment Label?" A bandit can collect outcomes only for actions that are actually assigned and taken up. If every applicant below a cut-off remains declined, the policy has no repayment observations for that region; Bayesian imputation does not change this fact. A safe offer experiment could generate evidence among a screened, eligible population near a decision boundary, while preserving affordability and legal safeguards. It estimates the effect or repayment experience for that specified offer and sampled population. Historical rejects outside the support region remain unsupported, and any inference about them should be presented as a sensitivity analysis rather than as observed default labels [12], [13].

The first two articles assign distinct roles to the methods. The risk models describe outcomes conditional on data, contract and assumptions; the experiment creates comparative evidence for a defined action; the bandit allocates among approved offers using logged probabilities; and the sequential controller evaluates repeated servicing choices only when actions affect future states. Article 3 examines why historical rejection prevents complete outcome labels, and article 4 moves the control frame to bank funding and balance-sheet exposures. Governance determines which actions are allowed, who reviews them, and how outcomes can be challenged. Capital and fairness analyses inform constraints and oversight but do not replace these functions. This sequence supports a practical agenda: learn from controlled offers and measured collection actions, preserve a standard-care comparison, and expand only where evidence supports the next decision.

A sequence, a ledger and an evaluation plan

The operational sequence begins with an evidence ledger. At each decision point the record should state what was known, when it was known, where it came from, whether it was verified, which entity and account it referred to, and how it was linked to the borrower or facility. A cash receipt can be recurring, seasonal, borrowed, transferred between related accounts, reversed or disputed. For a trading business, the distinction between customer receipts and owner injections can change the sustainability story. For a ring-fenced receivables stream, the record should distinguish invoice issuance, eligibility, obligor acceptance, dilution, assignment, collection account movement, remittance and waterfall allocation. This provenance allows analysts to measure the state as it existed at the decision date rather than quietly using later information that would not have been available to the policy.

The next record is a state transition table that connects an observation to the decisions staff can take. For example, a "missed instalment" state might branch to verified timing issue, unresolved payment allocation, dispute, hardship, emerging business weakness, or possible manipulation; each branch has its own evidence quality and response route. A ring-fenced facility may instead branch to payer delay, invoice dispute, concentration breach, servicer delay, account-control failure, or legal-enforceability concern. The table should identify the owner of each field, update frequency, missingness code, permissible use, retention trigger and correction pathway. A state label should not become an irreversible borrower attribute: when corrected information arrives, the system should support reassessment, preserve the audit trail and allow the person affected to understand the decision.

Relationship graphs, contextual bandits and sequential search

A temporal heterogeneous graph can represent the evidence needed to reconstruct a collection decision. Its nodes may include borrowers, operating businesses, beneficial owners, guarantors, accounts, buyers or other obligors, invoices, collateral, contracts, servicers, cash receipts and interventions. Typed edges can record ownership, control, invoice assignment, payment, security, servicing, communication and account movement. Each node and edge should carry source, confidence, observed-at and effective-at times, version, and correction history. This makes the graph a data and state representation; it need not be a graph

neural network. It can help trace whether cash came from a customer, a related party or a ring-fenced receivable, and whether the relevant buyer, account and waterfall links were in force at the decision date. Entity-resolution mistakes, shared-group label leakage, temporal leakage and access controls must be tested before graph-derived features enter a model. A compact encoding is:

$$x_t = \varphi(G_t) \tag{2.39}$$

Here G_t is the evidence graph available at time *t*, and φ is a documented feature or state transformation, which may be rules-based or learned.

Linear UCB and logistic Thompson sampling remain one-step contextual-bandit policies: they compare approved actions using the current context and observed reward. Graph-derived context can enrich x_t, or a validated graph model can provide predictions, but neither method by itself searches a whole sequence of future interventions. In a linear reward model, LinUCB uses a predicted mean plus an uncertainty bonus:

$$\mathsf{UCB}_t(x, a) = x_t(a)^\mathsf{T}\hat{\theta}_t + \alpha\sqrt{x_t(a)^\mathsf{T}A_t^{-1}x_t(a)} \tag{2.40}$$

Logistic Thompson sampling draws plausible parameters for a binary outcome model, then uses the sampled probabilities with loss, timing and cost estimates to construct a sampled risk-adjusted action value:

$$\tilde{\theta}_t \sim p(\theta \mid D_t), \qquad \tilde{p}_t(a) = \sigma\big(x_t(a)^\mathsf{T}\tilde{\theta}_t\big) \tag{2.41}$$

$$a_t = \operatorname*{arg\,max}_{a \in A_{\mathsf{safe}}} \widetilde{Q}_t(a) \tag{2.42}$$

The choice is made over a pre-approved safe set; maximizing a sampled default probability would be the wrong objective. Pólya–Gamma sampling is one computational method for logistic posterior draws, while PG-TS applies that idea to logistic contextual bandits [14].

Monte Carlo Tree Search enters only after the problem has a sequential state-transition model, a permitted action set and a sufficiently credible simulator or replay environment. A tree-search rollout can compare paths such as verify an invoice, contact the buyer, then review a dispute, against a permitted reminder followed by a hardship review; each branch leads to possible next states and outcomes. The search tree is a temporary unfolding of candidate action sequences, distinct from the persistent borrower–invoice–account evidence graph. A common node-selection score balances the estimated path value and visits:

$$U_{\mathsf{MCTS}}(s, a) = Q(s, a) + c\sqrt{\frac{\ln N(s)}{N(s, a)}} \tag{2.43}$$

A finite-horizon path value can be summarized as:

$$Q_H(s, a) = \mathbb{E}\left[\sum_{k=0}^{H} \gamma^k r_{t+k} \,\middle|\, s_t = s, a_t = a, \mathsf{model}\right] \tag{2.44}$$

Here Q is the mean rollout return, N records search visits, and c controls search allocation. In credit operations, this is offline computational search over a validated model, not live experimentation on borrowers.

Legal, affordability, hardship, dispute, contact-frequency, operational and human-review gates must constrain every step. MCTS should be compared with rules, dynamic programming or model-predictive control, and stress-tested for model error and unsupported trajectories [15].

"Graph Shapley" belongs to a separate interpretability step. For example, GraphSVX is a post-hoc local explanation method for graph-neural-network predictions: it estimates how selected features, nodes or graph components contribute to a particular model output under its coalition definition [16]. That attribution does not plan an intervention, estimate its causal effect, or explain why an MCTS rollout has greater long-term value than another path. If graph predictions inform a decision, show the prediction explanation alongside the competing action paths, expected outcomes, uncertainty, constraints and human rationale. Keep the action-value comparison distinct from an explanation of one model prediction, and record how graph perturbations affect the explanation.

The evaluation should be organized around a predeclared estimand. One option is the average effect of offering an approved contact action versus standard servicing among eligible cases, measured as incremental net cash realized within 60 days and sustained status at 90 days. Secondary outcomes might include contact burden, complaint rate, repeat arrears, hardship outcomes, time to resolution and staff effort. "Incremental" matters: cash that would have arrived anyway should not be credited to the intervention. "Net" matters: direct contact or servicing cost is deducted, while customer harm cannot simply be priced away. Separate cohort reporting should show the eligible population, assignment, delivery, compliance, loss to follow-up and outcome maturity so that a large nominal lift is not driven by a changing denominator.

For an adaptive policy, the assignment probability at each step becomes part of the data-generating process. A contextual bandit can explore among allowed actions and adapt to observed reward, but adaptive allocation can reduce simple interpretability and complicate variance estimation. An inverse-propensity estimator weights each observed reward by the inverse probability of its selected action; doubly robust estimators combine this weighting with an outcome model. These methods can be statistically efficient when their assumptions hold, but rare actions with tiny propensities can produce unstable weights, while deterministic assignment destroys overlap. The protocol should set minimum support, cap weight influence transparently, report effective sample size, use uncertainty intervals, and keep an unadapted randomized benchmark where feasible. Sequential doubly robust methods extend ideas to trajectories, but require correct logging at every decision time [17].

The concept chart shows the progression from evidence collection to a governed decision. It places legal, affordability, product and operational eligibility before any optimization, because an algorithm should compare only actions that the institution has already authorized for the defined population. The fixed pilot supplies a benchmark; the contextual bandit estimates one-step action value; and the later branch asks whether an action changes a borrower's future state or information enough to justify sequential control. The diagram also shows the stopping point: if the evidence does not support a more complex model, the lender can retain a validated one-step policy. This is a conceptual map, not an automatic deployment sequence; each transition requires evidence, review and an approved use case.

A worked example with deliberately hypothetical numbers

Consider a hypothetical Kenyan enterprise borrower with a KES 120,000 instalment due on day 30. The lender observes irregular customer receipts, a recent delayed payment and a high-confidence invoice accepted by one major buyer but payable on day 38. These are scenario assumptions, not empirical findings. Two policy-permitted actions are available: A, a neutral reminder and self-service payment link; and B, a respectful contact offering a trained review of payment timing and business cash evidence. The standard policy is reminder only. For a randomized pilot, eligible cases can be assigned 50:50 after hardship, complaint and active-dispute gates are applied. The question is whether the review offer increases durable repayment compared with the reminder, not whether the borrower's profile "deserves" pressure.

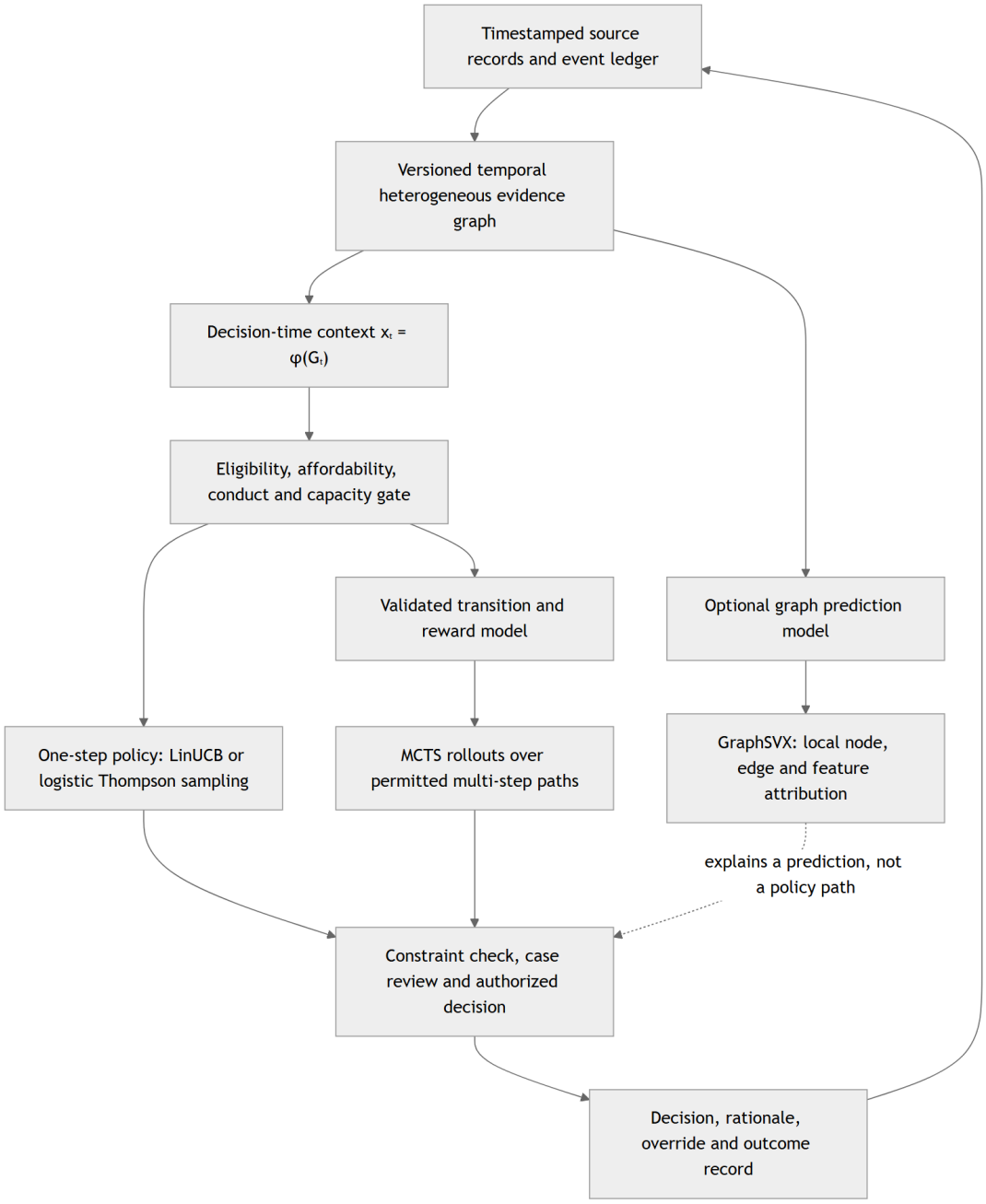


Figure 2: Temporal graphs, bandits, MCTS and GraphSVX

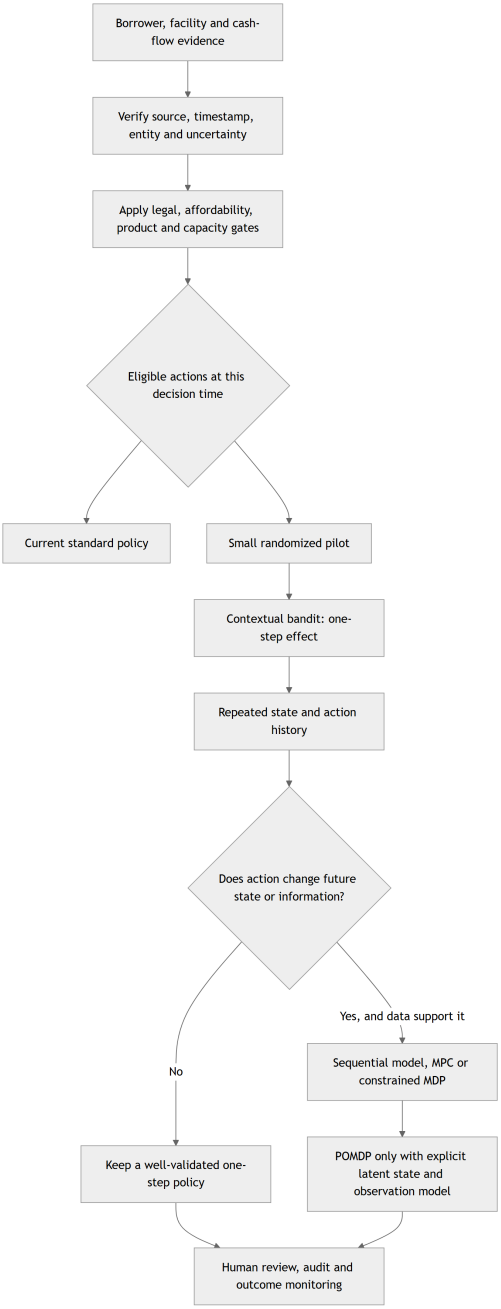


Figure 3: Evidence-to-sequential-control pathway

Suppose 100 eligible, hypothetical cases are evenly randomized. By day 60, 38 reminder cases and 46 review-offer cases have paid the scheduled instalment; by day 90, sustained payment is 34 and 41 respectively. With 50 assigned cases per arm, the day-60 rates are 76% and 92%, a difference of 16 percentage points; the day-90 rates are 68% and 82%, a difference of 14 percentage points. These are illustrative counts only; no confidence interval, statistical significance or business conclusion follows without sample-size planning, outcome definitions and observed data. Delivery may differ from assignment, and a case may pay through a channel the system did not initially link. The primary analysis should therefore be intention-to-treat, with delivery and acceptance reported as process outcomes and a secondary compliance analysis clearly labelled as such.

The next question is whether the review offer creates incremental value. Suppose, solely for illustration, the review process costs KES 250 per eligible case in staff time and operating expense, and additional verified cash collected by day 90 averages KES 1,100 per assigned case in the review group compared with standard policy. The illustrative average net benefit is:

$$\text{KES } 1{,}100 - \text{KES } 250 = \text{KES } 850 \quad \text{per assigned case} \tag{2.45}$$

before funding, capital, tax, servicing and borrower outcomes are considered. If the cash difference disappears after cohort adjustment or the 90-day cure reverses by day 120, the case for scaling weakens. If the intervention increases complaints, induces unaffordable payment sacrifices or places hardship cases under inappropriate pressure, a positive cash metric alone does not make the policy acceptable.

Now compare this with a sequential approach. At day 30 the reminder is sent; at day 38 the buyer has not paid; at day 40 an officer verifies that the receivable is genuine and the payment is scheduled; on day 42 the lender offers a contractually permitted adjustment or support referral; on day 60 the outcome is measured. The action at day 40 depends on new evidence generated by earlier events. A sequence model can compare action paths, such as reminder→review→standard repayment versus reminder→no additional contact, while accounting for cost and outcomes at each date. Yet an exact dynamic program would require transition estimates for these states and actions, a defined horizon and reliable support. If those are not available, a transparent rule plus a pilot may be a better operational answer than a simulated optimum.

The Kenyan legal and governance framework is part of the design

The Data Protection Act, 2019 creates duties for processing personal data. A company is not itself a natural-person data subject, but a business file can contain personal data about a sole proprietor, director, guarantor, employee or identifiable customer. Transaction-level cash evidence may also reveal private or sensitive circumstances depending on what it shows. The design must identify controller and processor roles, define the purpose and lawful basis, minimize the information collected, provide appropriate notices, maintain accuracy, limit retention, protect the data and enable the statutory rights that apply [10]. Consent is one possible legal basis in specified circumstances; it is not a universal substitute for purpose limitation, necessity, fairness or a valid processing condition.

Section 31 of the Act requires a data-protection impact assessment (DPIA) where processing is likely to result in high risk to the rights and freedoms of a data subject; section 35 addresses decisions based solely on automated processing that produce legal effects or significantly affect a person, subject to statutory exceptions and safeguards [10]. Regulation 49 of the General Regulations, 2021 identifies high-risk operations requiring a DPIA, while regulations 50–53 address its conduct, prior consultation, consideration and audit. Regulation 22 sets out safeguards for automated individual decision-making, including notice, meaningful information about logic, transparency and fairness measures, error prevention, appropriate statistical procedures, bias reduction and human intervention [11]. A decorative review button does not establish meaningful human involvement; the reviewer must have context, authority, time and the ability to change the result.

The other provisions matter across the data lifecycle. The Act's principles, rights, lawful-basis rules, security duties, breach obligations, sensitive-data conditions and cross-border-transfer rules apply according to the actual processing rather than the model label [10]. Data minimization affects feature design; accuracy affects entity matching and correction; purpose limitation affects whether data gathered for one service can be repurposed for scoring; retention affects trajectory storage; and security affects account-level feeds and access. A model-development project should define a controlled research dataset, access roles, pseudonymization where appropriate, linkage approvals, retention periods and a process for correcting source records. Removing direct names does not automatically make detailed transaction histories anonymous, especially where rare patterns can identify a person.

Credit-specific rules also require scope discipline. The Credit Reference Bureau Regulations address credit information and include safeguards relevant to automated scoring and adverse credit information, while the Digital Credit Providers Regulations, 2022 apply within their stated scope and exclude specified institution categories. The ODPC's digital-credit guidance provides further operational context [18], [19]. Before relying on a particular rule, the institution must classify itself, the product, the decision and the data flow. A bank, a digital credit provider, an MFI and a SACCO do not necessarily sit under the same sector rulebook. The article therefore treats consumer protection, credit reporting, banking supervision and data protection as connected controls, while leaving institution-specific legal conclusions to a documented Kenyan-law review.

Operational validation and model controls

Model validation should cover calibration and discrimination, but also stability, missingness, subgroup performance, decision thresholds, action availability, override frequency, data freshness, contact delivery, outcome maturity and rollback criteria. For a survival model, calibration at a defined horizon and competing-event treatment matter; for an MDP, transition calibration and value estimates matter; for a POMDP, belief quality and observation model matter. A model can have good ranking metrics and still recommend actions that do not improve outcomes. Independent validation should replay historical data only where the information timing is reconstructed, compare against simple rules and current policy, stress periods of economic change, and report performance by product and segment. Any simulator used for policy learning must be tested against observed distributions and known constraints before it informs a real decision.

A data-lineage lesson from mortgage performance files

Article 1 discusses the GSE data architecture as a template for dated origination and performance records. For sequential collections, the specific lesson is to preserve the information set at each decision: what the servicer knew before contact, what action was sent, what response followed, and whether the later payment can be linked to that action. Monthly mortgage histories are not event logs for Kenyan MSME collections, but they show why release dates, loan identifiers, outcome windows and corrections must be versioned [20], [21]. They also illustrate the selection boundary: published acquired mortgages do not reveal outcomes for every denied applicant. Use these datasets to validate time alignment and data lineage; estimate local collection effects only from properly governed Kenyan records and designs.

From borrower evidence to bank decisions

A borrower-level intervention experiment connects to the institution through clearly assigned responsibilities. The lending and servicing teams define the borrower, facility, decision date, permissible action and realised cash or customer outcome. Where repayment comes from an operating business, the evidence follows trading activity, working capital and the obligations supported by that cash. Where repayment comes from a ring-fenced receivables stream, it follows eligible receivables, legal transfer, payer performance, servicing continuity, controlled accounts, reserves and the contractual payment waterfall. Finance and risk teams translate reconciled outcomes into the institution's approved loss, capital and funding measures. An analytical workbench can present these records, model distributions and proposed actions to

authorised staff. Every handoff carries an agreed definition, source, timestamp, permitted purpose, responsible owner and reconciliation rule. Credit authority, model approval and customer communication remain explicit stages in the operating process, so the reader can follow how an experimental finding becomes evidence for a particular decision.

Article 5 develops the bank-controlled agentic workflow that connects these responsibilities across credit-offer experiments, sequential collections, reject inference and Treasury decisions. A restricted orchestration service retrieves a dated case view, invokes the approved analytical tools and assembles a proposal showing alternatives, credible intervals, risk-cap exceedance probabilities and policy checks. A generative model can help explain the evidence, while versioned statistical models calculate the quantities under review. Building on the bank-controlled deployment architecture in Part 4 of the *Underwrite for Collection* whitepaper, the authorised officer can inspect source records, obtain further information, change the proposed action or decline it before a separate execution service acts. The record then joins approval, customer communication, execution, settlement and later outcomes. Corrections and complaints return through the same case history, and validated learning advances through documented release gates. Kenya's Data Protection Act and General Regulations shape purpose, access, explanation, review and data handling throughout this workflow [10], [11].

Collection evidence can inform later loss measurement, but the connection needs a reconciled amount, date, probability, legal control, availability, currency, account and destination. Earlier cash may reduce an outstanding balance or change recovery timing; a modelled borrower-level effect does not become a portfolio accounting or capital result until the institution applies its approved measurement process. Expected credit loss under IFRS 9 is a probability-weighted present value of cash shortfalls using reasonable and supportable information; it is not identical to a simple PD×LGD×EAD expected-loss approximation or to economic and regulatory capital [1]. The analysis should keep borrower cash, expected loss, realized loss and capital measures separate, then document any bridge between them.

A practical development path

The first stage is descriptive and operational: reconcile identifiers, cash and servicing events; define a data dictionary; establish outcome windows; document existing actions and their authority; and measure missingness, correction delay and denominator drift. The second stage is a prespecified randomized pilot among eligible cases, with a current-policy comparator, power analysis, outcome maturity, stop criteria, user-facing notices and independent review. The third stage is heterogeneity analysis using context that was available at assignment, with uncertainty and subgroup safeguards. Only after repeatable evidence and stable operations should the team test contextual-bandit allocation among allowed actions. Adaptive assignment should retain a benchmark and record propensities and policy versions at each decision point.

The fourth stage asks whether a sequential model earns its complexity. The team should show that decisions recur, actions affect later states, meaningful state transitions can be measured, and historical or experimental data support the relevant action paths. Then compare interpretable dynamic programming, model-predictive control or constrained MDP methods before deep RL. A POMDP is justified only if the unobserved state matters, observations can be modelled and validated, and uncertainty improves decisions enough to warrant the additional complexity. A simulator can support stress testing and planning, but policy evaluation should include off-policy uncertainty, model misspecification and distribution shift. If evidence is too weak, a transparent rule with a manual review can remain the most defensible choice.

The fifth stage is governance and ongoing evaluation. Define who may approve a policy, who can pause it, who owns each data source, who reviews complaints and fairness findings, how adverse decisions are explained, what triggers recalibration, and when an old policy is retired. Track action rates, override reasons, contact failure, payment reversals, sustained cure, complaints, hardship referrals, data-correction requests, subgroup outcomes and economic conditions. Keep the record reproducible so that a reviewer can reconstruct the information and action set at a decision date. Change control should preserve prior versions

and support rollback. This turns a research result into a governed operating capability rather than treating deployment as the inevitable final step.

Conclusion

The transition from a one-step bandit to sequential credit control is a progression of evidence, not a change of algorithm name. A randomized pilot can estimate whether an approved action improves a defined outcome for an eligible population. A contextual bandit can adapt among such actions when assignment probabilities and delayed outcomes are managed. A sequential model becomes relevant when decisions change future borrower states or information; a POMDP becomes relevant when material state is partly hidden and observations support a validated belief model. Reject inference remains a separate counterfactual-identification problem, and Bayesian modelling cannot substitute for missing experimental support. The practical foundation is a clean lifecycle record that distinguishes operating-business cash from ring-fenced receivables and connects every decision to its evidence, authority, action and outcome.

For Kenyan application, privacy and data-protection duties shape the dataset, experiment and user journey from the beginning. The Data Protection Act, General Regulations and applicable credit rules must be mapped to the actual institution and processing. Hard eligibility gates protect rights and affordability; explainable records, meaningful human review, correction routes and independent validation support accountable decisions. The proposed research agenda is therefore concrete: define a local eligible population, preserve a standard-care benchmark, test support-oriented actions, measure durable repayment and borrower outcomes, and publish uncertainty. Only after that evidence exists should a team ask whether a more complex controller provides enough additional value to justify its cost, governance and operational risk.

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Publication note. Scenario figures in this article are explicitly hypothetical. This article describes a research and governance pathway; it does not report a completed Kenyan randomized trial, fitted collection controller or production deployment. The legal references should be checked against the institution's current classification and the authoritative consolidated instruments before publication.

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Mathematical appendix to Article 2

A2.1 Indices and model meanings

This appendix derives the mathematical specifications used in Article 2. Let i index a facility, t a decision time, τ an outcome horizon, $s(i)$ sector, $r(i)$ region and $c(i)$ origination vintage. D is the registered evidence. A policy π selects from the permitted action set $\mathcal{A}_t$. Model label L identifies fixed-horizon logistic parameters; H identifies separate hazard parameters. Normal distributions use mean and variance, Gamma distributions shape and rate. All posterior statements condition on the declared model, data and scenario. A credible interval describes uncertainty about a parameter or expected contrast; a predictive interval also includes future outcome variation.

A2.2 Bellman recursion

The finite-horizon value recursion is the fully observed special case of the control formulation in [22].

For terminal value g_T, stage reward r_t, horizon T and per-step discount γ, policy value is:

$$V_t^{\pi}(s) = \mathbb{E}^{\pi}\left[\sum_{j=t}^{T-1} \gamma^{j-t} r_j + \gamma^{T-t} g_T(S_T) \mid S_t = s\right]. \tag{A2.1}$$

Separate the first reward and apply iterated expectation. Under the Markov assumption, conditioning the continuation on the next state is sufficient:

$$V_t^{\pi}(s) = \sum_{a \in \mathcal{A}_t(s)} \pi_t(a \mid s) \int \left[r_t(s,a,s') + \gamma V_{t+1}^{\pi}(s')\right] P_t(ds' \mid s,a). \tag{A2.2}$$

Maximising at each permitted state gives:

$$\begin{aligned} V_t(s) &= \max_{a \in \mathcal{A}_t(s)} \int [r_t(s,a,s') + \gamma V_{t+1}(s')] P_t(ds' \mid s,a), \\ V_T(s) &= g_T(s). \end{aligned} \tag{A2.3}$$

The action set encodes eligibility and authority. The reward, transition and terminal condition describe one common decision problem. Discounting must match the time unit; irregular intervals require the adjustment below.

A2.3 POMDP prediction and observation update

The belief-state construction and observation update follow [22].

Let Z_t be latent state, O_t observation, $T_a(z' \mid z)$ the transition law, and $L_a(o \mid z')$ the observation likelihood. The belief conditions on observations and previous actions:

$$b_t(z) = \Pr(Z_t = z \mid O_{1:t}, A_{1:t-1}). \tag{A2.4}$$

The law of total probability gives a predicted belief; Bayes' rule incorporates the new observation:

$$
\begin{aligned}
\bar{b}_{t+1}(z') &= \sum_{z} T_{a_t}(z' \mid z) b_t(z), \\
p(o \mid b_t, a_t) &= \sum_{z'} L_{a_t}(o \mid z') \bar{b}_{t+1}(z'), \\
b_{t+1}(z') &= \frac{L_{a_t}(o_{t+1} \mid z') \bar{b}_{t+1}(z')}{\sum_{\tilde{z}} L_{a_t}(o_{t+1} \mid \tilde{z}) \bar{b}_{t+1}(\tilde{z})}.
\end{aligned}
\tag{A2.5}
$$

Replace sums by integrals for continuous state. The denominator must be positive under the model. If $\mathcal{B}(b, a, o)$ denotes the update and $\bar{r}(b, a)$ expected stage reward, the belief-space Bellman equation is:

$$
V_t(b) = \max_{a \in \mathcal{A}_t(b)} \left\{ \bar{r}(b, a) + \gamma \sum_{o} p(o \mid b, a) V_{t+1}(\mathcal{B}(b, a, o)) \right\}. \tag{A2.6}
$$

Information-gathering actions can improve continuation value through a changed belief. Their observation model, cost and permitted data collection are specified. The transition law requires credible action-response evidence.

A2.4 Semi-Markov timing

Let $U > 0$ be time to the next transition, $K(ds', du \mid s, a)$ its joint destination-duration distribution, and ρ a continuous discount rate. If $r(s, a, s', u)$ is the present value of rewards earned before and at that transition, conditioning gives:

$$
V(s) = \max_{a \in \mathcal{A}(s)} \int \left[r(s, a, s', u) + e^{-\rho u} V(s') \right] K(ds', du \mid s, a). \tag{A2.7}
$$

Cash received continuously enters r through its discounted integral over $[0, U]$. A finite horizon truncates durations at remaining time and uses a terminal value. Setting $U = 1$ and $\gamma = e^{-\rho}$ recovers the discrete recursion. This distinguishes otherwise equal receipts arriving at different dates.

A2.5 Complete logistic hierarchy

The hierarchical posterior and partial-pooling construction follow [9] (ch. 5).

For a mature default indicator at horizon τ:

$$
\begin{aligned}
D_i(\tau) &\sim \text{Bernoulli}(p_{i,\tau}), \\
p_{i,\tau} &= \frac{e^{\eta_i^L}}{1 + e^{\eta_i^L}}, \\
\eta_i^L &= \alpha^L + X_i^{\mathsf{T}} \beta^L + u_{s(i)}^L + v_{r(i)}^L + w_{c(i)}^L, \\
\mathcal{L}_L &= \prod_i p_{i,\tau}^{D_i(\tau)} (1 - p_{i,\tau})^{1 - D_i(\tau)}.
\end{aligned}
\tag{A2.8}
$$

The same named levels apply separately for $m \in \{L, H\}$:

$$
\begin{aligned}
u_s^m &= \sigma_u^m z_{us}^m, & z_{us}^m &\sim N(0,1),\\
v_r^m &= \sigma_v^m z_{vr}^m, & z_{vr}^m &\sim N(0,1),\\
w_c^m &= \sigma_w^m z_{wc}^m, & z_{wc}^m &\sim N(0,1),\\
\sigma_k^m &\sim \mathsf{HalfStudentT}(3, s_{0,k}^m), & k &\in \{u,v,w\},\\
\alpha^L &\sim N(0, s_\alpha^2), & \beta^m &\sim N(0, B^m).
\end{aligned}
\tag{A2.9}
$$

Raw effects are independent in this baseline specification. A dependence prior or identifying constraint is recorded when used. B^m is positive definite. Hyperprior scales are calibrated on the link scale through prior-predictive simulation. Additional institution, product or calendar levels each require their own index and prior. An immature or selectively observed binary label requires a model for observation and selection.

Multiplying likelihood and priors gives the posterior. Sharing a scale distribution allows partial pooling. A Gaussian local approximation illustrates it: if a group estimate $\hat{u}_s$ has variance v_s and prior $N(0, \sigma_u^2)$, completing the square yields:

$$
\mathbb{E}[u_s \mid \hat{u}_s, \sigma_u] = \frac{\sigma_u^2}{\sigma_u^2 + v_s}\hat{u}_s, \qquad \mathsf{Var}(u_s \mid \hat{u}_s, \sigma_u) = (v_s^{-1} + \sigma_u^{-2})^{-1}. \tag{A2.10}
$$

Large sampling variance produces stronger pooling. The logistic posterior itself uses the Bernoulli likelihood rather than this explanatory approximation.

A2.6 Hierarchical hazards and their likelihood

The proportional-hazards and partial-likelihood construction follows [23].

The hazard model uses separate coefficients and group effects:

$$
\eta_i^H(t) = X_i(t)^\top \beta^H + u_{s(i)}^H + v_{r(i)}^H + w_{c(i)}^H, \qquad \lambda_i(t) = \lambda_0(t) e^{\eta_i^H(t)}. \tag{A2.11}
$$

Time-varying predictors use information available just before the relevant time. The baseline hazard supplies the intercept role. A Bayesian piecewise-constant baseline can use $\lambda_{0\ell} \sim \mathsf{Gamma}(a_\ell, b_\ell)$ in interval ℓ. A Cox partial-likelihood fit instead estimates relative effects while leaving baseline estimation to a subsequent step.

From the hazard definition, $S(t+dt) = S(t)\{1 - \lambda(t)dt + o(dt)\}$. Rearranging gives $S'(t)/S(t) = -\lambda(t)$. Integration with $S(0) = 1$ yields:

$$
S_i(t) = \exp\left[-\int_0^t \lambda_i(u)du\right], \qquad f_i(t) = \lambda_i(t) S_i(t). \tag{A2.12}
$$

For $y_i = \min(T_i, C_i)$ and event indicator $\delta_i = 1\{T_i \le C_i\}$, conditionally independent censoring allows its likelihood factor to be separated:

$$
\mathcal{L}_H = \prod_i \lambda_i(y_i)^{\delta_i} \exp\left[-\int_0^{y_i} \lambda_i(u)du\right]. \tag{A2.13}
$$

At an untied event time, condition on one event occurring in risk set $\mathcal{R}(t_i)$. The common baseline cancels, giving:

$$\mathcal{L}_{\text{partial}} = \prod_{i:\delta_i=1} \frac{e^{\eta_i^H(t_i)}}{\sum_{j\in\mathcal{R}(t_i)} e^{\eta_j^H(t_i)}}. \tag{A2.14}$$

Ties require a declared method, and absolute risk needs a baseline estimate. For a piecewise baseline with d_ℓ events and adjusted exposure $E_\ell = \sum_i \int_{I_\ell} Y_i(t) e^{\eta_i^H(t)} dt$, its conditional likelihood is proportional to $\lambda_{0\ell}^{d_\ell} e^{-\lambda_{0\ell} E_\ell}$. Consequently:

$$\lambda_{0\ell} \mid \cdots \sim \text{Gamma}(a_\ell + d_\ell, b_\ell + E_\ell). \tag{A2.15}$$

This baseline-rate conjugacy is separate from logistic Pólya–Gamma inference.

These survival and hazard identities hold conditional on the fitted coefficients, frailties and baseline. A posterior survival prediction averages the conditional survival for each draw. For posterior parameter vector ϑ, the calculation is

$$\overline{S}_i(t \mid D) = \int \exp[-\Lambda_i(t;\vartheta)] p(\vartheta \mid D)\, d\vartheta \approx \frac{1}{B}\sum_{b=1}^{B} \exp[-\Lambda_i(t;\vartheta^{(b)})]. \tag{A2.16}$$

In general this differs from $\exp\{-\mathbb{E}[\Lambda_i(t;\vartheta) \mid D]\}$. By convexity, the former is at least the latter. For a new group, draws also integrate its new group effect; for an observed group they use its conditional posterior. A logistic model fitted independently at several horizons needs a joint horizon specification or an explicitly checked monotonicity restriction to ensure that cumulative default probabilities increase with the horizon. Its group labels alone do not impose that restriction.

For competing terminal causes, each hazard has its own baseline, coefficients and named hierarchy. Event-free survival and cumulative incidence are:

$$\begin{aligned} S_i(t) &= \exp\left[-\int_0^t \sum_k \lambda_{ik}(u) du\right], \\ F_{ik}(t) &= \int_0^t S_i(u-)\lambda_{ik}(u) du, \\ S_i(t) + \sum_k F_{ik}(t) &= 1. \end{aligned} \tag{A2.17}$$

The final identity follows by integrating $-S_i'(t) = S_i(t)\sum_k \lambda_{ik}(t)$. Default probability with competing exits is $F_{iD}(\tau)$; a single-event probability is $1 - S_i(\tau)$. Cure followed by re-default needs intermediate states. For multi-state intensity matrix $Q(t)$, off-diagonal entries are transition rates and each diagonal is minus its row sum. The transition matrix satisfies $\partial P(s,t)/\partial t = P(s,t)Q(t)$ with $P(s,s) = I$, preserving total probability.

A2.7 Expected loss, cash shortfall and capital

The cash-shortfall interpretation follows [1], and the regulatory/internal capital distinction follows [4].

For scenario weight w_j, contractual cash C_t^{contract}, expected receipt C_{tj}^{received} and accounting discount d_t, probability weighting and present-value linearity give:

$$\mathsf{ECL} = \sum_j w_j \sum_t d_t (C_t^{\text{contract}} - C_{tj}^{\text{received}}), \qquad \sum_j w_j = 1. \tag{A2.18}$$

The recognition purpose determines the horizon and scenario set. Twelve-month ECL includes lifetime shortfalls from defaults possible during the next twelve months. For $Z = 1\{\text{default}\}$ EAD LGD, conditioning gives:

$$\mathbb{E}[Z] = \mathsf{PD}\ \mathbb{E}[\mathsf{EAD}\ \mathsf{LGD} \mid \text{default}]. \tag{A2.19}$$

Replacing the conditional product expectation by fixed EAD and LGD produces the familiar product approximation. When both vary, conditional covariance matters. Recovery timing and discounting belong in the full cash-loss specification.

For portfolio loss L, define $\mathsf{VaR}_\alpha(L) = \inf\{l : \Pr(L \le l) \ge \alpha\}$. Expected shortfall can be defined for general distributions by:

$$\mathsf{ES}_\alpha(L) = \frac{1}{1-\alpha} \int_\alpha^1 \mathsf{VaR}_u(L)\, du. \tag{A2.20}$$

Internal capital may subtract $\mathbb{E}[L]$ from either declared tail measure. RAROC divides a consistently defined risk-adjusted income numerator by allocated economic capital. Each cost appears once. The article's illustration gives $1100 - 250 = 850$ and $850/5000 = 17\%$; a complete RAROC still specifies funding, expected loss, tax and capital allocation.

A2.8 Constraints and fairness

The conditional fairness rates follow [2].

Let $J(\pi) = \mathbb{E}^\pi[\sum_t \gamma^t r_t]$ and $C_k(\pi) = \mathbb{E}^\pi[\sum_t c_{kt}]$. Maximisation is subject to $C_k(\pi) \le B_k$ and permitted actions. Nonnegative multipliers give:

$$\mathcal{L}(\pi, \lambda) = J(\pi) - \sum_k \lambda_k \{C_k(\pi) - B_k\}, \qquad \lambda_k \ge 0. \tag{A2.21}$$

This assists optimisation of expectation constraints; feasibility is checked against the original problem. Hard eligibility remains in the action set. Strong duality requires further conditions.

For lawful evaluation group g, defined positive decision $\hat{Y} = 1$ and outcome Y, use:

$$\begin{aligned} \mathsf{TPR}_g &= \Pr(\hat{Y} = 1 \mid Y = 1, G = g), \\ \mathsf{FPR}_g &= \Pr(\hat{Y} = 1 \mid Y = 0, G = g), \\ s_g &= \Pr(\hat{Y} = 1 \mid G = g). \end{aligned} \tag{A2.22}$$

Equal opportunity compares TPRs; equalised odds compares both conditional rates. A selection-rate ratio is s_g / s_{g_0} when its denominator is positive. Empirical rates divide the relevant event count by its eligible denominator. Selective labels and small groups affect uncertainty and interpretation.

A2.9 Clayton copula

The dependence construction follows [3].

For uniform marginals and $\theta > 0$:

$$C_\theta(u, v) = (u^{-\theta} + v^{-\theta} - 1)^{-1/\theta}. \tag{A2.23}$$

Setting $u = v = q$ yields the lower-tail coefficient:

$$\lambda_L = \lim_{q \downarrow 0} \frac{C(q, q)}{q} = \lim_{q \downarrow 0} (2 - q^\theta)^{-1/\theta} = 2^{-1/\theta}. \tag{A2.24}$$

Near one, $C(q, q) = 1 - 2(1 - q) + o(1 - q)$, so $\lambda_U = \lim_{q \uparrow 1} \{1 - 2q + C(q, q)\}/(1 - q) = 0$. The Archimedean generator $\varphi(u) = (u^{-\theta} - 1)/\theta$ gives Kendall's tau:

$$\begin{aligned} \tau &= 1 + 4 \int_0^1 \frac{\varphi(u)}{\varphi'(u)} du \\ &= 1 - \frac{4}{\theta} \left(\frac{1}{2} - \frac{1}{\theta + 2} \right) = \frac{\theta}{\theta + 2}. \end{aligned} \tag{A2.25}$$

Variable orientation determines the economic meaning of the lower tail. The copula parameter is estimated separately from a hierarchy-scale prior's degrees of freedom.

A2.10 Pólya–Gamma derivation

The augmentation identity and conditional Gaussian calculation follow [5].

For binomial trials b_i, successes y_i, logit η_i and $\kappa_i = y_i - b_i/2$, the augmentation identity is:

$$\frac{e^{y_i \eta_i}}{(1 + e^{\eta_i})^{b_i}} = 2^{-b_i} e^{\kappa_i \eta_i} \int_0^\infty e^{-\omega_i \eta_i^2 / 2} p_{\mathsf{PG}(b_i, 0)}(\omega_i) d\omega_i. \tag{A2.26}$$

Exponential tilting gives $\omega_i \mid \eta_i \sim \mathsf{PG}(b_i, \eta_i)$; Bernoulli data use $b_i = 1$. Let Z be the expanded logistic design, including all sector, region and vintage indicators, and θ its stacked coefficients. Given scales, use $\theta \sim N(m_0, V_0)$ and $\Omega = \mathrm{diag}(\omega_i)$. Multiplying likelihood kernel and prior gives:

$$\log p(\theta \mid \cdots) = -\tfrac{1}{2} \theta^\mathsf{T} (Z^\mathsf{T} \Omega Z + V_0^{-1}) \theta + \theta^\mathsf{T} (Z^\mathsf{T} \kappa + V_0^{-1} m_0) + \text{constant}. \tag{A2.27}$$

Complete the square:

$$\begin{aligned} V_* &= (Z^\mathsf{T} \Omega Z + V_0^{-1})^{-1}, \\ m_* &= V_* (Z^\mathsf{T} \kappa + V_0^{-1} m_0), \\ \theta \mid \omega, \text{scales}, D &\sim N(m_*, V_*). \end{aligned} \tag{A2.28}$$

Scales, correlations, smoothing parameters and latent variables receive their own validated updates. Sharing the same named hierarchy does not change the hazard likelihood into this logistic likelihood.

When updating one coefficient block, retain the other predictor components as an offset c: $\eta = Z\theta + c$. The Gaussian precision stays $Z^{\mathsf{T}}\Omega Z + V_0^{-1}$, while its information vector becomes $Z^{\mathsf{T}}(\kappa - \Omega c) + V_0^{-1}m_0$. A sum-to-zero constraint can be implemented through a full-column-rank basis for the constrained effects; the design and Gaussian covariance are then expressed in that basis. For computation, factor the precision and solve linear systems instead of explicitly calculating its inverse. This retains the same distribution while making the numerical operations and residual checks explicit.

A2.11 Half-t mixture and scale updates

The scale prior follows [6], the normal–Gamma representation [7], and the inverse-Gamma auxiliary representation [8] (Result 5).

Let $\lambda \sim \text{Gamma}(\nu/2, \nu/2)$ and $q \mid \lambda \sim N(0, s_0^2/\lambda)$. Multiply their densities and integrate using $\int_0^\infty x^{a-1}e^{-bx}dx = \Gamma(a)b^{-a}$:

$$\begin{aligned} p(q) &= \frac{(\nu/2)^{\nu/2}}{\Gamma(\nu/2)\sqrt{2\pi}\, s_0} \int_0^\infty \lambda^{(\nu+1)/2-1} \exp[-(\nu + q^2/s_0^2)\lambda/2]\, d\lambda \\ &= \frac{\Gamma((\nu+1)/2)}{\Gamma(\nu/2)\sqrt{\nu\pi}\, s_0} \left(1 + \frac{q^2}{\nu s_0^2}\right)^{-(\nu+1)/2}. \end{aligned} \tag{A2.29}$$

Taking $\sigma = |q|$ doubles the symmetric density on the positive half-line. For $\nu = 3$:

$$p(\sigma) = \frac{4}{\pi\sqrt{3}\, s_0} \left(1 + \frac{\sigma^2}{3s_0^2}\right)^{-2}, \qquad \sigma > 0. \tag{A2.30}$$

The mixing Gamma shape and rate are both $3/2$. The half-t here is assigned to a hierarchy SD. Gaussian effects conditional on that SD give the pooling model.

In this normal–Gamma representation, conditioning on q gives $\lambda \mid q \sim \text{Gamma}((\nu+1)/2, (\nu+q^2/s_0^2)/2)$. When $|q|$ is used as the standard deviation of J centred Gaussian effects, its own conditional distribution also includes those effects' likelihood. For $\sigma > 0$, its log kernel given λ is

$$\log p(\sigma \mid u, \lambda) = -J \log \sigma - \frac{\sum_j u_j^2}{2\sigma^2} - \frac{\lambda\sigma^2}{2s_0^2} + \text{constant}. \tag{A2.31}$$

This specifies why the normal–Gamma identity by itself gives no Gaussian full conditional for the hierarchy standard deviation. The inverse-Gamma auxiliary representation below supplies a different route with explicit centred scale conditionals.

An auxiliary representation also permits scale updates. Define inverse-Gamma density $p(x \mid a, b) \propto x^{-a-1}e^{-b/x}$:

$$\sigma^2 \mid a \sim \text{IG}(\nu/2, \nu/a), \qquad a \sim \text{IG}(1/2, 1/s_0^2). \tag{A2.32}$$

Integration over a gives $p(\sigma^2) \propto (\sigma^2)^{-1/2}(\sigma^2 + \nu s_0^2)^{-(\nu+1)/2}$. Applying the Jacobian 2σ recovers the half-t density. For J independent effects $u_j \mid \sigma \sim N(0, \sigma^2)$:

$$\begin{aligned}\sigma^2 \mid a, u &\sim \mathsf{IG}\left(\frac{\nu+J}{2}, \frac{\nu}{a}+\frac{1}{2}\sum_j u_j^2\right), \\ a \mid \sigma^2 &\sim \mathsf{IG}\left(\frac{\nu+1}{2}, \frac{\nu}{\sigma^2}+\frac{1}{s_0^2}\right).\end{aligned} \tag{A2.33}$$

These conditionals use centred effects. A non-centred sampler conditions on different variables and uses the corresponding updates. Conditional conjugacy applies to specified blocks; the marginal posterior remains a mixture. Calibration, convergence, effective sample size and reference-fit agreement assess computational adequacy.

For example, hold non-centred raw effects z, the Pólya–Gamma variables and other parameters fixed. Write the current group's contribution as σv, where $v = Z_g z$ and $\eta = c + \sigma v$. Collecting powers in the augmented logistic likelihood gives

$$p(\sigma \mid z, \omega, \ldots) \propto \mathbf{1}\{\sigma > 0\} \exp\left[-\frac{\sigma^2}{2} v^\mathsf{T} \Omega v + \sigma v^\mathsf{T}(\kappa - \Omega c)\right] p(\sigma). \tag{A2.34}$$

Here $p(\sigma)$ is the selected positive-scale prior or its conditional auxiliary form. The centred inverse-Gamma update cannot be reused while holding z fixed. A sampler may use the correct centred Gibbs blocks, a validated non-centred method, or an explicitly derived interweaving scheme; effective sample size per elapsed second determines whether the extra representation helps the registered workload.

A2.12 Limit boundary and human-review uncertainty

The expected-utility comparison follows [9] (ch. 9). The declared payoff matrix determines the economic threshold.

Let $\bar{p} = \mathbb{E}[p \mid D]$, repayment gain $G \geq 0$, default loss $L > 0$ and separately defined denial cost $O \geq 0$. Linearity of expectation for fixed payoffs gives:

$$\begin{aligned}\mathbb{E}[V_1 \mid D] &= (1-\bar{p})G - \bar{p}L, \\ \mathbb{E}[V_0 \mid D] &= -(1-\bar{p})O, \\ \mathbb{E}[V_1 - V_0 \mid D] &= (G+O) - \bar{p}(G+O+L).\end{aligned} \tag{A2.35}$$

Set the final expression to zero:

$$p^* = \frac{G+O}{G+O+L}. \tag{A2.36}$$

For $G = 1200$, $O = 0$, $L = 18800$, the threshold is 6%. At $\bar{p} = 4\%$, expected extension value is $0.96(1200) - 0.04(18800) = 400$ KES.

This boundary compares one specified extension with its stated alternative, under linear monetary utility and fixed payoffs. The repayment probability concerns that extension's outcome; the denial cost is incurred in the counterfactual repayment state of the same extension. For a limit increase on an existing facility, measure changes in value relative to keeping the current limit, preserving the existing balance and expected recoveries in both alternatives. If the action changes utilisation, default risk, repayment timing, servicing cost or recovery, calculate the full action-specific posterior value rather than substituting its risk into a fixed-payoff threshold:

$$a^* = \arg\max_{a \in \mathcal{A}^{\text{permitted}}} \int\int U(v,a)\, p(v \mid \vartheta, D, a) p(\vartheta \mid D)\, dv\, d\vartheta. \tag{A2.37}$$

For linear $U(v,a) = v$ the comparison uses expected value; nonlinear utility requires the value distribution. The cap-exceedance gate below is a separately approved constraint, so it can route a case to review even when its mean-based financial comparison is favourable.

Posterior draws give the cap-exceedance estimate:

$$\hat{q}_{\text{cap}} = \frac{1}{B} \sum_{b=1}^{B} 1\{p^{(b)} > p_{\text{cap}}\}. \tag{A2.38}$$

Compare it with the approved ϵ to select a review route. Monte Carlo precision depends on effective sample size. If utilisation, gain or recovery is uncertain, preserve dependence through joint draws and compute $\Delta^{(b)} = V_1^{(b)} - V_0^{(b)}$. Quantiles supply its credible interval and $B^{-1} \sum_b 1\{\Delta^{(b)} > \delta\}$ estimates probability of worthwhile benefit. A causal Δ is a declared conditional or population-average potential-outcome contrast under an identified design. Separate action intervals do not determine uncertainty in their difference:

$$\text{Var}(V_1 - V_0) = \text{Var}(V_1) + \text{Var}(V_0) - 2\,\text{Cov}(V_1, V_0). \tag{A2.39}$$

For posterior uncertainty about two expected action values, paired draws from the same fitted model preserve their shared parameter uncertainty in this calculation. A variance for the realised individual treatment effect $Y(1) - Y(0)$ is a different target: a standard parallel-arm trial identifies the marginal outcome distributions but does not observe both outcomes for the same person. Its cross-potential-outcome covariance requires additional assumptions or bounds. The reported credible interval should therefore name its mean-effect or policy-value estimand.

Future cash prediction integrates both parameter and outcome variation:

$$p(C \mid D, a) = \int p(C \mid \theta, a) p(\theta \mid D) d\theta. \tag{A2.40}$$

The law of total variance decomposes predictive variance into $\mathbb{E}[\text{Var}(C \mid \theta, a) \mid D] + \text{Var}(\mathbb{E}[C \mid \theta, a] \mid D)$. This distinguishes the PD parameter interval, causal-contrast interval and predictive cash interval on the reviewer screen.

A2.13 Linear UCB and logistic Thompson sampling

For $r_j = x_j^\top \theta + \varepsilon_j$, minimise squared residuals plus $\lambda \|\theta\|^2$. Differentiation gives ridge normal equations:

$$A_t = \lambda I + \sum_{j<t} x_j x_j^\top, \quad b_t = \sum_{j<t} x_j r_j, \quad \hat{\theta}_t = A_t^{-1} b_t. \tag{A2.41}$$

Suppose a validated confidence ellipsoid is $(\theta - \hat{\theta}_t)^\top A_t (\theta - \hat{\theta}_t) \le \alpha_t^2$. Cauchy–Schwarz after transformation by $A_t^{1/2}$ gives:

$$\max_{\theta \text{ in ellipsoid}} x^\top \theta = x^\top \hat{\theta}_t + \alpha_t \sqrt{x^\top A_t^{-1} x}. \tag{A2.42}$$

The coefficient α_t depends on noise assumptions and coverage. For hierarchical logistic Thompson sampling, draw the full expanded coefficient vector:

$$\tilde{\theta}_t \sim p(\theta \mid D_t), \qquad \tilde{p}_t(a) = \{1 + e^{-z_t(a)^\top \tilde{\theta}_t}\}^{-1}, \qquad a_t = \arg\max_{a \in \mathcal{A}_{\text{safe},t}} \tilde{Q}_t(a). \tag{A2.43}$$

The design includes the named group levels; $z_t(a)$ here denotes that expanded row, separately from non-centred raw effects. The sampled binary probability enters a defined cash-and-cost value. The appropriate objective depends on whether the event means repayment, default or another outcome.

A2.14 MCTS and graph explanations

Given a first action, a declared subsequent rollout policy and simulator:

$$Q_H(s,a) = \mathbb{E}\left[\sum_{k=0}^{H} \gamma^k r_{t+k} \mid S_t = s, a_t = a, \text{rollout policy and model}\right]. \tag{A2.44}$$

A sample mean of simulated returns estimates this expectation. UCT allocates additional search visits through:

$$U(s,a) = \hat{Q}(s,a) + c\sqrt{\frac{\log N(s)}{N(s,a)}}. \tag{A2.45}$$

Unvisited permitted actions are initialised before division by their visit count. Search allocation is computational; model error is assessed independently against observed outcomes.

Here H is the last reward offset, so the displayed sum includes $H+1$ rewards. A specified terminal value adds $\gamma^{H+1} g(S_{t+H+1})$. If the application defines a horizon as H decisions instead, use offsets $0, \dots, H-1$ and terminal factor γ^H. Under partial observation, the search node is a belief or a retained history, and simulated observations update it through A2.3. The UCT constant c is scaled to the return units; scaling a cash reward without updating c changes the exploration trade-off.

For an explanation with M players and coalition value $v(S)$, the Shapley attribution is:

$$\phi_j = \sum_{S \subseteq \{1,\dots,M\} \setminus \{j\}} \frac{|S|!(M-|S|-1)!}{M!}[v(S \cup \{j\}) - v(S)]. \tag{A2.46}$$

The weight counts permutations whose predecessors of j are exactly S, divided by all $M!$ permutations. Averaging marginal contributions gives this formula. GraphSVX approximates attribution under its feature/graph and perturbation convention. It explains the selected prediction target; MCTS evaluates a defined action path.

For each ordering, the sum of players' successive contributions telescopes to $v(\{1,\dots,M\}) - v(\emptyset)$. Averaging proves the efficiency identity $\sum_j \phi_j = v(\{1,\dots,M\}) - v(\emptyset)$. A graph-explanation method can also use tree search to identify explanatory subgraphs, but that search space and coalition value differ from the controller's permitted action trajectories and economic return. Neither attribution construction supplies a causal intervention effect without the separate identification assumptions.

A2.15 Off-policy evaluation and pilot arithmetic

When behaviour propensity $e(a \mid x) > 0$ wherever target $\pi(a \mid x) > 0$, conditional expectation gives the importance-weight identity:

$$\mathbb{E}\left[\frac{\pi(A \mid X)}{e(A \mid X)} R \mid X\right] = \sum_a \pi(a \mid X)\mathbb{E}[R \mid X, A = a]. \tag{A2.47}$$

Adding an outcome model m gives the doubly robust score:

$$\psi = \sum_a \pi(a \mid X) m(X, a) + \frac{\pi(A \mid X)}{e(A \mid X)}[R - m(X, A)]. \tag{A2.48}$$

Correct propensities cancel outcome-model error in conditional expectation. A correct outcome model makes the residual mean zero. Causal interpretation also requires consistency, exchangeability and support. Cross-fitting and design-appropriate uncertainty estimation address fitted nuisance models and adaptive data.

For trajectories, define $\rho_t = \prod_{j=0}^{t} \pi(A_j \mid H_j)/e(A_j \mid H_j)$ and $\hat{V}_t(H_t) = \sum_a \pi(a \mid H_t)\hat{Q}_t(H_t, a)$. A sequential doubly robust score is:

$$\hat{V}_0(H_0) + \sum_{t=0}^{T-1} \gamma^t \rho_t [R_t + \gamma \hat{V}_{t+1}(H_{t+1}) - \hat{Q}_t(H_t, A_t)], \qquad \hat{V}_T = 0. \tag{A2.49}$$

Correct sequential weights convert each term to the target-policy distribution; the fitted continuation terms telescope under iterated expectation. Exchangeability and overlap are required at each decision. Weight truncation alters the estimator and its bias–variance trade-off.

In the illustrative 100-case pilot, each arm contains 50 cases. Day-60 rates are $38/50 = 0.76$ and $46/50 = 0.92$, a difference of 16 percentage points. Day-90 sustained rates are $34/50 = 0.68$ and $41/50 = 0.82$, a difference of 14 percentage points. Count differences of eight and seven are distinct from those rate differences. The net-cash calculation is $1100 - 250 = 850$ KES per assigned case. These are arithmetic checks on hypothetical inputs; empirical conclusions require observed data and the registered analysis.

A2.16 Sources and implementation status

Article 2 references [17], [24] support doubly robust evaluation; [2] fairness definitions; [1] ECL interpretation; [3], [4], [5] dependence, capital context and Pólya–Gamma inference; and [6], [7], [14], [15], [16] MCTS, graph attribution, Thompson sampling and scale priors. The auxiliary half-t representation above is checked by direct integration under the stated inverse-Gamma convention. The appendix expands the article's mathematical specification; fitted diagnostics, local treatment effects and deployed-system evidence are separate validation deliverables.

Further methodological reading

Additional sources retained from the article bibliography: [25].

Who Gets a Repayment Label?

Reject inference, selective labels and evidence for sustainable MSME credit

Underwrite for Collection · Article 3 of 5 · LinkedIn research article · Companion piece · 8 October 2026

Abstract

Credit institutions observe repayment only after an applicant receives and takes up a particular financing offer. A rejected applicant therefore does not have an observed repayment outcome under that offer, and a model cannot turn the missing counterfactual into fact by imputation. This article treats reject inference as an evidence-design problem: define the contract-specific outcome, reconstruct how historical approval selected the observed portfolio, map support and uncertainty, and build controlled evidence where a decision is safe and lawful. It develops hierarchical Bayesian logistic and survival models, selection assumptions, sensitivity analysis, Pólya–Gamma computation and a guarded offer pilot. A worked example shows what a small study can establish and how wide its uncertainty remains. The discussion connects applicant selection to business cash, ring-fenced receivables, Kenya's data-protection framework, Fannie Mae and Freddie Mac performance data, IFRS 9 and the *Underwrite for Collection* lifecycle.

This third article takes the evidence boundary raised in the first two pieces and examines it directly: lenders observe repayment after a contract is offered, accepted, funded and serviced, while rejected applicants have no realized repayment under the unissued contract. The first article explains how an RCT can create comparative evidence for approved offers; the second shows why repeated collection actions require action-time records and supported comparisons. Reject inference asks what can responsibly be learned outside those observed paths. The answer depends on the defined contract, the historical selection mechanism, overlap, assumptions and prospective evidence. This article develops those elements without repeating the earlier tutorials on bandits or sequential control, then identifies where the fourth article moves to bank balance-sheet decisions.

Research questions

1. In which Kenyan MSME segments and contract types do historical funded-loan records provide enough overlap to estimate repayment risk, and how do conclusions change under plausible unobserved-selection assumptions?
2. What randomized offer or repayment-schedule design can generate useful evidence near an approval boundary while preserving affordability, data-protection safeguards, human review and ordinary borrower rights?
3. How should borrower cash provenance, ring-fenced receivables, collective guarantees and realized collection outcomes enter a validated risk framework without treating imputed reject labels as observed repayment or conflating ECL with capital?

Start with the question that repayment data can answer

The practical question behind *Underwrite for Collection* is how a lender can support sustainable repayment across the full facility life. That question reaches back to origination because the amount, price, repayment calendar, security and servicing plan shape the cash burden that the borrower must meet. Historical performance files contain payment paths for loans that were approved, accepted and disbursed under previous policies. They do not contain repayment under a loan that was declined or never drawn. This is a selective-label problem: the decision process determines whose outcomes become observable. It is not an administrative gap that can be solved by assigning a simulated good-or-bad label to every rejected application.

For an MSME, a financing offer is inseparable from the operating context. A seasonal trader may not be able to meet the same monthly schedule as a service firm with regular invoices, even when their annual cash generation is similar. A borrower who needs working capital for stock has a different repayment mechanism from a business financing a productive asset. An invoice-backed facility may depend on eligible receivables, obligor payment, dilution, legal assignment, account control and waterfall rules rather than unrestricted enterprise cash. Thus the target is not an abstract "creditworthy" flag. It is the repayment and loss path associated with a specified product and contract, for a defined population, under a defined servicing process and time horizon.

The distinction among prediction, transport and causal policy analysis is central. Prediction asks how well the model estimates an outcome in a population with support in the development data. Transport asks how to adjust or qualify that prediction for a different target population. Causal evaluation asks what would happen under an alternative offer or policy. A model that ranks approved borrowers accurately can still be wrong for applicants outside the approval region. Reweighting may target a broader distribution only if selection is measured and overlap exists. A policy effect needs a defined intervention and a credible comparison group. The research should name which question it answers before reporting a performance metric, because a high AUC among funded loans does not establish the counterfactual repayment behavior of applicants who were never funded [1], [2], [3].

Define the potential outcome under a complete contract

Let X_i be applicant information available at the decision time, and let

$$S_i = 1 \tag{3.1}$$

indicate that the historical process financed applicant i. Let a denote a complete offer: principal or limit, price and fees, term, repayment schedule, grace period, collateral or guarantee conditions, disbursement requirements and servicing regime. Define

$$D_i^a(\tau) = 1 \tag{3.2}$$

if applicant i would default by horizon τ under offer a. The contract-specific risk target is

$$p_{i,\tau}(a) = \mathsf{Pr}(D_i^a(\tau) = 1 \mid X_i). \tag{3.3}$$

For an applicant who never received offer a, $D_i^a(\tau)$ is unobserved. If two applicants received different amounts or schedules, their outcomes are not automatically labels for the same intervention. Writing the offer explicitly prevents the analysis from treating credit risk as a property detached from contract design.

Take-up is another potential outcome. Let

$$T_i^a = 1 \tag{3.4}$$

if the applicant would accept and draw offer a. A lender may care about default among originated loans, take-up among eligible applicants, expected net contribution per applicant approached, or the effect of an offer policy on the broader cohort. These are different quantities with different denominators. An offer may change who accepts; a schedule may change both affordability and take-up; a guarantee may change recovery without changing the business's ability to pay. Therefore, observed default conditional on acceptance under a high-rate offer cannot be compared casually with observed default among accepted lower-rate

loans. Research should record the full chain—eligible, assigned, offered, accepted, originated, serviced, cured, prepaid or defaulted—so an analyst can identify which population a reported estimate describes.

The same contract specificity applies to time. A fixed 12-month default indicator may be appropriate for one question, while a lender planning amortization, expected loss or recovery may need the event timing. Default, cure, prepayment, refinancing, restructuring, write-off and liquidation change exposure and observation. A time-to-event model estimates event hazards only under its assumptions; it does not supply missing events for declined cases. A model that censors every prepayment as if it were unrelated to future default can answer the wrong question if prepayment is informative. The study must state the event definition, start date, horizon, competing events, censoring rule, and whether the target is default by any time, default before another event, or a net cash-shortfall measure.

What a hierarchical Bayesian model contributes

A hierarchical Bayesian logistic model can partially pool evidence across sectors, regions, vintages or product types. One specification is

$$D_i^a(\tau) \sim \text{Bernoulli}\big(p_{i,\tau}(a)\big), \tag{3.5}$$

with

$$\begin{aligned}\text{logit}(p_{i,\tau}(a)) = \alpha^L + u^L_{s(i)} + v^L_{r(i)} + w^L_{c(i)} \\ + f^L(X_i) + h^L(a, X_i)\end{aligned}. \tag{3.6}$$

The function f represents borrower characteristics and pre-specified nonlinearities; h represents offer and context interactions; and the group terms allow information sharing across related cohorts. Partial pooling can stabilize estimates for small segments, while the posterior represents uncertainty conditional on the likelihood, prior and model structure. Model checks should include prior sensitivity, calibration, posterior predictive checks, time-based validation, missingness, subgroup performance and sensitivity to contract definitions. Hierarchical structure is a way to share information, not a guarantee that a model extrapolates correctly beyond the historical support [4], [5].

A hierarchical Cox model can represent the time to a specified event:

$$\begin{aligned}\lambda_i(t \mid X_i, a) = \lambda_0(t)\exp\big(X_i^\top \beta^H + q^H(a, X_i) \\ + u^H_{s(i)} + v^H_{r(i)} + w^H_{c(i)}\big)\end{aligned}, \tag{3.7}$$

with survival

$$S_i(t) = \exp\left(-\int_0^t \lambda_i(u)\,\mathrm{d}u\right) \tag{3.8}$$

in a single-event formulation. It can show how risk changes over loan seasoning and help distinguish near-term from later exposure. If cure, prepayment and default compete, cause-specific hazards or a multi-state model may be needed. The model's term structure is estimated for observed contracts and populations; it does not by itself justify sequential price changes or prove an intervention is effective. For fixed-term loans, the price is ordinarily set at origination. The survival model may inform future monitoring, new offer cohorts or a permitted renewal decision, while collection actions after disbursement remain a distinct decision process [6].

The hierarchy retains sector s, region r and origination vintage c in both specifications. Model label L identifies the fixed-horizon logistic model; H identifies the event-hazard model. Each model has its own group effects, scales and likelihood. The following non-centred representation supplies every group level and its half-Student-t scale prior. The functions f^{L}, h^{L} and q^{H} use declared feature and interaction bases, whose coefficients belong to the corresponding β^m block; B^m is its positive-definite prior covariance. Normal uses variance as its second argument. The positive scale settings $s^m_{0,k}$ and $s\alpha$ are declared before fitting and assessed by prior-predictive checks. For a fully Bayesian piecewise-constant baseline hazard, interval ℓ receives a positive Gamma prior with specified shape $a\ell$ and rate $b\ell$. Cause-specific models repeat the structure with an additional event label. Explicit joint modelling is required if information is shared between outcome types [4], [5], [6].

$$u^m_s = \sigma^m_u z^m_{us}, \qquad z^m_{us} \sim N(0,1) \tag{3.9}$$

$$v^m_r = \sigma^m_v z^m_{vr}, \qquad z^m_{vr} \sim N(0,1) \tag{3.10}$$

$$w^m_c = \sigma^m_w z^m_{wc}, \qquad z^m_{wc} \sim N(0,1) \tag{3.11}$$

$$\begin{gathered}\sigma^m_k \sim \mathsf{HalfStudentT}(3, s^m_{0,k}), \\ k \in \{u,v,w\}, \quad m \in \{L,H\}\end{gathered} \tag{3.12}$$

$$\alpha^L \sim N(0, s^2_\alpha), \qquad \beta^m \sim N(0, B^m) \tag{3.13}$$

$$\lambda_{0,\ell} \sim \mathsf{Gamma}(\mathsf{shape} = a_\ell, \mathsf{rate} = b_\ell) \tag{3.14}$$

Model the selection process, then test its assumptions

Appendix A3.3 derives selection transport beginning with equation (A3.7). Equation (A3.16) supplies the explicit sensitivity model when unobserved selection remains relevant; Article 1's policy estimators appear in equations (1.10), (1.11).

Let

$$e(x) = \mathsf{Pr}(S = 1 \mid X = x) \tag{3.15}$$

be the historical financing-selection probability. If the recorded information captures all relevant causes of selection and potential repayment, one may consider a conditional exchangeability assumption such as $D^a(\tau) \perp\!\!\!\perp S \mid X$, together with overlap

$$0 < e(x) < 1 \tag{3.16}$$

in the target region. These conditions are strong and must be assessed against decision practice. If policy always rejected an entire score or business segment,

$$e(x) = 0 \tag{3.17}$$

there; no reweighting estimator can obtain outcomes from financed counterparts that do not exist. An estimated propensity close to zero can generate unstable weights, and an algorithmic smoothing prior does not turn unsupported data into observed evidence. A useful report therefore pairs predicted PD with a support label and clearly distinguishes observed, weakly supported and extrapolated regions.

Historical lending decisions may have used information that the modelling extract does not contain. Loan officers might have reviewed business continuity, inconsistent documents, supplier reliability or customer concentration; a credit committee may have recorded a policy exception in a separate system. If such information influenced both approval and repayment, conditioning on X does not make historical selection ignorable. The analysis should reconstruct available decision rules, application channels, manual overrides, policy versions, score cut-offs, declined reasons and economic periods. It should preserve the variables and timestamps actually available at each decision. If a source field cannot be recovered, the absence should be documented, because the selection process may have relied on it. A propensity model is useful only insofar as it represents the actual decision mechanism and shares support with the target population.

Sensitivity analysis makes hidden assumptions reviewable. One illustrative parameterization is

$$\begin{aligned}\operatorname{logit}\Pr(D^a = 1 \mid X, S = 0) = {} & \operatorname{logit}\Pr(D^a = 1 \mid X, S = 1) \\ & + \delta_g\end{aligned}, \tag{3.18}$$

where δ_g represents a segment-specific difference in risk associated with unobserved selection information. Analysts can vary δ_g across plausible values, compare resulting expected loss and approval policies, and show where the preferred action changes. A joint Bayesian selection-outcome model can also describe dependence between approval and repayment, but its posterior depends on identifying restrictions, priors, exclusions or other information that links the two processes. The practical aim is not to invent a "true" hidden adjustment; it is to show whether a conclusion survives credible assumptions. Parameter uncertainty and selection uncertainty should be reported separately so a narrow posterior conditional on a strong assumption is not mistaken for certainty about the counterfactual.

Inverse-probability weighting can help target the historical sample to an applicant population when selection is conditionally explainable and overlap is adequate. If an applicant with context x has financing probability $e(x)$, observations can be weighted to represent the intended population. Yet extreme weights can make a small number of loans dominate the estimate, and clipping them changes the estimand or introduces bias that must be disclosed. Outcome-regression and doubly robust approaches combine a model for repayment with a model for selection or action assignment; under standard conditions, they can remain consistent if one nuisance model is correct, but they do not solve unmeasured confounding and no-overlap together [7], [8]. Reports should show propensity distributions, weight truncation sensitivity, effective sample size, support by segment, outcome missingness and uncertainty. "Doubly robust" describes a property under assumptions; it is not a guarantee that an arbitrary reject-inference process is robust.

Pólya–Gamma helps compute a model; it does not reveal missing outcomes

Pólya–Gamma augmentation can make Bayesian logistic regression computationally convenient. For binary outcome $y_i \in \{0, 1\}$ with predictor

$$\eta_i = x_i^{\mathsf{T}}\beta, \tag{3.19}$$

define

$$\kappa_i = y_i - \frac{1}{2} \tag{3.20}$$

and introduce latent

$$\omega_i \sim \mathsf{PG}(1, \eta_i). \tag{3.21}$$

Conditional on ω, the likelihood is quadratic in β; with a Gaussian prior, the regression coefficients have a Gaussian conditional distribution, with covariance:

$$\Sigma^* = \left(X^\mathsf{T}\Omega X + \Sigma_0^{-1}\right)^{-1} \tag{3.22}$$

and a corresponding posterior mean [9]. Hierarchical models add group-level parameters and additional sampling steps. This structure can support posterior inference for a default model or historical selection model, provided the sampler mixes and converges, posterior predictive checks are satisfactory and results are tested for prior sensitivity. Store these diagnostics with the fitted version and link them to the decision for which the model is being evaluated.

The computational identity should not be overextended. It can simplify sampling from a posterior conditional on a stated model, but it cannot identify a repayment label that was never observed. If a Gibbs sampler draws y_i for a rejected application from Bernoulli(p_i) and then feeds that draw back into the same model, the result is model-generated synthetic data. It may illustrate consequences under assumptions or support a sensitivity analysis, but it does not add empirical repayment evidence. A posterior can be precise conditional on an assumed selection mechanism while the actual counterfactual remains weakly identified. Label provenance should therefore distinguish observed outcomes, administrative corrections, externally verified events, model predictions and simulated values. Only observed or appropriately adjudicated outcomes should be used as factual repayment records in a lender's performance ledger.

The compact Pólya–Gamma expression uses an expanded design matrix containing the intercept, approved borrower and offer-interaction bases, sector indicators, region indicators and vintage indicators. Its coefficient vector stacks the logistic coefficients and group effects; the Gaussian prior covariance conditions on the current hierarchy scales. Thus the short matrix formula represents the same complete logistic hierarchy rather than dropping its group levels. The survival model uses those named levels in its own hazard likelihood and posterior calculation. Versioned design records preserve every column, reference level, group mapping, constraint and prior setting. The appendix derives the coefficient conditional distribution and explains the separate scale updates, so the computational shortcut and the human-review uncertainty summary remain connected to the fitted specification.

Generate evidence through a guarded offer design

The alternative to unsupported imputation is prospective evidence collection when a decision is genuinely uncertain and an offer is safe, affordable and lawful. Define a narrow study population that passes identity, eligibility, ability-to-repay, product and other required policy checks. Within that group, a randomized offer or repayment-schedule comparison can estimate outcomes for the specified action. A design might compare a standard monthly schedule with a cash-flow-aligned schedule, or randomize a controlled invitation to reconsider an application after a policy review. The randomized action is not a license to expand credit to applicants who fail safeguards. It should be approved by the institution's responsible-lending, legal, data-protection, model-risk and research governance functions, and the control group should retain standard service and rights.

The protocol should specify the estimand, population, offer terms, assignment mechanism, comparator, follow-up horizon, outcome definitions, sample-size assumptions, treatment noncompliance, missing-data plan, interim monitoring, stop rules and human review. Keep the point-in-time application record, proposed

contract, policy version, assignment probability, final offer, staff override and reason, acceptance, disbursement, servicing history, repayment, arrears, cure, restructuring, recovery and loss. A randomized offer estimates the effect of the offer policy, including take-up, if analyzed by assignment. It does not automatically estimate the effect of origination among those who accept. If an encouragement is used as an instrument for take-up, its assumptions and local-complier interpretation must be made explicit. The assignment unit may be a borrower, account, branch or group depending on spillovers and operational delivery.

Adaptive allocation can follow once a fixed pilot establishes data quality, outcome latency, safe action sets and support. A mixture policy might be written

$$\pi_t(a \mid x) = (1 - \varepsilon_t)\widehat{\pi}_t(a \mid x) + \varepsilon_t q(a \mid x), \tag{3.23}$$

where $\widehat{\pi}_t$ is the current estimated best action and q preserves controlled exploration. In credit, the allowed action set should be determined by law, affordability, credit policy and capacity before the bandit acts. Propensities should be logged after all eligibility and human gates, and the design should retain a randomized benchmark or valid holdout. Bandit adaptation can reduce assignment to actions that look inferior, but it can also starve an arm of evidence and amplify noise when outcomes are delayed. The adaptive method must use inference appropriate to changing allocation, rather than standard fixed-randomization formulas without adjustment [10].

A worked example: four defaults among eighty funded pilot loans

Consider a hypothetical design with **1,000** applicants who were declined under an earlier policy but pass a new minimum affordability and eligibility screen for one specified product. Suppose **100** are randomly selected to receive the offer, **80** accept and draw, and all 80 have complete twelve-month follow-up. Four default by the pre-defined twelve-month endpoint. The observed funded-loan rate is

$$\frac{4}{80} = 5\%. \tag{3.24}$$

A Wilson 95% interval for a binomial proportion is approximately **2.0%–12.2%**. These numbers are illustrative, not portfolio findings or a recommended approval policy. They show that the study has produced evidence for a defined offer and pilot-eligible population, while leaving material uncertainty about its risk. The denominator is 80 originated loans, not the 100 offered and not all 1,000 screened applicants [11].

The randomized offer also creates information about offer take-up and outcomes among the assigned population if the study includes an appropriate comparator and captures the defined outcome for all assigned applicants. The intention-to-treat comparison can measure the effect of making the offer available on net contribution per eligible applicant, take-up, or a later arrears indicator. The 900 applicants not offered a loan and the 20 offered applicants who did not draw account for 920 applicants without funded-loan repayment observations under this pilot contract. Comparing the 80 acceptors with a historical approved pool is not a causal comparison; acceptance and historical approval both reflect selection. A report should state which quantity is estimated, show attrition and noncompliance, and avoid stating that all applicants previously rejected have a 5% PD. The pilot's evidence is local to the design and does not travel automatically to a different rate, product, sector or economic period.

If the four defaults occurred in months 2, 5, 10 and 12, and some other loans cured or prepaid before month 12, a survival or multi-state analysis could describe time to event and competing exits. The small sample would make segment-specific hazards highly uncertain. A hierarchical model might pool groups, but its prior and risk structure would matter substantially; reporting the 5% point estimate alone would conceal that. The next study could grow the sample only within suitable eligibility and governance boundaries, with matured cohorts and ongoing loss monitoring. A separate sensitivity analysis could ask how estimates change if the

pilot population still differs from historical rejects in unrecorded ways. Neither step turns the estimate into a universal rejected-applicant default rate.

Quantity	Hypothetical value	Interpretation
Screened and eligible population	1,000	Passed documented pre-randomization safeguards
Applicants assigned an offer	100	Randomly selected in this example
Offers accepted and originated	80	Take-up count
Twelve-month defaults	4	Matured outcomes under this offer
Funded-loan default estimate	5%	4 ÷ 80 for the pilot offer and population
Approximate 95% Wilson interval	2.0%–12.2%	Sampling uncertainty under binomial approximation

Connect the evidence to business cash and receivables

For an operating business, the repayment analysis should distinguish customer receipts from owner injections, inter-account transfers, refinancing, asset sales and one-off grants. Those flows may be valid sources of payment but they answer different questions about sustainability. The lender needs an entity-resolved view of revenue, working capital, taxes, payroll, suppliers, debt obligations, owner withdrawals and seasonal cycles. An applicant may repay an instalment through a transfer from a related account while the business itself is weakening; another may have temporary timing pressure because a verified customer invoice will settle after the due date. A risk model that treats all credits as recurring operating cash can misclassify these situations. The data ledger should preserve source, account ownership, transaction date, reversal status and verification level, subject to privacy and purpose limits.

For financing a ring-fenced receivables stream, the core exposure is identified assets and the collections legally dedicated to repayment. Evidence should cover invoice eligibility, underlying supply, obligor acceptance, concentration, dilution, set-off, dispute, assignment, perfection, account control, servicing continuity, collection timing, reserves and waterfall distributions. The collections dedicated to the receivables facility can be strong even where the originator's unrestricted business cash is weak, and the reverse can also occur. Where the worked structure uses a special-purpose vehicle (SPV), the analysis follows the receivables held by that vehicle and the collections available under its waterfall. Reject inference for an operating-business facility therefore needs a separately specified outcome and selection model when applied to financing ring-fenced receivables. The offer, unit of analysis, obligor, legal rights and observed repayment event differ. A trial should record whether cash came from ordinary trading, an assigned invoice, a guarantor, a reserve, owner support or recovery action so that a "repayment" label preserves its economic source.

Collective guarantee arrangements and chama-based screening add group information and group-level risks. Peer screening may provide contextual evidence about trading continuity, member history and local knowledge; peer monitoring may identify deterioration after disbursement. At the same time, members can be exposed to correlated shocks, social pressure, privacy loss, inaccurate peer reports and unclear liability. The research design should distinguish individual debt from group guarantee obligations, identify who supplies and can correct information, explain what members can see, and define the procedure for disputes and guarantee calls. Statistically, members share an environment and may affect one another's outcomes; the no-interference assumption can fail. Cluster-level randomization or explicit network and group effects may be needed. Peer knowledge should augment documented underwriting, not act as an unreviewable label or substitute for a member's rights.

What the GSE data can show about selection

The GSE files are useful here for their visible population boundary. A performance dataset begins after a defined acquisition and eligibility process, so it supports analysis of included loans but cannot reveal the potential repayment of applicants or loans excluded before acquisition. That is the same distinction a reject-inference study must preserve: document who entered the file, which policy or acquisition cut-offs applied, and what exclusions shaped the observed labels. Loan vintages, identifiers, reporting dates and later corrections also matter because selection can occur at several stages, not only at the original application decision [12], [13]. Article 1 covers the broader data architecture; in this article the focus is how recorded coverage limits counterfactual claims. These mortgage files are not Kenyan MSME evidence, so local estimates require locally governed data and prospective designs.

Kenyan data protection and credit rules shape the study

The Data Protection Act, 2019 sets processing principles in section 25 and data-subject rights in section 26. Section 31 requires a data-protection impact assessment where processing is likely to result in high risk to rights and freedoms; section 35 addresses decisions based solely on automated processing that produce legal effects or significantly affect a person, subject to statutory exceptions and safeguards [14]. A company is not itself a natural-person data subject, but business files may identify a sole proprietor, director, guarantor, employee or customer. Transaction-level records can reveal personal or sensitive circumstances. The project should classify controller and processor roles, purpose, lawful basis, notices, data sources, minimisation, accuracy, retention, security, rights handling, access controls, linkage and cross-border transfers before model construction.

Regulation 22 of the Data Protection (General) Regulations, 2021 gives operational safeguards for automated individual decision-making, including notice, meaningful information about the logic, measures to prevent errors and bias, suitable statistical procedures, and a route to human intervention and expression of the data subject's point of view. Regulation 49 identifies processing activities requiring a DPIA, and regulations 50–53 address assessment, prior consultation, consideration and audit [15]. The assessment should distinguish a model that prioritizes a case for human review from one that automatically declines an application, changes a credit limit or triggers a collection outcome. Human review should be substantive: the reviewer needs information, expertise, authority and time to change the recommendation. The workflow should preserve correction and complaint routes. A DPIA identifies and mitigates processing risks; it does not by itself establish every lawful basis or permission needed for the project.

Credit-specific rules depend on the institution and product. The Credit Reference Bureau Regulations govern defined credit-information processes. The Digital Credit Providers Regulations, 2022 apply within their statutory scope and include provisions on responsible lending, debt collection and complaint handling [16], [17]. A bank, microfinance institution, SACCO, digital credit provider, intermediary and chama do not necessarily have the same regulator, permissions or duties. Before a pilot, classify the entity and determine which banking, microfinance, SACCO, credit reporting, consumer-protection and data-protection provisions apply. The research file should record which rule or policy governs eligibility, offer, collection contact, credit reporting, automated decision-making and appeals. Kenyan counsel and compliance teams should verify application against current instruments; a research paper should not generalize a rule for one provider category to the entire credit market.

Fairness and outcome measurement

Fairness diagnostics should be selected according to the decision and the error that matters. Equal opportunity compares true-positive rates across groups; equalized odds compares true-positive and false-positive rates conditional on the outcome [18]. Neither means equal accuracy, and neither resolves all questions about access, price, affordability, service quality, collection burden, recovery or privacy. The labels them-

selves are selective: repayment is observed under prior offers and servicing, and an intervention can change the outcome. Protected characteristics may be restricted or sensitive, though a carefully governed fairness assessment may need lawful access to some attributes. Reports should state group definitions, sample sizes, label source, outcome horizon, uncertainty, thresholds, support gaps and the action taken when a disparity appears. Excluding sensitive fields or adding an adversarial constraint does not by itself eliminate proxy effects or prove legal compliance.

IFRS 9, expected loss and capital remain distinct

Expected credit loss, pricing expected loss, economic capital and regulatory capital are related but non-interchangeable. IFRS 9 uses probability-weighted discounted cash shortfalls over the applicable horizon using reasonable and supportable information; 12-month ECL is not simply the cash shortfall expected to occur during the next twelve months [19]. The approximation

$$\mathrm{EL} \approx \mathrm{PD} \times \mathrm{LGD} \times \mathrm{EAD} \tag{3.25}$$

can explain an expected-loss component under simplifying assumptions, but it is not a complete ECL calculation or an economic-capital measure. Economic capital may be defined using a selected tail-loss measure and horizon, while regulatory capital follows applicable prudential rules. Reject-inference sensitivity can inform a risk discussion, but model-imputed labels should not be inserted into a ledger as observed Stage 1-to-Stage 2 transitions or used to state an actual provision amount. Accounting and regulatory outputs must follow their separate methods and approvals.

A staged evidence program

The first stage reconstructs historical decision-making: policy versions, score cut-offs, manual overrides, offer terms, application channels, applicant data available at the time, and the path from approval to acceptance, disbursement and servicing. It measures overlap, missingness, outcome maturity, group support, data corrections and changes in market conditions. The second stage validates fixed-horizon PD and survival models within the funded population, using time-based holdouts, calibration, competing-event checks and subgroup diagnostics. The third stage varies assumptions about unrecorded selection information and reports where estimates or decisions change. These steps show which segments can be addressed with historical evidence and which rely mostly on extrapolation. They also help the institution design a prospective study whose action and target population match the decision it needs to make.

The evidence pathway below keeps the observed and unobserved branches separate. Historical modelling describes supported experience, while sensitivity analysis makes assumptions visible; a safeguarded prospective design can add outcomes only for a defined eligible population and contract. The path does not collapse model-generated values into the repayment ledger. That separation matters for audit, fairness, accounting and future policy evaluation because the same applicant may face different outcomes under different contract terms. The flow also makes a review point explicit: if the proposed study cannot establish an eligible population, lawful data use, meaningful comparison and protection of ordinary rights, the process returns to design rather than progressing to adaptive policy.

The fourth stage develops a controlled offer or schedule pilot among applicants who meet minimum eligibility and affordability requirements, with a current-policy comparator, a prespecified outcome, appropriate statistical power and governance approval. The fifth stage examines heterogeneous effects and outcomes by product, sector, season and relevant groups while preserving uncertainty and privacy safeguards. Contextual allocation may be considered after fixed randomization establishes reliable outcomes, sufficient overlap and an operationally safe action set. Adaptive assignments need accurate logged propensities, a benchmark and inference that accounts for response-adaptive allocation. The final stage follows financed cases into

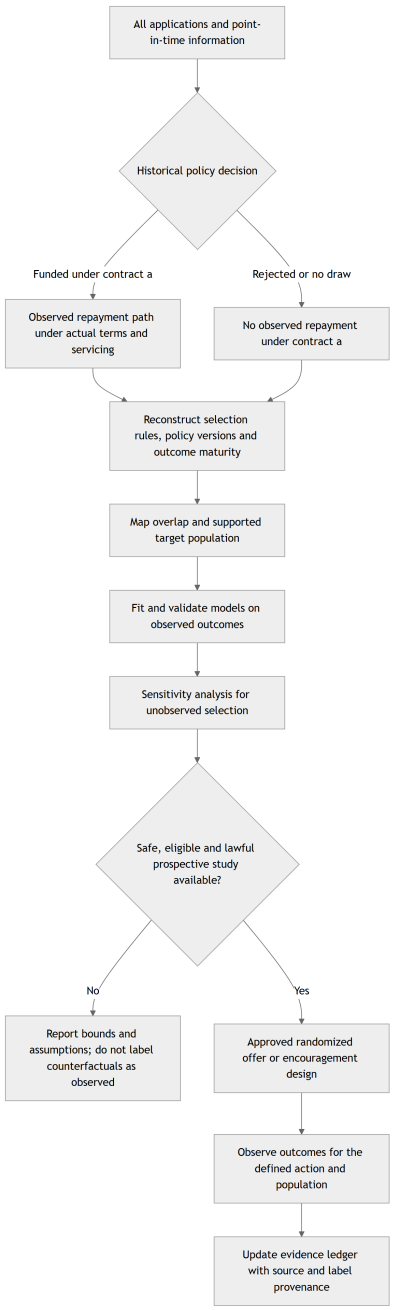


Figure 4: Reject-inference evidence pathway

borrower cash, servicing, cure, recovery and loss, distinguishing enterprise repayment from receivables proceeds, guarantee payments and owner support. This gives model users a credible basis for future decisions without claiming that one experiment resolves every rejected application.

Conclusion

Reject inference is best handled as a transparent account of what is observed, what is assumed and what new evidence can be generated. Define repayment under a specific offer and horizon; reconstruct the historical selection process; map overlap; test selection assumptions; and separate model uncertainty from uncertainty about unrecorded decision information. Hierarchical Bayesian logistic and survival models can pool supported data and describe risk over time. Pólya–Gamma augmentation can support computation for logistic models. Neither creates repayment outcomes for loans never made. Randomized offers can generate evidence for carefully screened populations and specified contracts, while adaptive bandits can allocate only among approved actions with logged probabilities and suitable inference.

For *Underwrite for Collection*, this method connects application, contract, operating cash, monitoring, collection and recovery. The evidence unit differs between a going-concern business loan and a ring-fenced receivables stream, and it differs again for a collective guarantee. Kenyan privacy, data-protection and sector rules determine what data can be processed and how people can challenge consequential decisions. Fannie Mae and Freddie Mac provide valuable examples of cohort design and performance-data documentation, while local evidence must establish local MSME outcomes. The most credible next step is a governed support map followed by a narrow, affordable study with clear outcomes and a standard-care comparator. The result can improve future lending decisions while stating its limits plainly.

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Publication note. The pilot numbers are hypothetical and do not describe a Kenyan portfolio or propose an approval rule. This article is a research design discussion, not a fitted model, legal opinion or claim that reject inference has been solved.

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Article 3 mathematical appendix: selective labels, posterior inference and evidence boundaries

This appendix derives the statistical relationships used in Article 3. A repayment outcome is indexed by contract, population and horizon. The calculations distinguish an observed loan outcome, an estimate for a supported population and a sensitivity calculation for unobserved outcomes. The examples are hypothetical and use KES for currency. The common hierarchical structure retains sector, region and origination-vintage effects so that its uncertainty can be carried into the later decision and operating workflows.

A3.1 Define the offer, funding and outcome separately

Let X_i be the information available before applicant i is considered for a complete contract a. Let $U_i(a)$ indicate whether that applicant would accept and draw the offer. Let $D_i^a(\tau)$ indicate default by horizon τ under origination of that specified contract and servicing regime. The latter is a potential outcome under a funding intervention. If the target is an offer policy rather than a funding intervention, also define an applicant-level outcome $Y_i^{\text{offer}}(a)$, whose meaning includes non-take-up. These definitions avoid assigning a factual loan-default label to somebody who never took a loan.

The principal targets are

$$\begin{aligned} p_a(x) &= \Pr\{D_i^a(\tau) = 1 \mid X_i = x\}, \\ t_a(x) &= \Pr\{U_i(a) = 1 \mid X_i = x\}, \\ p_a^{\text{takers}}(x) &= \Pr\{D_i^a(\tau) = 1 \mid X_i = x, U_i(a) = 1\}, \\ v_a(x) &= E[Y_i^{\text{offer}}(a) \mid X_i = x]. \end{aligned} \tag{A3.1}$$

These are respectively funding-intervention risk, offer take-up, risk among that offer's takers, and value per applicant receiving the offer policy. Equal numerical values are possible but are not assumed. The contract a includes limit or advance, price and fees, term, repayment dates, guarantees, security and servicing. Historical outcomes under different contracts require explicit action adjustment or contract-specific strata.

For the derivations below, let $S_i = 1$ denote financing under the specified contract a with the target outcome observed. Selection can be decomposed further into approval, delivery, acceptance, disbursement and outcome observation. If the historical data contain several offers, replace S by $S_a = 1\{S = 1, A = a\}$ and use the joint selection probability for that contract. A propensity for generic approval alone does not adjust changes in contract or take-up.

A3.2 Why observed labels do not identify missing outcomes by themselves

The selective-label and reject-inference problem follows [1], [2], [3].

Write

$$e(x) = \Pr(S = 1 \mid X = x), \qquad m_s(x) = \Pr\{D^a(\tau) = 1 \mid X = x, S = s\}. \tag{A3.2}$$

The law of total probability gives

$$p_a(x) = e(x)m_1(x) + \{1 - e(x)\}m_0(x). \tag{A3.3}$$

Historical financed loans estimate $m_1(x)$ where supported; the full application file can estimate $e(x)$. The unobserved $m_0(x)$ remains needed for the all-applicant target. With only the fact that a binary probability lies between zero and one,

$$e(x)m_1(x) \le p_a(x) \le e(x)m_1(x) + 1 - e(x). \tag{A3.4}$$

At $e = 0.60$ and $m_1 = 0.05$, these identification bounds are 3% to 43%. They are logical bounds under the defined outcome, not a 95% uncertainty interval. Additional justified restrictions can narrow them; a narrow fitted-model interval alone does not alter their identification basis.

An observed-data likelihood makes the information gap explicit. Under a selection-ignorable outcome model with parameter θ, an unobserved binary outcome contributes

$$\sum_{d=0}^{1} p_\theta(x)^d \{1 - p_\theta(x)\}^{1-d} = p_\theta(x) + 1 - p_\theta(x) = 1. \tag{A3.5}$$

Thus its missing label contributes no direct repayment-likelihood factor. In a correctly specified data-augmentation algorithm, drawing that missing label is a computational device whose integration returns this same likelihood; it does not enlarge the observed sample. Treating one imputed draw as an additional fixed observation during refitting can instead add artificial information.

For non-ignorable selection, let $e_d(x) = \Pr(S = 1 \mid D^a = d, X = x)$. At a given x, the observed categories have probabilities

$$\begin{aligned} \Pr(S = 1, D = 1 \mid x) &= e_1(x)p_a(x), \\ \Pr(S = 1, D = 0 \mid x) &= e_0(x)\{1 - p_a(x)\}, \\ \Pr(S = 0 \mid x) &= 1 - e_1(x)p_a(x) - e_0(x)\{1 - p_a(x)\}. \end{aligned} \tag{A3.6}$$

Only two category probabilities are independent, while p_a, e_1, e_0 supply three unknown quantities at that x. Without extra restrictions or information, different combinations produce the same observed distribution. A joint selection-outcome model makes those restrictions explicit through a functional form, measured predictors, a defensible exclusion variable, prior information or prospective observations.

Source connection: Lakkaraju and colleagues, "The Selective Labels Problem," motivates the observation boundary; Crook and Banasik, "Does Reject Inference Really Improve the Performance of Application Scoring Models?", motivates testing how reject-inference assumptions affect results.

A3.3 Transport from financed applicants to a defined target

Assume consistency for the declared contract, conditional selection exchangeability

$$D^a(\tau) \perp\!\!\!\perp S \mid X, \tag{A3.7}$$

and positive selection probability $e(X) > 0$ wherever the target applicant distribution has mass. Then $m_0(x) = m_1(x) = m(x)$ and the target mean default probability is $\mu_a = E_X[m(X)]$. The usual stronger overlap statement $0 < e(X) < 1$ permits comparison of selected and non-selected applicants; estimating the all-applicant mean from outcomes fundamentally requires a positive chance of a selected analogue in each target region.

For a complete applicant sample, the inverse-selection identity follows by conditioning:

$$\begin{aligned} E\left[\frac{SD}{e(X)}\middle| X\right] &= \frac{\Pr(S = 1 \mid X)}{e(X)} E[D \mid X, S = 1] \\ &= m(X). \end{aligned} \tag{A3.8}$$

Consequently,

$$\hat{\mu}_{\mathsf{IPW}} = \frac{1}{N}\sum_{i=1}^{N}\frac{S_i D_i}{\hat{e}(X_i)} \tag{A3.9}$$

targets all N applicants, under the stated conditions and a consistent selection model. Multiplication by S_i means the summand is zero when the outcome is absent; no default value is imputed into the underlying record.

If the target is specifically the historically unfinanced population, use its covariate distribution:

$$\begin{aligned} E[D^a \mid S=0] &= \frac{E[\{1-e(X)\}m(X)]}{\mathsf{Pr}(S=0)} \\ &= \frac{E\left[S\frac{1-e(X)}{e(X)}D\right]}{\mathsf{Pr}(S=0)}. \end{aligned} \tag{A3.10}$$

The selected observations receive odds weights $(1-e)/e$, rather than the all-applicant weights $1/e$. If only a financed file exists and the applicant distribution or selection probabilities are unknown, neither target is automatically recovered. An independently sampled target population can instead be handled with a properly estimated density ratio; that ratio needs information about both populations.

Numerical example. Suppose the target has 100 applicants in each of two strata. Financing probabilities are 0.5 and 0.25, giving 50 and 25 financed observations. Observed default counts are 5 and 5. The raw financed-loan estimate is $10/75 = 13.333\%$. The IPW numerator is $5/0.5 + 5/0.25 = 30$, giving $30/200 = 15\%$. This equals the equally weighted stratum calculation $(0.10 + 0.20)/2$. The first calculation describes the selected portfolio; the second describes the target under exchangeability. Neither calculation corrects an unrecorded selection variable that also affects default.

At $e(x) = 0$, the reciprocal weight is undefined, expressing the absence of outcome support. Near-zero probabilities generate concentrated weights. Report the weight distribution and the diagnostic $(\sum_i w_i)^2/\sum_i w_i^2$ alongside uncertainty. Truncating weights changes the calculation and requires an explicit sensitivity or revised target interpretation.

A3.4 Doubly robust selection adjustment

The augmented estimator adapts the doubly robust construction in [7], [8] to the stated selection assumptions.

For working outcome model $\widehat{m}(X)$ and working selection model $\hat{e}(X)$, define

$$\hat{\mu}_{\mathsf{DR}} = \frac{1}{N}\sum_i \left[\widehat{m}(X_i) + \frac{S_i}{\hat{e}(X_i)}\{D_i - \widehat{m}(X_i)\}\right]. \tag{A3.11}$$

Conditioning on X under exchangeable selection gives

$$\begin{aligned} E\left[\widehat{m}(X) + \frac{S}{\hat{e}(X)}(D-\widehat{m}(X))\middle| X\right] &= \widehat{m}(X) + \frac{e(X)}{\hat{e}(X)}\{m(X)-\widehat{m}(X)\}, \\ \mathsf{Bias}(X) &= \left(\frac{e(X)}{\hat{e}(X)} - 1\right)\{m(X)-\widehat{m}(X)\}. \end{aligned} \tag{A3.12}$$

The conditional bias is zero if either working model is correct. It need not be zero when selection exchangeability fails, since then $E[D \mid S = 1, X]$ differs from the target conditional mean. Cross-fitting fits nuisance functions away from the evaluated observation; its purpose is to control overfitting effects, with inference requiring suitable regularity and convergence rates. Observed model accuracy in the financed subset cannot establish the exchangeability condition.

For unequal contract assignments, a more complete residual term uses the probability of selection and action jointly. With $I_i(a) = 1\{S_i = 1, A_i = a\}$ and $e_a(x) = \Pr\{I_i(a) = 1 \mid X_i = x\}$,

$$\hat{\mu}_a = \frac{1}{N} \sum_i \left[\widehat{m}_a(X_i) + \frac{I_i(a)}{\widehat{e}_a(X_i)} \{D_i - \widehat{m}_a(X_i)\} \right]. \tag{A3.13}$$

Further right-censoring or missing follow-up requires its own outcome-observation treatment. The Article 1 appendix derives the corresponding action-policy estimator and observation weights. Those weights adjust separate mechanisms and must correspond to the decision and data actually delivered.

Source connection: Dudík, Langford and Li, "Doubly Robust Policy Evaluation and Learning," gives the policy version of this augmentation logic; Wang, Agarwal and Dudík, "Optimal and Adaptive Off-Policy Evaluation in Contextual Bandits," discusses its evaluation setting and overlap challenges.

A3.5 Derive log-odds sensitivity for unobserved selection

Let δ_g be a declared sensitivity parameter for segment g. Write

$$\operatorname{logit} m_0(x) = \operatorname{logit} m_1(x) + \delta_g. \tag{A3.14}$$

Since $\operatorname{logit} p = \log\{p/(1-p)\}$, exponentiating yields

$$\frac{m_0(x)}{1 - m_0(x)} = e^{\delta_g} \frac{m_1(x)}{1 - m_1(x)}. \tag{A3.15}$$

Solving for the unobserved-group risk gives

$$m_0(x; \delta_g) = \frac{e^{\delta_g} m_1(x)}{1 - m_1(x) + e^{\delta_g} m_1(x)}. \tag{A3.16}$$

Thus e^{δ_g} is the assumed rejected-to-financed odds ratio within that segment and context. It is not a parameter identified merely by observing financed-loan defaults. At $m_1 = 0.05$, choices $\delta = \log(1/2), 0, \log 2$ give 2.5641%,5%,9.5238% for the unfinanced-group risk. If $e = 0.60$, corresponding all-applicant risks are

$$0.6(0.05) + 0.4m_0 = 4.0256\%, \quad 5.0000\%, \quad 6.8095\%. \tag{A3.17}$$

For posterior draws $m_1^{(b)}(x)$ and $e^{(b)}(x)$, transform each draw at each fixed sensitivity value δ and aggregate over the target covariates:

$$\mu_a^{(b)}(\delta) = \frac{1}{N} \sum_i \left[e^{(b)}(X_i) m_1^{(b)}(X_i) + \{1 - e^{(b)}(X_i)\} m_0^{(b)}(X_i; \delta) \right]. \tag{A3.18}$$

The quantiles across b quantify posterior uncertainty conditional on a chosen δ. Variation across δ shows sensitivity to the identification assumption. These two sources of uncertainty should remain visible to the reviewer. A prior for δ can be used if its external basis and influence are stated; its posterior concentration does not become observed reject performance.

A3.6 The common sector, region and vintage hierarchy

The multilevel prior and pooling structure follow [4], [5].

For a fixed horizon and contract, let the linear predictor be

$$\begin{aligned} D_i^a(\tau) \mid p_i &\sim \text{Bernoulli}(p_i), \\ p_i &= \text{logit}^{-1}(\eta_i), \\ \eta_i &= \alpha^L + u^L_{s(i)} + v^L_{r(i)} + w^L_{c(i)} + f^L(X_i) + h^L(a, X_i). \end{aligned} \tag{A3.19}$$

Here $s(i), r(i), c(i)$ index sector, region and origination vintage. Superscript L identifies logistic parameters; superscript H identifies hazard parameters. f^L contains declared borrower transformations, while h^L contains offer effects and interactions. A design row $z_i^\top$ can collect these terms so that $\eta_i = z_i^\top \theta$. Group effects require identifiability constraints or proper centred priors together with a declared intercept convention. Reference levels or sum-to-zero constraints must be implemented consistently in scoring and estimation.

One non-centred scale specification is

$$\begin{aligned} u_s^m &= \sigma_u^m \tilde{u}_s^m, & \tilde{u}_s^m &\sim N(0,1), \\ v_r^m &= \sigma_v^m \tilde{v}_r^m, & \tilde{v}_r^m &\sim N(0,1), \\ w_c^m &= \sigma_w^m \tilde{w}_c^m, & \tilde{w}_c^m &\sim N(0,1), \\ \sigma_k^m &\sim \text{HalfStudentT}(\nu = 3, s_{0,k}^m), & k &\in \{u, v, w\}, \quad m \in \{L, H\}, \\ \alpha^L &\sim N(0, s_\alpha^2), & \beta^m &\sim N(0, B^m), \\ \lambda_{0,\ell} &\sim \text{Gamma}(a_\ell, b_\ell). \end{aligned} \tag{A3.20}$$

Each σ_k^m is a family scale shared across that effect's levels within model m. Normal's second argument is variance or covariance; Gamma uses shape and rate. The declared positive $s_{0,k}^m, s_\alpha, a_\ell, b_\ell$ and positive-definite B^m complete the prior specification. β^m includes the coefficients of the declared feature and action-interaction bases. Prior-predictive checks assess implications on the appropriate log-odds or log-hazard scale. A vintage random walk or autoregression is an optional extension with its own innovation-scale prior and explicit temporal structure; exchangeable vintage effects do not imply that time-series dependence automatically exists.

Why pooling depends on information. In a Gaussian analogue, suppose a noisy group statistic obeys $\overline{y}_j \mid u_j \sim N(u_j, s^2/n_j)$ with $u_j \sim N(0, \tau^2)$. Multiplying the two normal kernels gives posterior precision $n_j/s^2 + 1/\tau^2$ and mean

$$E[u_j \mid \overline{y}_j] = \frac{\tau^2}{\tau^2 + s^2/n_j} \overline{y}_j. \tag{A3.21}$$

The shrinkage weight approaches one as the group's data become precise and approaches zero as the prior group scale becomes small. For $\tau^2 = 0.04$, $s^2 = 1$, $\overline{y}_j = 0.7$, a group of 10 gives posterior mean 0.2;

a group of 100 gives 0.56. This is an exact Gaussian demonstration of pooling, not an exact closed-form logistic posterior. The augmented logistic model obtains analogous conditional Gaussian calculations within its full sampler.

Posterior draws preserve uncertainty about group scales and correlations where those are modelled. A sector–region–vintage subgroup with little information may depend strongly on pooling and prior choices. Human review should therefore display its support and sensitivity alongside its credible interval. To estimate a causal offer effect through $h(a, X)$, the model still needs the experiment or adjustment assumptions that identify the offer comparison.

A3.7 Half-Student-t scale mixtures and their computational role

The auxiliary scale representation follows [20], Result 5.

Use a Gamma distribution with shape and rate. For $\nu > 0$ and scale $s_0 > 0$, let

$$\lambda \sim \text{Gamma}(\nu/2, \nu/2), \qquad q \mid \lambda \sim N(0, s_0^2/\lambda). \tag{A3.22}$$

Multiply the normal and Gamma densities, retaining terms depending on q and λ:

$$p(q, \lambda) \propto \lambda^{(\nu+1)/2-1} \exp\left[-\frac{\lambda}{2}\left(\nu + \frac{q^2}{s_0^2}\right)\right]. \tag{A3.23}$$

The Gamma integral $\int_0^\infty \lambda^{a-1} e^{-b\lambda} d\lambda = \Gamma(a)/b^a$ gives

$$p(q) = \frac{\Gamma((\nu+1)/2)}{\Gamma(\nu/2)\sqrt{\nu\pi}\, s_0}\left(1 + \frac{q^2}{\nu s_0^2}\right)^{-(\nu+1)/2}. \tag{A3.24}$$

This is a centred Student-t density. Setting $\sigma = |q|$ yields density $2p(\sigma)$ for $\sigma > 0$, the half-t prior. For $\nu = 3$, the Gamma shape and rate are both 3/2; the Student-t degrees of freedom are 3. The normal variance is s_0^2/λ, so its standard deviation is $s_0/\sqrt{\lambda}$.

In the hierarchy, a group effect is $u_j = \sigma z_j$ with $z_j \sim N(0, 1)$. Conditional on σ, the effect is normal; assigning a half-t to σ defines a scale mixture for u_j, but its marginal law is not automatically the same Student-t density just derived for q. The scale-mixture identity therefore needs its variable labels. It supplies a flexible positive scale prior and useful parameterisations, while coefficient conjugacy in the binary likelihood comes from the separate Pólya–Gamma augmentation below.

Source connection: Gelman, "Prior Distributions for Variance Parameters in Hierarchical Models," motivates calibrated half-t scale priors and sensitivity analysis. The Stan User's Guide, "Reparameterizing a Student-t Distribution," documents the normal–Gamma representation and its computational use.

A3.8 Derive the Pólya–Gamma Gaussian conditional posterior

The augmentation identity follows [9].

For observed binary y_i and log-odds η_i, the Bernoulli likelihood is

$$p(y_i \mid \eta_i) = \frac{e^{y_i \eta_i}}{1 + e^{\eta_i}}. \tag{A3.25}$$

Use $1 + e^{\eta} = 2e^{\eta/2}\cosh(\eta/2)$ and define $\kappa_i = y_i - 1/2$. Then

$$p(y_i \mid \eta_i) = \frac{1}{2} e^{\kappa_i \eta_i} \cosh^{-1}(\eta_i/2). \tag{A3.26}$$

The Pólya–Gamma Laplace-transform identity for $\omega \sim \mathsf{PG}(1,0)$ is

$$E[e^{-\omega\eta^2/2}] = \cosh^{-1}(\eta/2). \tag{A3.27}$$

Substitution rewrites the likelihood as

$$p(y_i \mid \eta_i) = \frac{1}{2} e^{\kappa_i \eta_i} \int_0^{\infty} e^{-\omega_i \eta_i^2/2} p_{\mathsf{PG}(1,0)}(\omega_i)\, d\omega_i. \tag{A3.28}$$

The tilted conditional distribution is $\omega_i \mid \eta_i, y_i \sim \mathsf{PG}(1, \eta_i)$. Conditional on the latent ω_i, its contribution to the log-likelihood is quadratic: $\kappa_i\eta_i - \omega_i\eta_i^2/2$.

Let Z be the full design matrix for the common hierarchy, $\Omega = \mathsf{diag}(\omega_i)$, and $\kappa = (\kappa_i)$. Conditional on hierarchy scales, choose a proper Gaussian coefficient prior $\theta \sim N(m_0, Q_0^{-1})$, where Q_0 is its precision matrix. Combine likelihood and prior:

$$\log p(\theta \mid \omega, y, \mathsf{scales}) = -\frac{1}{2}\theta^{\top}(Z^{\top}\Omega Z + Q_0)\theta + \theta^{\top}(Z^{\top}\kappa + Q_0 m_0) + \mathsf{constant}. \tag{A3.29}$$

Completing the square gives

$$\begin{aligned} V_* &= (Z^{\top}\Omega Z + Q_0)^{-1}, \\ m_* &= V_*(Z^{\top}\kappa + Q_0 m_0), \\ \theta \mid \omega, y, \mathsf{scales} &\sim N(m_*, V_*). \end{aligned} \tag{A3.30}$$

For the article's compact notation, substitute the expanded $Z = X$, stacked $\theta = \beta$ and $Q_0 = \Sigma_0^{-1}$, obtaining its displayed covariance. The expansion includes sector, region and vintage indicators, with covariance conditional on their current scales. If fixed offsets c are retained outside the updated block, use $Z^{\top}(\kappa - \Omega c) + Q_0 m_0$ in the conditional mean. This accounts for other predictor components while updating a subset of coefficients.

One conditional-update check. In an intercept-only model with nine outcomes and one event, $\sum_i \kappa_i = 1 - 9/2 = -3.5$. Suppose the current latent draws sum to 2 and the intercept prior is $N(0, 4)$. Then

$$V_* = (2 + 1/4)^{-1} = 4/9, \qquad m_* = (4/9)(-3.5) = -14/9. \tag{A3.31}$$

The next coefficient draw is therefore $N(-1.5556, 0.4444)$ conditional on those latent variables. This single update is not the marginal fitted posterior. Repeated validated updates of latent variables, coefficients, scales and any correlation parameters generate the marginal posterior draws used for reporting. In software, factor and solve the positive-definite precision matrix rather than explicitly forming its inverse whenever feasible; compare numerical residuals and Monte Carlo diagnostics against the declared tolerances.

The hierarchy-scale updates are separate from this Gaussian block. Conditional conjugacy improves a defined computational step; elapsed time, effective sample size, convergence, calibration and decision stability

still need measurement. The approved scoring artefact can store posterior draws or validated summaries so that a reviewer receives uncertainty without refitting the full model during every decision.

Source connection: Polson, Scott and Windle, "Bayesian Inference for Logistic Models Using Pólya–Gamma Latent Variables," supplies the augmentation identity and Gaussian conditional calculation.

A3.9 Survival likelihood, hierarchy and default incidence

The hazard and partial-likelihood framework follows [6].

For event time T_i, censoring time C_i, observed time $t_i = \min(T_i, C_i)$ and event indicator δ_i, specify

$$\lambda_i(t \mid a) = \lambda_0(t) \exp\{X_i^\top \beta^H + q^H(a, X_i) + u^H_{s(i)} + v^H_{r(i)} + w^H_{c(i)}\}. \tag{A3.32}$$

Using the same sector, region and vintage indexing does not require the logistic and survival coefficients to be numerically identical; they describe different likelihoods and time targets. Any joint sharing needs an explicit joint model. For a time-constant predictor η_i, cumulative hazard is $\Lambda_i(t) = e^{\eta_i}\Lambda_0(t)$ and survival is $S_i(t) = \exp[-\Lambda_i(t)]$.

Under conditionally independent censoring, the event component of the likelihood is

$$L = \prod_i \lambda_i(t_i)^{\delta_i} S_i(t_i) = \prod_i \lambda_0(t_i)^{\delta_i} \exp\{\delta_i \eta_i - e^{\eta_i}\Lambda_0(t_i)\}. \tag{A3.33}$$

At distinct event times and conditional on one event in the risk set $\mathcal{R}(t_i)$, the baseline hazard cancels from the probability that unit i is that event:

$$\Pr(i \text{ fails} \mid \mathcal{R}(t_i), \text{one event}) = \frac{e^{\eta_i}}{\sum_{j \in \mathcal{R}(t_i)} e^{\eta_j}}. \tag{A3.34}$$

Multiplying these probabilities gives the Cox partial likelihood. Ties need a declared approximation or exact treatment. Absolute default probabilities additionally require an estimated or modelled baseline hazard. The fully Bayesian implementation here uses the H hierarchy and the piecewise baseline Gamma priors in A3.6. Let d_ℓ be the number of events in interval ℓ, and let $e_{i\ell}$ be subject i's observed time at risk in that interval. Conditional on hazard predictors, the likelihood kernel for the interval rate is $\lambda_{0,\ell}^{d_\ell} \exp[-\lambda_{0,\ell} \sum_i e_{i\ell} e^{\eta_i}]$. Multiplying by its Gamma prior gives

$$\lambda_{0,\ell} \mid \text{event data}, \eta \sim \text{Gamma}\left(a_\ell + d_\ell, b_\ell + \sum_i e_{i\ell} e^{\eta_i}\right). \tag{A3.35}$$

This is a conditional baseline update within the joint posterior; the hazard coefficient and scale updates follow their own likelihood blocks. A partial-likelihood coefficient fit alone does not provide all components of a Bayesian cash-flow forecast.

For competing terminal causes, λ_{ik} is cause-specific and

$$L_i = \left\{\prod_k \lambda_{ik}(t_i)^{1\{\delta_i = 1, J_i = k\}}\right\} \exp\left[-\int_0^{t_i} \sum_k \lambda_{ik}(u)\, du\right]. \tag{A3.36}$$

The contract-specific default incidence is

$$F_{iD}(\tau) = \int_0^\tau \exp\left[-\int_0^u \sum_k \lambda_{ik}(v)\,dv\right] \lambda_{iD}(u)\,du. \tag{A3.37}$$

A twelve-month logistic target for default before competing termination should be calibrated to this same event definition if compared with a survival-derived probability. Cure and restructuring often need intermediate states because a loan can later default. Article 1's appendix derives the corresponding Aalen–Johansen transition estimator and distinguishes cause-specific from subdistribution hazards.

A3.10 The guarded pilot and the Wilson interval

The score-inversion interval follows [11].

In the article's example,1,000 applicants pass the study eligibility screen,100 receive a random offer,80 accept and draw, and four of those 80 default by the declared horizon. The three directly observed proportions have distinct meanings:

$$\widehat{\Pr}(\text{offered}) = 100/1{,}000 = 10\%, \quad \widehat{\Pr}(\text{draw} \mid \text{offered}) = 80/100 = 80\%, \quad \hat{p} = 4/80 = 5\%. \tag{A3.38}$$

There are 900 applicants not offered the contract,20 offered but not funded, and 920 without an observed funded-loan repayment path. This distinction keeps assignment and take-up visible. An ITT effect additionally needs the specified comparator and an applicant-level outcome measured under both assignments.

For the funded-loan descriptive rate, suppose $K \sim \text{Binomial}(n, p)$ with independent, exchangeable outcomes and complete follow-up. Under the score test, retain candidate values of p satisfying

$$\frac{(\hat{p} - p)^2}{p(1-p)/n} \le z^2, \qquad z = z_{1-\alpha/2}. \tag{A3.39}$$

Multiplying through and collecting powers of p gives

$$(n + z^2)p^2 - (2n\hat{p} + z^2)p + n\hat{p}^{\,2} \le 0. \tag{A3.40}$$

The interval between the two quadratic roots can be written

$$\frac{\hat{p} + z^2/(2n)}{1 + z^2/n} \;\pm\; \frac{z}{1 + z^2/n}\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}. \tag{A3.41}$$

Substituting $n = 80$, $\hat{p} = 0.05$ and $z = 1.96$ produces centre 0.07061888 and half-width 0.05100564, hence

$$[0.01961324,\ 0.12162453], \tag{A3.42}$$

or approximately 1.96% to 12.16%. This reproduces the article's rounded 2.0%–12.2%. It is a frequentist score confidence interval for the defined funded sample, not a posterior credible interval or a whole-reject-population guarantee. Correlated group loans, unequal risk, incomplete follow-up or adaptive recruitment require an uncertainty method matched to those features.

For comparison, an explicitly chosen beta prior gives a Bayesian binomial calculation. If $p \sim \text{Beta}(a_0, b_0)$, multiplication by the likelihood yields

$$p \mid K = k, n \sim \text{Beta}(a_0 + k, b_0 + n - k). \tag{A3.43}$$

An equal-tail credible interval is given by the 0.025 and 0.975 quantiles of that posterior. Under the illustrative Jeffreys prior $a_0 = b_0 = 1/2$, the posterior is $\text{Beta}(4.5, 76.5)$ and its mean is $4.5/81 = 5.5556\%$. The prior, posterior mean and interval should be labelled; they describe funded pilot risk under the binomial model. A hierarchy extends this calculation to sector, region and vintage variation and carries the added parameter uncertainty into its draws.

Source connection: Wilson, "Probable Inference, the Law of Succession, and Statistical Inference," derives the score-interval approach; DOI: https://doi.org/10.1080/01621459.1927.10502953. Its confidence-interval interpretation is retained separately from the Bayesian illustration.

A3.11 Adaptive offers preserve an explicit observation process

The adaptive-allocation setting follows [10].

After a suitable fixed-design pilot, a registered mixture allocation can use

$$\pi_t(a \mid x) = (1 - \epsilon_t)\hat{\pi}_t(a \mid x) + \epsilon_t q(a \mid x). \tag{A3.44}$$

Because both component policies sum to one, so does the mixture. If $q(a \mid x) > 0$ and ϵ_t has a positive floor, the mixture preserves a calculable probability for that eligible action. With two eligible offers, uniform q and $\epsilon_t = 0.2$, a deterministic current favourite receives 0.9 and the alternative 0.1. Outcomes still arrive only for delivered actions and, for funded-loan repayment, for acceptors who originate. Logged assignment probabilities support offer-policy evaluation; they do not by themselves eliminate acceptance selection or disclose repayment under an unissued loan. The Article 1 appendix supplies the expectation proofs and delayed-outcome conditions.

Under group guarantees, let $Y_i(\mathbf{A}_g)$ depend on the vector of assignments within group g. The ordinary notation $Y_i(A_i)$ is then incomplete unless an interference restriction is justified. Cluster randomisation can compare group-level assignment policies. An exposure-mapping design can instead define quantities such as the proportion of other members assigned a support treatment and identify effects under explicit network assumptions. The randomisation unit, analysis unit and customer-rights process must therefore agree with the guarantee structure.

A3.12 Expected loss, cash shortfall and tail capital

The accounting cash-shortfall interpretation follows [19].

Let D indicate default over a specified horizon, E exposure at default and G loss given default as a fraction of exposure. The random credit loss is $L = DEG$, so

$$\begin{aligned} E[L] &= \text{Pr}(D = 1)E[EG \mid D = 1] \\ &= \text{PD}\left[E[E \mid D = 1]E[G \mid D = 1] + \text{Cov}(E, G \mid D = 1)\right]. \end{aligned} \tag{A3.45}$$

The familiar product PD $\times$ EAD $\times$ LGD follows when the exposure and severity definitions already absorb their joint expectation, or when fixed values or appropriate conditional-independence approximations justify dropping the covariance. Wrong-way dependence between drawdown and recovery makes that approximation material. For fixed EAD KES 100,000, LGD 40% and PD 5%, the simplified expected loss is KES

2,000. Applying the sensitivity risks 2.5641% and 9.5238% changes that calculation to KES 1,025.64 and KES 3,809.52; these are assumption-based scenarios rather than booked provisions.

For an accounting cash-shortfall calculation, let C_t denote contractual amounts due, $R_t^{(s)}$ expected receipts including eligible recoveries in scenario s, w_s scenario probabilities summing to one, and d_t^{EIR} the applicable effective-interest discount factors. A schematic probability-weighted amount is

$$\mathsf{ECL} = \sum_s w_s \sum_t d_t^{\mathsf{EIR}} \{C_t - R_t^{(s)}\}. \tag{A3.46}$$

The contractual cash flows, scenarios, discount rate, horizon, modifications and recoveries follow the applicable IFRS 9 measurement requirements. Twelve-month ECL concerns lifetime cash shortfalls associated with default events possible during the next twelve months; it is not simply twelve months of missing instalments. A default-time expression illustrating that distinction is

$$\mathsf{ECL}_{12} = \int_0^{12\text{ months}} E[\text{discounted lifetime shortfall} \mid T_D = u]\, dF_D(u), \tag{A3.47}$$

with the credit model and accounting definition aligned to the relevant default and shortfall conventions. Receipts occurring after twelve months can enter the conditional lifetime shortfall for a default within the next twelve months.

An economic-capital convention may instead use the loss tail:

$$\mathsf{VaR}_\alpha(L) = \inf\{\ell : \Pr(L \le \ell) \ge \alpha\}, \qquad \mathsf{EC}_\alpha = \mathsf{VaR}_\alpha(L) - E[L]. \tag{A3.48}$$

For a single stylised loan with loss KES 40,000 with probability 5% and zero otherwise, expected loss is KES 2,000 and $\mathsf{VaR}_{0.99} = 40{,}000$, giving KES 38,000 under this convention. A portfolio calculation requires dependence, concentration, exposure and recovery modelling; applicable regulatory capital is determined by its own rules. The example isolates the mathematical difference between the mean and the tail, while the research keeps simulated reject outcomes separate from actual ledger events.

Source connection: IFRS Foundation, "IFRS 9 Financial Instruments," supplies the accounting objective and applicable horizon. The expectation and quantile equations above expose the assumptions behind the article's simplified product and tail-risk distinctions.

A3.13 Evidence delivered to the human reviewer

For a candidate offer or limit, retain the posterior draws $p_a^{(b)}(x)$, the population-support label, the declared sensitivity grid and any identified action-effect draws. A 95% equal-tail risk interval is $[q_{0.025}(p_a \mid D), q_{0.975}(p_a \mid D)]$. A policy-cap probability can be calculated as

$$\widehat{\Pr}(p_a > p_{\mathsf{cap}} \mid D) = \frac{1}{B} \sum_{b=1}^{B} 1\{p_a^{(b)} > p_{\mathsf{cap}}\}. \tag{A3.49}$$

The Monte Carlo uncertainty in this estimated probability depends on the effective sample size of the correlated posterior indicator draws, not simply the raw number of stored draws. A credible interval for a causal effect additionally depends on its identified treatment design. Predictive intervals include future outcome variation as well as parameter uncertainty. The interface should retain these labels, the contract and horizon, data cutoff, hierarchy and prior version, sensitivity assumptions, permitted actions and approval history.

This supplies uncertainty that is useful for an actual review decision and a computational record that can be reproduced.

A3.14 Classification diagnostics in a selectively observed portfolio

Let r_i be a default-risk score, with larger values indicating greater predicted default risk. The area under the receiver operating characteristic curve (AUC) for the financed, observed population has the pairwise interpretation

$$\begin{aligned}\mathsf{AUC}_{S=1} = {} & \mathsf{Pr}(r_i > r_j \mid D_i = 1, D_j = 0, S_i = S_j = 1) \\ & + \tfrac{1}{2}\,\mathsf{Pr}(r_i = r_j \mid D_i = 1, D_j = 0, S_i = S_j = 1).\end{aligned} \tag{A3.50}$$

Its sample estimate averages the ranking indicator over every observed default–non-default pair. It measures ranking in the selected sample, rather than calibration or performance among unfinanced applicants. A target-population weighted ranking estimate needs the same selection-exchangeability and support conditions as the earlier transport calculation, and remains sensitive to unsupported regions.

For a declared threshold c, write $\widehat{D} = 1\{r \geq c\}$ and let G denote an evaluation group. The observed-population rates are

$$\mathsf{TPR}_{g,S=1} = \mathsf{Pr}(\widehat{D} = 1 \mid D = 1, G = g, S = 1), \qquad \mathsf{FPR}_{g,S=1} = \mathsf{Pr}(\widehat{D} = 1 \mid D = 0, G = g, S = 1). \tag{A3.51}$$

Equal opportunity compares the first rate; equalized odds compares both. A positive default flag is different from a positive approval decision, so the recorded label convention belongs in every report. If target rates are estimated with selection weights, derive each as a ratio of weighted joint and conditioning-event probabilities. For example,

$$\widehat{\mathsf{TPR}}_g = \frac{\sum_i S_i \hat{e}(X_i)^{-1} 1\{G_i = g, D_i = 1, \widehat{D}_i = 1\}}{\sum_i S_i \hat{e}(X_i)^{-1} 1\{G_i = g, D_i = 1\}}. \tag{A3.52}$$

This ratio requires a positive supported denominator and suitable uncertainty analysis. Zero observed defaults in a small group do not establish a perfect default-detection rate. Report group sizes, the observed or transported population, label maturity, posterior or sampling uncertainty and the sensitivity assumptions. The selection derivations explain why an apparently strong aggregate AUC or fairness statistic can coexist with uncertainty about rejected applicants.

Source connection: Hardt, Price and Srebro, "Equality of Opportunity in Supervised Learning," defines the group-error criteria. The pairwise AUC identity and the weighted ratios follow from the definitions of the relevant conditional probabilities.

Can Reinforcement Learning Help Run a Bank's Balance Sheet?

From KESONIA and FTP to constrained ALM, IRRBB and ALCO decisions

Underwrite for Collection · Article 4 of 5 · LinkedIn research article · 8 October 2026

Abstract

A bank's Treasury decisions recur: funding matures, deposits move, rates change, loans reprice, collateral is called and liquidity buffers evolve. This makes asset-liability management (ALM) a natural setting for stochastic control. Reinforcement learning (RL) can help compare sequences of funding, investment and hedging decisions under uncertain scenarios, but only when the decision problem is defined around the bank's actual balance sheet, policy limits and governance. This article develops that proposition from the Kenyan reference-rate and credit-pricing setting. It distinguishes KESONIA, a market benchmark, from internal funds transfer pricing (FTP); connects earnings and economic-value measures; illustrates behavioural deposit assumptions, liquidity and hedge collateral; and sets out a constrained research pathway from deterministic planning to simulation and carefully evaluated policies. The worked numbers and diagrams are illustrative, not a bank forecast or a production recommendation.

Research questions

1. How should a Kenyan bank connect KESONIA-based customer pricing, internal FTP, curve construction and behavioural deposit assumptions so that product profitability reconciles with Treasury funding costs and ALCO risk measures over the facility lifecycle?
2. Under what data, simulation, constraint and governance conditions can stochastic control or reinforcement learning improve Treasury decisions relative to transparent baselines while preserving NII, EVE, liquidity, capital, collateral and operational limits?
3. Which data-protection and model-governance controls are required when Treasury analytics use identifiable deposit, borrower, guarantor or employee data, and when does a workflow move from institutional optimization to a potentially significant automated decision about an individual?

How this article extends the sequence

The first three articles focus on credit decisions at borrower and applicant level. Article 1 asks how randomized comparisons and contextual bandits can test credit offers; Article 2 follows repeated servicing actions and evolving borrower states; Article 3 examines what repayment data can and cannot tell us when historical approval policies create selective labels. This fourth article changes the unit of analysis from a borrower, account or application to an institution's balance sheet. It carries forward the sequence's core discipline—define the action, information available at the time, outcome, constraints and evidence—but applies it to Treasury and ALCO decisions. A loan portfolio may appear in both settings, yet a collection intervention and a bank funding or hedge decision have different owners, time horizons, risk limits and evidence requirements. The methods connect; the decisions remain distinct.

The borrower models pass their uncertainty into this balance-sheet view through versioned portfolio cashflow scenarios. The logistic and survival specifications developed earlier retain sector, region and origination-vintage effects, together with their model-specific coefficient and scale distributions. A portfolio simulation should sample each shared effect once for a scenario and apply that draw to the borrowers belonging to that group; otherwise independently resampling a common sector shock for every loan can understate aggregate uncertainty. Contract schedules, exposure paths, prepayment, collections and recoveries then determine dated cash, with separately specified dependence where needed. Treasury receives scenario cash flows, uncertainty summaries, modelling dates and data-support indicators. It can combine them with funding, deposit and market scenarios while preserving the distinction between an estimated borrower outcome

and an authorised Treasury action. The hierarchy therefore remains traceable from individual evidence to the liquidity and earnings distribution presented to ALCO.

Why balance-sheet control is a sequential problem

ALM coordinates assets, liabilities, capital and off-balance-sheet positions over time. A Treasury team does not make one isolated decision and then observe a fixed result. It reconsiders funding as wholesale instruments approach maturity, deposit balances change, lending growth departs from plan, market rates move, collateral values shift and liquidity forecasts are updated. Each action changes later choices: issuing term funding now can reduce refinancing risk while increasing current cost; entering a swap can reshape interest-rate sensitivity while creating margin and counterparty exposures; changing product pricing can alter both deposit retention and future asset growth. These links make the balance sheet a dynamic system, with actions today changing the state and feasible choices tomorrow.

A useful control model begins with a state vector, x_t, that summarizes information available at decision time t. It could include contractual repricing and maturity ladders, currency and basis exposures, deposit balances and estimated behavioural parameters, loan pipeline, funding spreads, liquid assets, collateral capacity, current NII and EVE sensitivities, capital and policy headroom. Market conditions and customer behaviour evolve under uncertainty, so the state transition is written $x_{t+1} \sim P(\cdot \mid x_t, a_t, w_{t+1})$, where a_t is the approved decision and w_{t+1} represents new shocks or observations. The state must be sufficient for the intended comparison and measurable on the bank's operating timetable. A convenient feature vector is not automatically a reliable economic state.

Actions are not unrestricted switches. They may include setting approved funding volumes and tenors, investment maturity, hedge type and notional, eligible transfer prices, balance-sheet growth limits or contingent-funding steps. The allowed set $A(x_t)$ must be specified by governance, mandate, operational readiness, legal entity, market access, counterparty limits, accounting designation and risk appetite. Some controls are continuous, such as a hedge notional within an authorized range; others are discrete, such as selecting among approved funding tenors. A model should identify actions that are unavailable, unsuitable or unaffordable before optimization. It should also represent implementation lag: a funding issue or hedge approved today may settle later, at a price that differs from the assumed one.

The bank's objective is not one scalar unless leadership has made the trade-offs explicit. Shareholder return, NII stability, economic value, liquidity resilience, capital use, customer pricing and strategy can conflict across horizons. A control objective can combine measurable financial outcomes, but it must show the reporting horizon and weights, and should present separate constraints and dashboards so that a high expected return cannot conceal a weak liquidity position. ALCO supplies the institution's risk appetite and approved strategy; Treasury executes within delegated authority; independent risk validates exposure and limits; Finance owns accounting and performance bridges; and the board remains accountable for its oversight. A learned policy can inform this process, but it cannot set institutional risk appetite by optimizing a reward function.

Economic value, earnings and the risks that connect them

Two essential IRRBB views answer different questions. Economic value of equity (EVE) measures how interest-rate changes alter the present value of expected banking-book cash flows. Net interest income (NII) measures how rates, repricing, volumes and behavioural responses alter earnings over a chosen horizon. EVE gives a longer-run value sensitivity; NII gives a near- to medium-term earnings view under a stated balance-sheet scenario. Basel's framework treats IRRBB as risk to both capital and earnings and explains that measurement depends on rate shocks, customer options, non-maturity-deposit assumptions, business replacement assumptions and accounting treatment [1], [2]. Reporting both views matters because a strategy that improves one can worsen the other.

EVE can be represented as:

$$\begin{aligned}\Delta\text{EVE} = \text{EVE}\big(&\text{shocked curves, behavioural cash flows}\big)\\ &- \text{EVE}\big(\text{base curves, behavioural cash flows}\big)\end{aligned}. \tag{4.1}$$

In a simplified fixed-cash-flow approximation:

$$\Delta\text{EVE} \approx -D_{\text{eff}} \times \text{EVE}_0 \times \Delta r, \tag{4.2}$$

where D_{eff} is an effective duration and Δr is a parallel rate move. This approximation is useful for intuition only: real banking books have multiple curves, basis risks, floors, caps, embedded options, non-parallel shocks and cash flows that respond to rates. NII under scenario *s* over horizon *H* can be written:

$$\begin{aligned}\text{NII}_s(H) = &\sum_{t=1}^{H}\big[\text{interest income}_{t,s} - \text{interest expense}_{t,s}\big],\\ &+ \text{relevant fee and hedge effects}\end{aligned} \tag{4.3}$$

with a consistent classification policy. A baseline comparison must state whether the balance sheet runs off, remains constant or grows dynamically, since those assumptions can materially change results [2].

Credit spread risk in the banking book also deserves a distinct lens. A bond or loan may lose economic value because its credit spread widens even when the risk-free curve is unchanged; changes in rates can also affect borrower affordability, deposits and credit quality through separate channels. ALM should avoid treating all valuation movements as pure interest-rate risk or assuming that a hedge of a risk-free curve eliminates spread risk. The control model should report key-rate and basis sensitivities, credit-spread measures where relevant, non-parallel shocks, option effects and concentration by currency and portfolio. It should also identify which exposures are behavioural rather than contractual, and show the sensitivity of reported outcomes to those assumptions. These outputs turn a headline "rates up" result into a risk explanation ALCO can act on.

Curves, KESONIA and the internal price of funding

The discount and projection curves are inputs to measured cash flows, pricing and valuation. A curve-building process begins with dated, quality-controlled instrument quotations, conventions, day counts, calendars and market cut-offs. It selects instruments appropriate to each curve's purpose, bootstraps discount factors or zero rates, specifies interpolation and extrapolation, and reprices the input instruments as a validation test. Projection curves for floating cash flows and discount curves for present value may differ; basis between indices and tenor points also matters. A model should store the source, timestamp, convention, interpolation choice and fallback for every curve node. In a market with sparse term observations, the gap between observed quotations and modelled long tenors must be visible as uncertainty, not hidden by a smooth chart.

KESONIA is Kenya's transaction-based average rate for unsecured overnight interbank KES transactions, administered and published daily by the Central Bank of Kenya [3]. CBK's revised Risk-Based Credit Pricing Model anchors variable-rate loan pricing to KESONIA compounded in arrears, with the premium K covering lending-related costs, shareholder return, borrower risk and other relevant costs; the published total cost of credit also includes fees and charges [4], [5]. This customer-facing reference and pricing framework should be mapped accurately. KESONIA is a market benchmark, while FTP is the bank's internal method for assigning funding and liquidity economics to business units and products. FTP may use a term structure of funding costs and behavioural adjustments; it is not a synonym for KESONIA or a substitute for the customer-pricing components required by the CBK model.

A coherent architecture can link the layers without blending their purposes. Treasury's external funding and market curves feed an approved internal FTP methodology; the business line receives a matched-maturity and liquidity-aware transfer charge or credit; credit risk, operating costs, capital and margin are assessed under the applicable product-pricing framework; and the customer contract identifies its benchmark, premium, fees and reset terms. Management overlays or strategic incentives should be separately visible from measured funding economics. This allows ALCO to understand how choices affect balance-sheet shape while Finance can reconcile internal profitability to general-ledger results. If a market rate, curve construction or customer benchmark changes, the bank should preserve the effective date and affected contracts, methods and approvals so it can reproduce both pricing and performance.

A worked FTP illustration

Consider a purely hypothetical one-year KES variable-rate business loan of KES 10,000,000 with a customer rate set at 13.0% annualized. Suppose the approved internal funding curve assigns a 9.4% one-year marginal funding rate, a 0.6% liquidity term premium and 0.3% operating allocation, while the applicable expected credit cost is 1.2% and allocated capital return is 0.8%. A simplified annual contribution is:

$$13.0\% - 9.4\% - 0.6\% - 0.3\% - 1.2\% - 0.8\% = 0.7\%, \tag{4.4}$$

before taxes, unmodelled costs and any additional adjustments. The arithmetic is only an illustrative bridge; an actual product requires institution-approved bases, measurement conventions, amortization, repricing and accounting treatment. It does not assert that a Kenyan bank uses these exact components or values.

Suppose KESONIA compounded over the customer's reset period rises by 100 basis points while the contractual premium stays fixed. The customer rate may reset according to the product's documented terms, but the internal profitability analysis must also update its funding, behavioural, liquidity and risk assumptions. If the bank's internal one-year funding curve rises by 120 basis points and the loan's effective repricing lags by one month, the mismatch can compress interim margin. The size and timing depend on cash-flow dates, balance amortization, repricing caps or floors, benchmark conventions, funding strategy and the chosen internal FTP method. The purpose of the example is to identify the questions to calculate and reconcile, not to imply that an overnight index and one-year bank funding costs move one-for-one.

The business case should be carried from origination through its life. At each review date, it can compare realized interest income and fees with the customer benchmark and margin; the approved FTP debit or credit; actual and expected funding costs; liquidity and capital allocations; credit losses and servicing costs; and the business unit's net contribution. New production, early repayment, delinquency, deposit migration, hedges and non-linear options can all change later economics. The result is a versioned profitability path rather than a single rate at loan approval. A robust implementation records both the original pricing decision and subsequent repricing or assumption changes, allowing management to distinguish model variance from customer behaviour, market movements, strategic subsidy and operational error.

Non-maturity deposits and behavioural replication

A non-maturity deposit has no contractual term that fixes its future balance or repricing date, yet portions of the book may remain stable and respond slowly to market rates. The bank can estimate deposit beta (the change in the customer deposit rate relative to a change in the market reference rate), balance decay, migration and withdrawal behaviour from longitudinal data. These quantities vary by product, segment, channel, customer type, rate regime and liquidity conditions. A single portfolio average can mask material differences between operational accounts, rate-sensitive savings, transactional balances and large corporate deposits. Estimation therefore needs stable definitions, suitable observations across different environments, confidence intervals and scenario testing; behavioural lives should not be selected solely because they make a preferred hedge or EVE result appear attractive.

A replication portfolio approximates the interest-rate profile of deposits through a set of modelled repricing tenors or instruments. It can help allocate interest-rate risk internally, compare product economics and assess hedging needs, but it is a model of observed behaviour rather than a legal maturity of the deposit. If customers move quickly when market rates rise, a long assumed behavioural duration may overstate stable funding value and understate NII sensitivity. The model should test alternative betas, decay rates, floors, migration channels and stress responses, and should separate stable core balance from volatile balances where the data support that distinction. Any replication model should be monitored against realized deposit repricing and runoff, with limits and a documented process for recalibration.

For a simple illustration, assume a KES 100 million deposit portfolio earns customers an average 3.0% while a market rate is 10.0%. The observed average beta over the past year might be 0.35, implying an estimated 35 basis-point customer-rate change for each 100 basis-point market move over that sample period. That statistic is descriptive, not a promise that next year's beta will be 0.35. If the bank's stress scenario assumes beta 0.60 and 20% balance migration, the expected expense and cash-flow profile differs sharply from a static-balance assumption. ALM should show these alternatives as scenarios, link each to sample period and estimation method, and ask what actions remain feasible if behaviour changes. This is especially important when a model-informed hedge relies on deposit stability that has not been observed in comparable conditions.

Liquidity and funding: the constraint that changes the optimum

Liquidity strategy cannot be reduced to a target ratio. The Basel Liquidity Coverage Ratio (LCR) compares eligible high-quality liquid assets (HQLA), after applicable haircuts and caps, with net cash outflows in a specified 30-day stress; the NSFR relates available stable funding to required stable funding over a longer structural horizon [6], [7]. The ratios are important but their calculation depends on legal entity, currency, encumbrance, counterparty, contractual maturity and run-off or funding factors. A bank also needs cash-flow ladders, intraday measures, concentration limits, stress scenarios, survival horizons and a contingency funding plan. The LCR stress scenario explicitly includes deposit outflows, wholesale funding loss, collateral requirements and committed facility draws, showing why a hedge or funding strategy must be evaluated for liquidity effects as well as price risk [6].

For a stylized ratio illustration, assume a bank reports KES 120 billion of eligible HQLA and KES 100 billion of 30-day net cash outflows; the simplified LCR is 120%. If a proposed hedge strategy or funding change increases stressed net outflows by KES 8 billion and also requires KES 5 billion of liquid assets to be encumbered or unavailable, the illustrative ratio becomes:

$$\frac{120 - 5}{100 + 8} = 106.5\%. \tag{4.5}$$

This is not a regulatory calculation: actual HQLA eligibility, haircuts, caps, inflow limits, cash-flow factors, currency treatment and national rules must be applied. The example makes the interaction visible: an action that looks profitable in a rate model can use liquidity capacity that has value precisely when stress occurs.

Structural funding decisions also affect lending growth and internal business incentives. A long-term asset funded by short-term liabilities may generate a higher current margin than a matched-tenor structure, but it can increase refinancing exposure and sensitivity to funding market disruption. A term funding premium should reflect the cost and stability of the funding profile, while reserves and liquidity buffers should be assessed under stress and legal-entity transferability. Funds raised by one subsidiary may not be immediately transferable to another; collateral may be pre-positioned, operationally available only after a delay, or subject to restrictions. These details affect the feasible state and action set. The Treasury controller should therefore represent intragroup limits and operational capacity explicitly, not assume that consolidated liquidity is instantly fungible.

Hedging, collateral and margin calls

A hedge can lower one risk measure while adding another. An interest-rate swap may reduce a fixed-versus-floating repricing mismatch and improve EVE or NII under a selected shock, but it creates counterparty exposure, basis and hedge-accounting considerations, and collateral calls under market moves. A hedge that is attractive in an average scenario can become hard to fund during a joint rates, spread and liquidity shock. The bank should calculate potential future exposure, initial and variation margin needs where applicable, eligible collateral, haircuts, concentration, settlement timing and cash buffers alongside market-risk benefit. LCR treatment and liquidity stress should reflect collateral outflows using the applicable rules, and internal stress tests should include larger or faster calls than the base case [6], [8].

A simple hedge comparison should place measures side by side. For example, a hypothetical swap might reduce a parallel-up-shock ΔEVE loss from KES 2.0 billion to KES 1.2 billion, while raising modeled peak collateral needs from KES 1 billion to KES 4 billion under a joint stress. The decision depends on the bank's appetite, available liquidity, hedge basis, counterparty terms and time path—not simply the 0.8 billion EVE benefit. These illustrative figures are not estimates for any institution. The comparison should include market value and earnings effects, collateral and liquidity, counterparty limits, accounting consequences, operational capacity, unwind cost and the impact of simultaneous customer deposit withdrawals. ALCO can then review the trade-off rather than seeing a single optimized hedge ratio.

Constrained stochastic control rather than reward-only optimization

The Treasury recursion in equation (A4.26) extends the control form introduced in equation (2.1) with Treasury-specific actions, cash flows and feasible sets. Its belief update is equation (A4.27); Appendix A2.3 derives the borrower-level belief construction.

A finite-horizon control objective can be written as maximizing the expected discounted value of outcomes,

$$J(\pi) = \mathbb{E}_{\pi}\left[\sum_{t=0}^{T} \gamma^t r(x_t, a_t, x_{t+1})\right], \tag{4.6}$$

subject to:

$$\begin{aligned} a_t \in A(x_t), \qquad & \mathsf{LCR}_t \geq \ell_t, \\ \mathsf{NSFR}_t \geq n_t, \qquad & \mathsf{capital}_t \geq c_t, \\ |\Delta\mathsf{EVE}_t| \leq e_t, \qquad & \mathsf{liquidity\ stress}_t \leq q_t \end{aligned} \tag{4.7}$$

and applicable counterparty, currency, concentration and operational limits. Here π is a policy mapping information to action, γ is a discount factor, and thresholds ℓ_t, n_t, c_t, e_t and q_t are approved risk limits or internal guardrails. This is a schematic formulation, not a full regulatory model. Hard legal, contractual and prudential constraints belong in a verified feasible-action gate, not merely as penalties that an optimizer may choose to violate in exchange for a larger expected reward.

A reward could represent a transparent combination of risk-adjusted earnings, funding cost, customer pricing effects and implementation cost over specified horizons. However, any scalar objective can hide value judgments: how much near-term NII is worth a unit of EVE sensitivity, how liquidity resilience is valued, whether customer repricing is delayed, and how low-probability stress is weighted. The model card and ALCO paper should therefore show the component outcomes, scenario distributions, tail measures, constraint headroom and sensitivity to weights. If an objective changes, the policy must be re-evaluated and approved as a new version. A "risk penalty" is not a substitute for keeping the bank inside formal limits, nor does a positive expected reward demonstrate that a strategy is resilient under adverse scenarios.

Constrained Markov decision processes, model-predictive control and stochastic programming are practical alternatives or companions to deep RL. A stochastic program can optimize decisions over a scenario tree with explicit constraints; model-predictive control can re-optimize at each review as new forecasts arrive; and a constrained MDP can represent repeated decisions and long-run resource limits. Deep stochastic control may approximate policies in richer state spaces, but it should be compared against interpretable deterministic and stochastic baselines. The 2023 deep-treasury study demonstrates a research approach using simulated yield-curve scenarios and dynamic funding/investment decisions, while also emphasizing simplifications and the need for thorough validation; its method is a candidate for study, not evidence that an AI policy is ready for a bank [9]. ALCO oversight remains necessary because the model cannot define the bank's mandate.

A POMDP can be useful where important economic states are only partially observed: the true persistence of deposit balances, future customer repricing response, or latent stress regime may be uncertain. The model then maintains a belief distribution b_t over possible states, updated from observed rates, flows, customer behaviour and forecast errors. But a belief state is only as useful as the observation process, timing and validation behind it. If account feeds arrive late, deposit definitions change, or a rate series has a methodology break, the uncertainty is not solved by a recurrent neural network. An institution may first improve data lineage and model the uncertainty through scenario ranges, Bayesian updating or robust optimization. A POMDP should be used when partial observability changes decisions materially and the model's belief update has demonstrable value.

A practical sequence for AI and Treasury research

The first stage is to define the balance-sheet decision and the control owner. Choose a narrow question—such as allocating new fixed-rate assets across approved tenors while preserving a liquidity buffer—rather than "optimize Treasury." State the decision frequency, time horizon, legal entity, currency, instruments, forecast sources, constraints and the baseline used today. Reconstruct what decision-makers knew when prior actions were taken and retain policy, override and execution records. This first step tests whether the research has decision-grade data and a feasible action set. If Treasury cannot replay a historical decision from timestamped curves, cash-flow positions, limits and scenario assumptions, a sophisticated agent would produce an impressive answer to an unreproducible question.

The second stage builds deterministic and stochastic baselines. Reprice contractual cash flows, validate curve construction, reconcile FTP allocations and compare run-off, constant-balance-sheet and dynamic NII projections. Stress rate level, slope, basis, spread, deposit beta and runoff; test available funding and collateral capacity; and verify that the model reproduces known portfolio sensitivities. The third stage is simulation validation: compare simulated market and balance-sheet paths with historical distributions, stress episodes, expert scenarios and known financial identities. Assess tail coverage, autocorrelation, cross-curve dependence, deposit behaviour and structural breaks. A simulator that generates convenient rate paths but misses funding crises, liquidity feedback or deposit migration can train a policy to exploit its defects.

The fourth stage evaluates candidate policies offline against approved historical decisions and a common set of scenarios. It should report the gap between logged actions and proposed actions, extrapolation beyond historical support, uncertainty in value estimates, tail outcomes and constraint breaches under model error. Importance weighting or doubly robust estimators can be unstable where the policy being evaluated chooses actions rarely seen in historical data; a high point estimate is not enough without overlap and uncertainty analysis. Policy comparisons should include simple rule-based strategies, current practice, robust or risk-sensitive optimization and a transparent no-change option. The fifth stage tests a shadow policy: it generates recommendations without executing them, while Treasury and independent risk review the recommendation, available information, constraints, sensitivity and implementation feasibility.

Only after that should a narrowly scoped pilot be considered, and only where experimentation is institutionally permissible and operationally safe. Market-wide Treasury actions can affect liquidity, capital, customer

rates and other firms; randomly exploring actions on the live balance sheet is not equivalent to a borrower offer pilot. A safer progression can use historical replay, stress simulation, shadow operation and controlled parameter adjustments approved through normal mandate and risk governance. If a pilot is used, it requires an explicit treatment definition, safe range, risk limits, monitoring, stop conditions and accountable human sign-off. No model should take live actions outside approved authorities or explore merely to improve its own future reward.

Diagram 1: from market data to approved ALCO action

The following flow separates data construction, measurement and recommendation from approval and execution. It highlights validation checkpoints: source quality and curve conventions are checked before valuation; balance-sheet assumptions are explicit before the scenario engine runs; and policy outputs pass through hard limits and delegated authority before a trade or pricing decision is made. The diagram also makes the feedback path visible. Actual funding cost, deposit behaviour, collateral calls and realised NII return to model monitoring, while accounting, risk and Treasury reconcile results using the same effective-dated decision record. This is the intended boundary for decision support: an AI component can compare strategies and explain trade-offs, but ALCO and delegated officers retain approval under the bank's governance.

Diagram 2: constrained sequential control loop and governance boundary

The second diagram shows the repeated control loop. At each decision date the bank updates its observable state and uncertainty, computes scenario outcomes, applies an approved feasibility gate, and sends a recommendation for review. If observations are incomplete, the model should widen uncertainty, request data remediation or decline to recommend rather than quietly imputing certainty. If a limit is near breach or the policy is outside its validated domain, the loop falls back to an approved contingency plan. This makes the action pathway auditable and resilient to model failure. It is especially important for Treasury because an action can change later liquidity, collateral and pricing states, and a sequence that looks optimal in the simulator may be infeasible after settlement or customer response.

Data protection and automated decision safeguards

Treasury optimization may use mostly institution-level or market data, but its workflow can also process personal data: identifiable depositors, sole proprietors, borrowers, guarantors, employees, counterparties or customer-level transaction and repricing histories. Whether a dataset is personal data depends on identifiability and processing context, not on whether an AI model is described as "Treasury." The Kenyan Data Protection Act requires lawful, fair, transparent and purpose-limited processing, data minimization, accuracy, retention control and appropriate security; it also establishes data-subject rights and controller/processor duties [10]. Teams should map fields and purposes, controller and processor roles, access, retention, linkage and cross-border handling before training or operational use. An aggregated portfolio statistic can reduce exposure but does not automatically remove re-identification risk from detailed histories.

Section 31 of the Act requires a data-protection impact assessment where processing is likely to result in high risk to rights and freedoms; section 35 concerns decisions based solely on automated processing, including profiling, with legal effects or significant effects on a data subject, subject to the section's conditions and safeguards [10]. Regulation 22 of the Data Protection (General) Regulations, 2021 sets safeguards for automated individual decision-making, including notice, meaningful information about the logic, fairness, error prevention, appropriate statistical procedures, bias mitigation and human intervention; regulations 49–53 provide the DPIA process [11]. These provisions should be mapped to the actual decision and personal data involved. An aggregate ALCO optimization recommendation will not automatically be an automated decision about an individual; a customer-specific pricing, deposit treatment or credit action may raise different questions if it has significant effect and is made solely by automated processing.

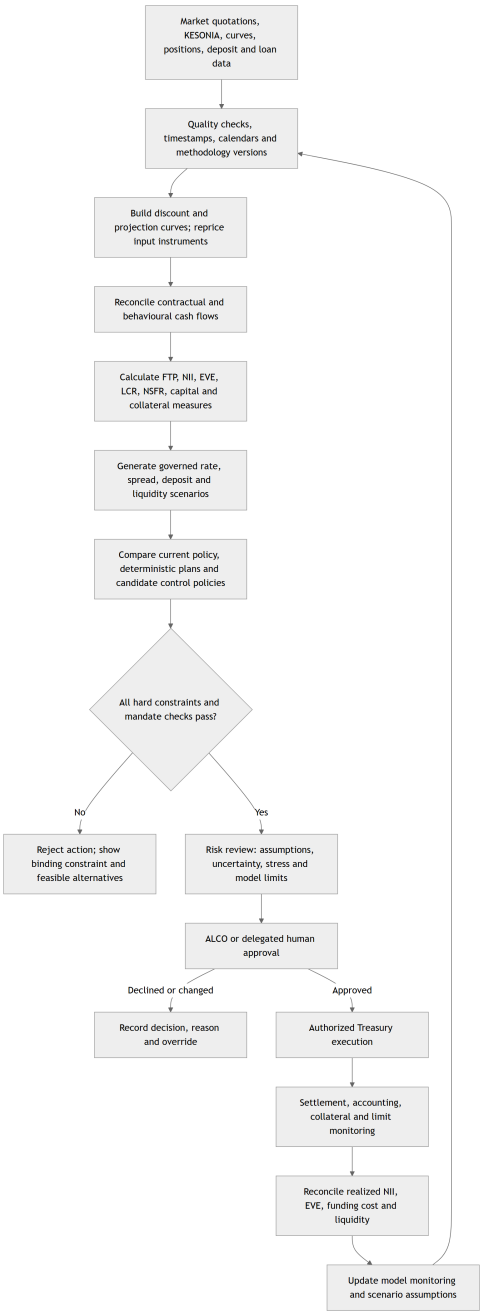


Figure 5: Treasury data to ALCO approval

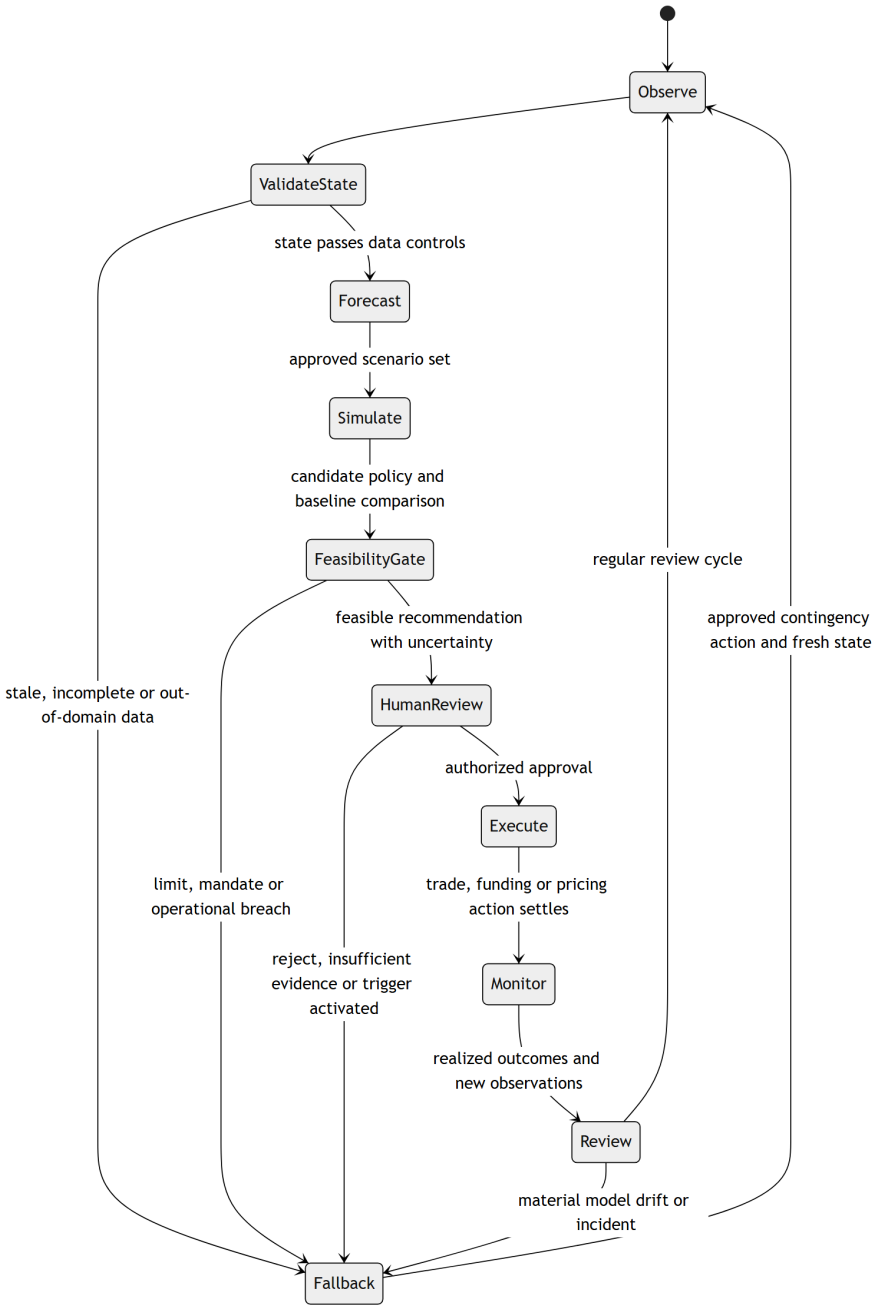


Figure 6: Treasury constrained control loop

The data-protection control path should specify lawful purpose and basis, notices, data minimization, correction routes, security, retention and meaningful human review where required. Where a model uses transaction histories to estimate deposit stability, it should distinguish the balance-sheet variable needed from unnecessary details of individual transactions. When customer-level information is linked across products, the institution must assess whether that linkage is necessary and compatible with the disclosed purpose, and whether a less granular or de-identified input would suffice. DPIA screening should occur early enough to shape the system rather than being a late approval document. For a consequential individual decision, a human reviewer needs enough context and authority to reconsider the recommendation; merely clicking "approve" without examining the basis does not supply meaningful review.

Model risk, ALCO and implementation evidence

Model governance should define ownership for data, curve construction, behavioural assumptions, scenario generation, objective weights, constraint implementation, code, validation and production monitoring. Independent validation should test conceptual soundness, implementation, outcomes and intended use, including challenger methods and stress beyond the training distribution. It should examine model drift, deposit beta stability, forecast error, curve repricing, basis effects, constraint logic, scenario coverage, policy support and override patterns. Explainability should show which state variables and constraints drove a recommendation, alongside the effect of alternative assumptions and feasible actions. A model that cannot explain whether a recommendation came from a maturity mismatch, liquidity pressure, deposit response, curve basis or reward weight is difficult to govern, even if its predictions are statistically accurate.

ALCO papers should contain the decision question, the current position, scenario set, baseline and candidate strategy; separate NII, EVE, LCR, NSFR, capital and collateral outcomes; state key behavioural and curve assumptions; and identify what the model does not represent. They should report central and tail outcomes, uncertainty, limit headroom, customer or business-line consequences, stress response, execution costs, accounting impacts and fallback plan. The minutes should record the selected policy, any rejected alternatives, rationale, owner, approval authority, limits, review date and conditions for stopping or revisiting the action. A reproducible decision record should link model and data versions to the approved action and later realized outcomes. This makes a Treasury recommendation reviewable months later, after rates or management priorities have changed.

A path toward operational use begins with decision-grade data and transparent baselines, not with a deep-learning agent. The institution can first create an integrated curve and cash-flow view, reconcile FTP, build behavioural deposit monitoring and produce consistent stress reporting. Next it can add scenario comparison and optimization with explicit constraints, then run shadow recommendations against ALCO decisions and analyze where they differ. If the differences improve risk-adjusted outcomes under robust tests and remain within governance, the bank may consider a restricted decision-support pilot. This sequence offers practical value even if the eventual model is not reinforcement learning. It also improves the foundation for an institutional analytics workbench: data contracts, definitions, scenario lineage and approved outputs should be reusable without embedding an unapproved execution policy into a reporting layer.

Conclusion

Reinforcement learning and stochastic control offer a useful research lens for Treasury because funding, pricing, hedging and liquidity decisions recur under uncertainty and affect future choices. Their practical contribution depends on a sound ALM representation: dated curves and cash flows, observable decision states, explicit behavioural assumptions, matched-maturity FTP, separate earnings and value measures, liquidity and capital constraints, and a complete collateral view. KESONIA provides an external overnight market reference in the Kenyan pricing framework; it does not replace an internal funding curve or determine the bank's FTP method. The worked cases show why Treasury should measure how a decision changes customer rates, business-line economics, EVE, NII, funding, liquidity and collateral together.

A credible AI program begins with deterministic and stochastic baselines, checks simulator realism, evaluates policies offline, runs shadow recommendations, and preserves human approval under the institution's ALCO and risk governance. Hard limits remain hard constraints, while uncertainty and model boundaries are visible in the decision paper. Data protection applies wherever personal data enter the process, with the Act and General Regulations mapped to the actual controller, purpose and decision; section 35 is relevant to qualifying individual decisions based solely on automated processing, not every portfolio-level optimization. This fourth article extends the first three from offers, collection and selective outcomes to the bank's own balance sheet. It extends the analysis to an institutional scale, preparing for Article 5's bank-controlled agentic workflow, which connects the analytical tools to evidence access, human review, authorised execution and monitored outcomes.

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Publication note. All numerical cases in this article are hypothetical and designed to illustrate calculation structure. This article presents a research and decision-support architecture, not a fitted Kenyan bank model, legal opinion, approved Treasury policy or claim of production performance. Basel standards are international reference material; institution-specific obligations and domestic implementation must be checked against the applicable Kenyan prudential instruments and the bank's approved policies.

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Mathematical appendix to Article 4

A4.1 Scope, units and notation

This appendix derives Article 4's economic-value, earnings, pricing, deposit, liquidity, collateral and control formulations. Numerical inputs are hypothetical. Additional assumptions introduced to complete a calculation are identified explicitly. A symbol is local to the section in which it is defined; amounts are in Kenyan shillings (KES), rates are annual decimal rates, and a basis point is 10^{-4} of a rate.

Symbol	Meaning and unit
$t = 0, \dots, T$	Decision or cash-flow index; the timetable states its actual dates
τ_t	Accrual fraction of a year for interval t
s	A declared economic and behavioural scenario
$P_s(0,t)$	Discount factor to date t, dimensionless
$CF^A_{t,s}, CF^L_{t,s}$	Asset receipts and liability payments, KES
$B^A_{t,s}, B^L_{t,s}$	Relevant average accrual balances, KES
$r^A_{t,s}, r^L_{t,s}$	Asset and liability annual interest rates
x_t, a_t, w_{t+1}	Decision state, action and next exogenous shock
π, γ	Policy and one-period discount factor
$\mathbb{E}$, Pr	Expectation and probability under the specified model

Currencies are converted using a declared convention before aggregation. Accrual profit, cash availability and present value are separate outputs. External funding expense and internal FTP are reconciled rather than charged twice to the consolidated bank.

A4.2 From dated cash flows to EVE and duration

The value and earnings risk perspectives follow [1], [2].

Let $N_{t,s} = CF^A_{t,s} - CF^L_{t,s}$ be the net banking-book cash flow, augmented by relevant off-balance-sheet flows. With one reporting currency,

$$\mathsf{EVE}_s = \sum_{t=1}^{T} P_s(0,t) N_{t,s}, \qquad \Delta\,\mathsf{EVE}_s = \mathsf{EVE}_s - \mathsf{EVE}_0\,. \tag{A4.1}$$

The scenario affects both discount factors and behavioural flows. For a single parallel continuously compounded yield y, fixed N_t, and elapsed time u_t in years,

$$V(y) = \sum_t N_t e^{-y u_t}, \qquad V'(y) = -\sum_t u_t N_t e^{-y u_t}. \tag{A4.2}$$

Define $D(y) = -V'(y)/V(y)$ when $V(y) \neq 0$. Taylor expansion gives

$$V(y+\Delta y) - V(y) = V'(y)\Delta y + \frac{1}{2} V''(y)(\Delta y)^2 + O((\Delta y)^3), \tag{A4.3}$$

$$\Delta V \approx -D(y)V(y)\Delta y. \tag{A4.4}$$

This is the article's first-order duration approximation. A finite-difference effective duration, allowing the valuation engine to re-estimate behavioural cash flows, is

$$D_{\text{eff}} = \frac{V(y-h) - V(y+h)}{2hV(y)}, \tag{A4.5}$$

where $h > 0$ is a small yield shock. Net EVE can be close to zero or negative, making duration normalised by EVE unstable. In that case report dollar sensitivity $V'(y)$ and direct scenario changes. With a vector of curve nodes y, the corresponding first-order expression is $\Delta V \approx \nabla V^{\mathsf{T}} \Delta y$. Separate spread and basis shocks add their own derivatives. This prevents a single parallel duration from standing in for all rate, basis, option and credit-spread effects.

A4.3 Earnings over an explicit horizon

For scenario s, a simple accrual representation is

$$\text{NII}_s(H) = \sum_{t=1}^{H} \tau_t \left(B^A_{t,s} r^A_{t,s} - B^L_{t,s} r^L_{t,s} \right) + \sum_{t=1}^{H} J_{t,s}. \tag{A4.6}$$

Here $J_{t,s}$ contains only fees or hedge effects classified within the institution's stated NII definition, including any required effective-interest treatment. Other fee income and valuation effects remain separately classified. Each interest term has units KES times annual rate times years, hence KES. Balances and rates can change with amortisation, repricing, deposit migration and new business. A run-off scenario sets new production to zero; constant-book and dynamic scenarios define their replacement rules. These assumptions explain why two NII projections can differ even when they share the same market-rate shock.

For an action comparison, compute $\Delta\,\text{NII}_s = \text{NII}_s^{\text{candidate}} - \text{NII}_s^{\text{baseline}}$ using the same scenario and accounting perimeter. Principal receipts affect liquidity and balances; they are not interest income.

A4.4 Curves, overnight compounding and FTP

The customer benchmark and pricing conventions follow [3], [4], [5].

A fixed-cash-flow instrument with quoted value V^{quote} supplies an equation

$$V^{\text{quote}} = \sum_{j=1}^{m} CF_j P(0, t_j). \tag{A4.7}$$

If earlier discount factors have already been obtained and $CF_m \neq 0$, bootstrapping solves

$$P(0, t_m) = \frac{V^{\text{quote}} - \sum_{j=1}^{m-1} CF_j P(0, t_j)}{CF_m}. \tag{A4.8}$$

Floating instruments require projected coupons as well as discount factors. Under a single-curve, simple-forward convention,

$$F(0;t_{j-1},t_j)=\frac{P(0,t_{j-1})/P(0,t_j)-1}{\tau_j}. \tag{A4.9}$$

A multi-curve framework uses a separately calibrated projection curve in the forward formula and the appropriate discount curve for valuation. Repricing checks compare each reconstructed value with its quotation under specified tolerances.

For overnight observations k_j, each applying to d_j calendar days, and an applicable day-count denominator B, an illustrative compounded rate over the accrual period is

$$\tau=\sum_j \frac{d_j}{B}, \qquad k_{\text{comp}}=\frac{\prod_j(1+k_j d_j/B)-1}{\tau}. \tag{A4.10}$$

The contract determines observation dates, holidays, lags, any floors and how the premium is applied. Where it specifies an annual additive premium K, a schematic rate is $r^{\text{customer}}=k_{\text{comp}}+K$. The benchmark is an input to contractual pricing; an internal FTP curve separately allocates the bank's funding and liquidity economics. CBK's revised pricing framework and the product's actual terms determine the customer-facing application.

For an amortising product, period FTP charges are

$$\text{FTP}_t=B_t^{\text{FTP}}f_t^{\text{FTP}}\tau_t, \tag{A4.11}$$

where B_t^{FTP} is the approved chargeable balance and f_t^{FTP} the corresponding transfer rate. An equivalent flat transfer rate, when that reporting convention is useful, satisfies

$$\bar{f}=\frac{\sum_t d_t B_t^{\text{FTP}}f_t^{\text{FTP}}\tau_t}{\sum_t d_t B_t^{\text{FTP}}\tau_t}, \tag{A4.12}$$

with declared reporting discount factors d_t. The weighting reflects cash-flow timing and balances; a one-year rate cannot automatically price every later exposure in a revolving facility.

A4.5 Reproduce the KES 10 million contribution bridge

Assume the balance remains KES 10,000,000 for one year and every quoted component uses that same balance and annual basis. Treat the 9.4% funding component and 0.6% liquidity term component as non-overlapping allocations. Then

$$\begin{aligned}\text{Contribution}&=10{,}000{,}000(0.130-0.094-0.006-0.003-0.012-0.008)\\&=10{,}000{,}000(0.007)=70{,}000\ \text{KES}.\end{aligned} \tag{A4.13}$$

The components are KES 1,300,000 customer interest, KES 940,000 funding, KES 60,000 liquidity allocation, KES 30,000 operating allocation, KES 120,000 expected credit cost and KES 80,000 capital-return allocation. Their residual is KES 70,000. The capital-return allocation is a management profitability charge; the expected credit cost is a pricing estimate. Mapping them to accounting expense requires the institution's approved methodology.

To quantify the article's repricing-lag example, add the following assumptions: twelve equal months, unchanged balance, funding cost immediately up 120 basis points for all twelve months, customer rate up 100

basis points only after the first month, and all other components unchanged. Incremental annual contribution is

$$\Delta \text{ Contribution} = 10{,}000{,}000 \left(0.010\frac{11}{12} - 0.012\right) = -28{,}333.33 \text{ KES}. \tag{A4.14}$$

The new contribution is KES 41,666.67. Of the reduction, KES 20,000 reflects the additional 20-basis-point annual mismatch and KES 8,333.33 the one-month customer repricing lag. An amortising or capped contract requires period-by-period recalculation.

For internal reconciliation, business-line contribution subtracts the FTP charge and Treasury receives the corresponding internal credit. Summing the two units eliminates the internal transfer; external funding expense remains in consolidated profit. Expected and realised credit costs are compared on consistent definitions rather than deducted together as if they were separate losses.

A4.6 Deposit beta, migration and replication

Deposit beta is a rate sensitivity. In a simple sample regression,

$$\Delta r_t^{\text{dep}} = \alpha + \beta \Delta r_t^{\text{market}} + \varepsilon_t, \tag{A4.15}$$

where α is an intercept and ε_t the residual. With an intercept, least-squares estimation gives

$$\widehat{\beta} = \frac{\sum_t (\Delta r_t^{\text{market}} - \overline{\Delta r^{\text{market}}})(\Delta r_t^{\text{dep}} - \overline{\Delta r^{\text{dep}}})}{\sum_t (\Delta r_t^{\text{market}} - \overline{\Delta r^{\text{market}}})^2}. \tag{A4.16}$$

This descriptive estimate needs uncertainty and regime checks. For the article's KES 100 million deposits, initial rate 3%, and a one-percentage-point market increase, beta 0.35 implies a rate of 3.35% and annual interest expense KES 3.35 million at unchanged balances. Under beta 0.60, the rate becomes 3.60%. If 20% migrates out immediately, remaining expense is $80{,}000{,}000(0.036) = 2{,}880{,}000$ KES, alongside a KES 20 million liquidity outflow. If the bank replaces that balance at annual rate f, total expense is $2{,}880{,}000 + 20{,}000{,}000 f$ KES. For an explicitly added illustration $f = 11\%$, it is KES 5.08 million.

A replication portfolio can approximate repricing cash flows through weights w_j on approved tenors j. One transparent fitting objective is

$$\min_{w} \sum_{t \in \mathcal{T}} \left(z_t - \sum_j w_j g_{jt} \right)^2, \qquad w_j \geq 0, \quad \sum_j w_j = 1, \tag{A4.17}$$

where z_t is the selected deposit-rate or cash-flow target and g_{jt} the corresponding instrument response. The measurement target, balance perimeter and out-of-sample period must be fixed before fitting. This approximation describes behaviour; withdrawal remains available according to the deposit contract.

A4.7 Liquidity ratios and the article's 106.5% result

The illustrative ratio definitions follow [6], [7].

At a simplified level,

$$\text{LCR} = \frac{H}{O - \min(I, 0.75O)}, \tag{A4.18}$$

where H is eligible HQLA after the applicable adjustments, O weighted stressed outflows and I eligible weighted inflows. The displayed 75% cap is the Basel standard convention, subject to applicable provisions; an institution applies its actual domestic requirements and classifications. Inputs must avoid double-counting pledged assets or the same cash movement.

Article 4 supplies net outflows directly: $H = 120$ and $N = 100$, in KES billions. After the stated changes, $H' = 115$ and $N' = 108$; therefore

$$\text{LCR}_0 = \frac{120}{100} = 1.20, \qquad \text{LCR}_1 = \frac{115}{108} = 1.0648148 \approx 106.5\%. \tag{A4.19}$$

This assumes the KES 8 billion increase is already the net-outflow increment after all applicable classification and cap calculations. The KES 5 billion reduction is an independently identified loss of available eligible HQLA. A real hedge package must reconcile their overlap rather than applying both numbers mechanically.

The structural funding ratio is

$$\text{NSFR} = \frac{\text{ASF}}{\text{RSF}}, \qquad \text{ASF} = \sum_i a_i L_i, \qquad \text{RSF} = \sum_j b_j A_j + \text{RSF}^{\text{off}}. \tag{A4.20}$$

L_i are eligible funding and capital amounts, a_i their available-stable-funding factors, A_j asset amounts, b_j required-stable-funding factors, and RSF^{off} the relevant off-balance-sheet requirement. The article provides no NSFR balance inventory or factor assignments, so it supports this calculation structure rather than a numerical NSFR. Numerical implementation requires institution-specific classification, residual maturities and the applicable rules.

A4.8 Hedging, collateral and cash availability

The article's hedge reduces the selected EVE loss from KES 2.0 billion to KES 1.2 billion: an improvement of KES 0.8 billion. Peak modelled collateral needs rise from KES 1 billion to KES 4 billion: an increase of KES 3 billion. The two differences have distinct horizons and meanings; their subtraction would not produce a well-defined net benefit.

For a simplified collateral agreement, let V_t be the derivative mark-to-market value to the bank and H_t a posting threshold. Required collateral posted by the bank can be represented schematically as

$$C_t^{\text{req}} = \max(-V_t - H_t, 0) + IM_t, \qquad \Delta C_t = C_t^{\text{req}} - C_{t-1}^{\text{held}}, \tag{A4.21}$$

where IM_t is applicable initial margin and held collateral is adjusted for returns, haircuts and settlement. Minimum transfer amounts, netting, independent amounts and eligible collateral terms must be included when applicable. A signed positive ΔC_t is an additional posting in this simplified representation.

A cash ledger over period t is

$$K_{t+1} = K_t + \text{receipts}_t + \text{funding}_t - \text{payments}_t - \text{drawings}_t - \text{collateral posted}_t + \text{collateral returned}_t\,. \tag{A4.22}$$

Every item uses actual availability dates. Peak collateral demand is $\max_t C_t^{\text{req}}$, while liquidity stress may measure $\max_t \max(K_t^{\text{minimum}} - K_t, 0)$. These path measures expose a funding problem that can coexist with improved EVE sensitivity.

A4.9 Finite-horizon stochastic control and Bellman recursion

The recursion follows the dynamic-programming framework in [12], connected to bank balance-sheet control in [9].

Represent the next state by $x_{t+1} = F_t(x_t, a_t, w_{t+1})$. Integrating over the conditional shock distribution defines the transition kernel

$$K_t(dx' \mid x, a) = \Pr\{F_t(x, a, w_{t+1}) \in dx' \mid x, a\}. \tag{A4.23}$$

The article's conditional notation involving both a state kernel and w_{t+1} can be interpreted through this transition mechanism. A finite-horizon objective with terminal value g is

$$J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{T} \gamma^t r_t(x_t, a_t, x_{t+1}) + \gamma^{T+1} g(x_{T+1}) \right]. \tag{A4.24}$$

Setting $g = 0$ gives the displayed article objective. The policy π selects an allowed action using information available at t, and $0 < \gamma \leq 1$ discounts consistently with the period length. Rewards and terminal values require one declared unit and economic accounting convention.

Split the first reward from later rewards and condition on x_{t+1}:

$$V_{T+1}(x) = g(x), \tag{A4.25}$$

$$V_t(x) = \max_{a \in \mathcal{A}_t^{\text{feas}}(x)} \int \left[r_t(x, a, x') + \gamma V_{t+1}(x') \right] K_t(dx' \mid x, a). \tag{A4.26}$$

This is the Bellman recursion under the Markov and measurability assumptions. The feasible set includes mandate, counterparty, currency, concentration and operational restrictions. It also includes the modelled LCR, NSFR, capital, EVE and cash-stress checks. For example, LCR $\geq \ell_t$, NSFR $\geq n_t$, $C_t^{\text{capital}} \geq c_t$, $|\Delta \text{ EVE}| \leq e_t$ and CashStress $\leq q_t$, with each threshold's units declared. A ratio-based capital constraint uses a ratio threshold; an absolute-capital constraint uses KES.

If feasibility is tested across a scenario set Ω_t, write $g_k(x, a, \omega) \leq 0$ for every required constraint k and $\omega \in \Omega_t$. A statistical chance constraint, $\Pr(g_k > 0) \leq \epsilon_k$, is a separate modelled risk control and does not relax a binding legal limit. Finite scenario coverage establishes compliance within those scenarios; it does not prove compliance for every future state. An empty feasible set invokes the defined contingency and escalation process.

A4.10 Partial observation, predictive cash and policy evaluation

The belief-state update follows [12].

For hidden economic state z_t, let $b_t(z) = \Pr(z_t = z \mid h_t)$ be the belief after observed history h_t. With hidden-state transition $K(z' \mid z, a)$ and observation likelihood $O(y \mid z', a)$,

$$b_{t+1}(z') = \frac{O(y_{t+1} \mid z', a_t) \sum_z K(z' \mid z, a_t) b_t(z)}{\sum_u O(y_{t+1} \mid u, a_t) \sum_z K(u \mid z, a_t) b_t(z)}. \tag{A4.27}$$

Replace sums by integrals for continuous states. The denominator is the predictive probability of the observation. A belief-state Bellman equation uses expected reward $\bar{r}(b, a)$ and the update map $\mathcal{B}$:

$$V_t(b) = \max_{a \in \mathcal{A}_t^{\text{feas}}(b)} \left\{ \bar{r}_t(b, a) + \gamma \sum_y \Pr(y \mid b, a) V_{t+1}(\mathcal{B}(b, a, y)) \right\}. \tag{A4.28}$$

Behavioural parameters, deposit persistence and latent stress states require an identified observation process and validation. Cash forecasts supplied by borrower hierarchies preserve shared sector, region and vintage effects. For total portfolio cash $C = \sum_i C_i$,

$$\operatorname{Var}(C) = \sum_i \operatorname{Var}(C_i) + 2 \sum_{i<j} \operatorname{Cov}(C_i, C_j). \tag{A4.29}$$

Scenario generation must retain those dependencies and any declared common shocks. Summing independently generated customer distributions would generally discard them.

For M common simulated scenarios, a paired policy comparison is

$$\widehat{\Delta J} = \frac{1}{M} \sum_{m=1}^{M} \left(G_m^{\text{candidate}} - G_m^{\text{baseline}} \right), \tag{A4.30}$$

where G_m is the discounted return. Its Monte Carlo standard error is the sample standard deviation of paired differences divided by $\sqrt{M}$ when scenarios are independent. This quantifies simulation sampling error, separately from parameter, simulator and policy-identification uncertainty. All comparisons also report constraint breaches, tail outcomes and implementation feasibility.

Sources supporting this appendix

The financial context follows Article 4's cited primary materials: the IRRBB standard [1], LCR and NSFR definitions [6], [7], revised customer-pricing framework [4], and bank Treasury control study [9]. Algebraic examples above are fully specified illustrations. Applicable Kenyan prudential requirements and institution-approved measurement conventions determine operational use.

Further methodological reading

Additional sources retained from the article bibliography: [13].

Bringing the Methods into the Bank

Agentic AI, Bayesian evidence and accountable credit and Treasury decisions

Underwrite for Collection · Article 5 of 5 · LinkedIn research article · 8 October 2026

Abstract

Credit experiments, sequential collection policies, selection analysis and Treasury optimisation become useful when a bank can connect their results to an authorised operating decision. This fifth article develops that connection through a bank-controlled agentic workflow: approved evidence enters registered analytical tools, the outputs form a reviewable proposal, an accountable officer authorises an action, and execution returns as a reconciled customer and cash outcome. Bayesian credible intervals, exceedance probabilities and predictive distributions help reviewers understand uncertainty alongside point estimates. A worked hypothetical case distinguishes uncertainty about default probability, incremental intervention benefit and future available cash. The design extends the research environment described in Part 4 of the *Underwrite for Collection* whitepaper into an operating architecture covering branch access, restricted agent tools, model releases, customer correction, meaningful human review and Kenyan data-protection obligations. Its practical objective is to make each decision traceable from the information available at the time to the action delivered and the outcome subsequently observed.

Research questions

1. How can a bank-controlled agentic workflow connect offer experiments, sequential collections, selection diagnostics and Treasury analysis while preserving purpose-bound data access, separate decision authority and reconciled execution?
2. How should credible intervals, exceedance probabilities and predictive cash distributions be presented to human reviewers so that uncertainty changes limit, servicing and funding decisions constructively?
3. What evidence and operating tests should govern progression from a research sandbox to shadow use and bounded deployment under Kenya's Data Protection Act and General Regulations?

The fifth step connects the first four

Article 1 developed experiments for credit offers, Article 2 followed the borrower through sequential servicing decisions, Article 3 examined the missing outcomes created by historical approval, and Article 4 considered the bank's balance-sheet decisions. Their shared operational question is now straightforward: who can act on these analyses, using which evidence, with what controls, and how will the institution know what happened? Part 4 of the whitepaper proposed bringing approved research methods into the bank, building purpose-specific evidence views and promoting complete analytical artefacts through controlled stages [1]. This article develops that foundation into a practical agentic workflow. Readers can follow the architecture directly: the bank owns the source records, decision mandates, customer relationship and execution systems; analytical services prepare evidence and alternatives; accountable people and approved policies determine the permissible action.

A branch example makes the connection tangible. An MSME asks for a higher revolving limit while one instalment is overdue and a large customer invoice remains unpaid. The relationship manager needs a current view of operating cash, existing obligations, the invoice's collectability and the borrower's next payment dates. Collections needs to know whether supportive contact or a revised arrangement is appropriate. Credit needs to understand risk and affordability under the requested limit. Treasury needs aggregate information about likely drawings and receipts across the portfolio. These questions share some events but produce different decisions. An agentic workflow can gather the relevant evidence, request approved calculations

and assemble coherent proposals for each owner, maintaining the links between them as payments, new information and human decisions arrive.

Deploy the workflow inside an accountable environment

In this design, branch employees use authenticated applications connected to bank-controlled services over private connections. Analytical workloads may run on premises or in an approved private-cloud environment, with distinct research, validation and production accounts. The institution records where processing, model inference, backups, logs and support access occur. Approved research code works against purpose-specific views, while stable pseudonymous identifiers link facilities, payments, interventions and outcomes. The identity mapping is held separately and accessed only by permitted operational services. This makes longitudinal analysis possible while keeping direct identifiers out of routine research tasks. The deployment specification also covers identity management, encryption, key ownership, service availability, disaster recovery and controlled export of results. Each arrangement is evaluated against the institution's actual processing and applicable requirements.

The agent harness is the coordinator: software that receives a task, checks permissions, invokes approved tools and returns a structured result. A generative model can help organise case evidence, draft explanations and highlight missing information. Statistical services calculate risk, uncertainty, treatment estimates and financial scenarios; a policy service defines permissible actions. Model hosting and harness hosting are separate choices, determined by available deployment arrangements, licences, security assessments and processing contracts. A bank may deploy a small set of specialist agents for evidence preparation, analysis and explanation, with explicit handoffs and narrow permissions. Shared memory is limited to the registered task, and access to customer cases expires. This keeps the proposed architecture understandable enough for technology, model risk, compliance and business owners to examine together.

A useful division of labour is an evidence agent that retrieves permitted records, an analysis agent that invokes the registered statistical services, and an explanation agent that prepares the reviewer's narrative. A deterministic verifier checks quoted values against tool receipts and flags unsupported statements. Each component receives only the information needed for its task, and the bank records the handoffs. The same arrangement can be implemented as one orchestrator with several tools when that is easier to operate. The design choice should follow measured quality, latency, cost and failure containment. Human model validation and business approval remain assigned to accountable institutional roles, so additional software agents improve task organisation while the bank's governance structure stays explicit.

External information arrives through a controlled gateway. Benchmark rates, market quotations and regulatory publications are collected from approved sources, checked for freshness and saved with source, timestamp and document version. Web text and uploaded documents are treated as evidence; permission settings remain under the bank's configuration controls. The gateway screens outbound requests so that an agent retrieving a public curve does not also transmit customer records. Where an approved external processor is used, the contract, transfer basis, permitted purposes and handling of prompts, outputs and support logs enter the deployment assessment. These controls also address generative-AI risks such as fabricated references and malicious instructions embedded in retrieved content. NIST's Generative AI Profile provides a useful voluntary framework for evaluating provenance, testing and the interaction between people and AI systems [2].

Give each analytical tool a defined operating contract

The four tool families retain their distinct purposes. The offer service records eligibility, experimental assignment, offer delivery, take-up and the relevant outcome window. The collection service compares permitted actions using the current evidence, action history, customer circumstances and validated transition or reward models. The selection-analysis service examines which populations and actions have repayment evidence,

then reports support, assumptions and sensitivity. The Treasury service works with reconciled portfolio cash-flow measures, instrument records and approved curves to assess funding, liquidity and risk. Each tool returns a typed result containing its inputs, units, horizon, model version, uncertainty and completion status. The harness can combine these results into a case packet while preserving the boundaries between an observed transaction, a model estimate, a proposed action and an authorised instruction.

That contract begins with time. A payment can occur on one date, settle on another and become visible to an analytical service later. An invoice can be disputed after it originally entered the evidence set. A corrected ownership link can change which transfers count as operating cash. The evidence service therefore preserves economic time, institutional availability time and correction history, allowing reviewers to reconstruct what was known when an earlier decision was made. Data-quality flags travel with features and proposals. For operating businesses, the view separates customer revenue, financing, owner contributions and internal transfers. For ring-fenced receivables, it includes eligibility, dilution, account control, servicing and collection availability. These distinctions help the bank maintain the cash-provenance discipline established in the earlier articles throughout deployment.

Show the reviewer three kinds of uncertainty

Equations (5.1), (5.2), (5.3) have separate derivations and worked calculations in Appendices A5.1–A5.4. The expected-value limit boundary they complement is equation (2.35).

A posterior mean is a useful summary, but a decision packet should also show how uncertain the relevant quantity remains. For a proposed limit, the reviewer may need the distribution of twelve-month default probability under the specified model and action. A credible interval describes posterior uncertainty about that probability, conditional on the observed data, model and prior. An exceedance probability answers a policy-facing question: how much posterior probability lies above the bank's approved risk threshold? The screen should identify the population, horizon, action, data cutoff and supporting evidence beside these summaries. It should also show sensitivity to material assumptions and the consequences of an incomplete record. The uncertainty becomes operational when it changes the requested evidence, review level or permitted action.

Uncertainty about incremental intervention benefit answers a different question. Suppose a supportive collection action is compared with standard servicing. Its relevant quantity might be the additional net cash recovered over ninety days for a defined eligible population, with treatment costs included consistently. A credible interval for this difference and the posterior probability that it exceeds a worthwhile-benefit threshold help assess the decision. Their causal interpretation depends on the experiment or identification assumptions, overlap, treatment delivery and outcome definitions developed in Articles 1–3. The prior and hierarchical model influence estimation and pooling; the design supplies the basis for interpreting the comparison as an effect. A reviewer should therefore see both the numerical uncertainty and the evidence supporting transfer of the estimated effect to the current case.

A predictive distribution for future cash answers the operational question of what may actually become available. It combines uncertainty about parameters with variability in future payments, drawings, prepayments, defaults and recoveries under the stated model. A facility can have a relatively well-estimated default probability while its next month's available cash remains volatile. Treasury may therefore need a probability of a cash shortfall, while collections needs the chance of a payment arriving within a promised window. The distribution should state whether cash is gross or net, which expenses and reversals are included, and when money becomes usable. The same packet can present all three uncertainty measures below, with each retaining its own definition and decision purpose.

$$q_p = \mathsf{Pr}(p_{12}(a) > p_{\mathsf{max}} \mid D) \tag{5.1}$$

$$q_{\Delta} = \Pr\big(\Delta(a, a_0) > \delta \mid D, \text{identification assumptions}\big) \tag{5.2}$$

$$q_C = \Pr(C_{30}(a) < B_{30} \mid D, a) \tag{5.3}$$

Here D denotes the registered evidence; a is the candidate action and a_0 the comparison action. The quantity $p_{12}(a)$ is the modelled twelve-month default probability and p_max its policy threshold. The quantity Δ is the incremental outcome under the stated causal comparison, while δ is the bank's declared minimum worthwhile benefit. $C_{30}(a)$ is future available cash over thirty days and B_{30} the corresponding required amount. The three q quantities are probabilities calculated from their respective distributions. They need not share one approval threshold or one escalation rule. A bank can require additional review for material default-risk uncertainty, retain a supportive intervention with credible benefit, and maintain a liquidity buffer against uncertain receipt timing in the same case.

Make Bayesian computation feasible and reviewable

The hierarchical modelling developed earlier can be trained in a controlled research environment. Pólya–Gamma augmentation gives logistic models conditionally Gaussian coefficient updates when the relevant latent quantities and prior scales are fixed [3]. Where a Student-t prior is represented through latent scales, those variables also belong to the conditional inference structure; the exact representation depends on whether the prior is placed on coefficients or on hierarchical standard deviations. The model package records this choice, its hyperparameters and sensitivity checks. Computation is evaluated using convergence diagnostics, effective sample sizes, posterior predictive checks and comparison with an independent reference. The deployment plan then specifies how the approved posterior is represented for scoring and how much approximation error is acceptable for the decision quantities.

At operational time, a registered service can evaluate an approved set of posterior draws or a validated approximation to produce credible intervals, exceedance probabilities and predictive summaries. The generative assistant receives these typed outputs and their definitions for explanation. Latency, reproducibility and numerical agreement are tested against the reference calculation, including sparse groups, unseen categories, missing evidence and values near a decision threshold. A compact approximation may support routine cases, while complex or uncertain cases follow a slower review path. The model registry records the scoring representation, expiry, approved population and fallback. This separation gives the bank computational flexibility while allowing an independent reviewer to reproduce how the uncertainty displayed on the officer's screen was obtained.

The registered model schema makes the hierarchy explicit for every call. A borrower record identifies the sector, region and origination vintage used by the approved model. Logistic and hazard services retain separate coefficient sets and outcome definitions, each with the Gaussian conditional effects and calibrated half-Student-t scale priors developed in Article 2. Their non-centred representation and posterior artefacts are recorded with the fit. If a model adds institution, product or calendar effects, the schema names those levels and their priors as well. An unseen group follows the approved predictive treatment, with its additional uncertainty visible to the reviewer. The orchestration service preserves model labels when combining risk, causal-effect and cash-flow summaries, allowing staff to trace a decision quantity to the exact hierarchy, likelihood and evidence that produced it.

Build Kenyan data-protection duties into the workflow

The Data Protection Act, 2019 supplies the legal foundation for personal-data processing. Sections 25, 26, 29 and 30 address principles, rights, notice and lawful processing; section 31 addresses high-risk impact

assessment and section 35 significant solely automated decisions. Sections 39–43 cover retention, correction, protection by design, organisational measures and breach response. Sections 44–47 concern sensitive data, while sections 48–49 address cross-border transfers [4]. The practical design records purposes and access roles for each tool, gives customers usable information and maintains routes for correction and review. Business records receive this assessment where they contain personal data about identifiable owners, guarantors, employees or customers. The bank also documents institutional roles, registration requirements and the applicable credit, consumer-protection and supervision framework.

The General Regulations make that assessment operational. Regulation 22 addresses automated individual decisions, including meaningful information about logic, consequences, statistical procedures, errors, bias and human intervention. Regulation 21 concerns sharing; regulations 23–25 cover handling policies and processor arrangements; regulation 19 concerns retention. Regulations 27–36 develop protection by design, regulations 39–48 address transfers and regulations 49–53 address DPIA processes [5]. These provisions inform the review screen, processor contract, retention schedule and release assessment. The proposed workflow makes the reviewer's evidence and authority visible, and gives a customer a usable route to express their position. Application of section 35 is assessed against the actual decision, its effect and the degree of automation, with any applicable statutory conditions recorded for that use case [4].

The release calendar incorporates the statutory assessment and review routes. Section 31 requires a DPIA before high-risk processing, and subsection (5) specifies submission of its report sixty days before processing; subsection (3) requires prior consultation with the Commissioner where the assessment indicates high risk [4]. Regulation 51 addresses the Commissioner's consultation period after receipt [5]. These are separate planning milestones, with any resulting conditions incorporated into the release decision. For a significant solely automated individual decision, the bank assesses the section 35 exceptions, including necessity for a contract, statutory authorisation with safeguards, or consent. It records the applicable written notification and the customer's route to request reconsideration or a new decision with substantive human involvement. The assigned reviewer can examine the case afresh and change its outcome.

Processor and hosting controls extend through the full service chain. The bank documents written processing instructions, authorised subprocessors, access by support staff, retention of prompts and logs, incident reporting and exit arrangements. Regulation 25 governs third-party processor authorisation and contracts, alongside the processor's continuing responsibility [5]. Overseas processing is assessed under sections 48–49 and the transfer regulations; section 49 adds consent and confirmation of appropriate safeguards for sensitive personal data processed outside Kenya [4]. Section 50 and regulation 26 require attention to specified localisation purposes when applicable. This assessment covers inference, backup, telemetry and remote support as well as the main database. Hosting on a private server is therefore one element of the bank's documented arrangement, together with the people, services, locations and purposes involved.

Turn an analysis into a meaningful human decision

A proposal packet contains the requested action, feasible alternatives, evidence cutoff, unresolved facts, model and policy versions, financial consequences and uncertainty. It also shows the relevant affordability, contractual, hardship, dispute, contact and exposure checks. The officer can inspect supporting source records, request clarification, change assumptions, select an allowed alternative, defer the case or decline the proposal. Review requires competence, sufficient time and authority over the result. The interface should make uncertainty and missing evidence as visible as the preferred recommendation, with clear explanations of the consequences of waiting. The retained record captures the officer's reasoning and any changes to the proposed action. A review-quality programme can then assess whether staff used the evidence meaningfully and whether escalation rules are operating as intended.

The packet also needs a stable operational identity. Its case number connects the customer and facility references, data snapshot, proposed amount, currency, effective date, approval expiry and permitted execution conditions. It includes the reviewer's identity, mandate reference, recorded rationale and any required sec-

ond approval. Numerical outputs retain their model run identifiers and the evidence needed to reproduce them; customer communications retain the approved template and final delivered version. If the customer supplies a corrected invoice after approval but before execution, the service can determine whether the change invalidates the packet and requires reconsideration. This makes approval an instruction about a specific action under specific conditions, with a clear route for handling material changes.

Credit and Treasury follow their own mandates within this common structure. A credit officer or committee may approve a facility or contractual adjustment within delegated authority, with customer acceptance where required. A servicing team executes permitted contact and support actions. Treasury evaluates funding and hedge proposals against dealing mandates, available collateral, counterparty limits and ALCO-approved risk appetite; Finance and independent Risk retain their accounting and validation responsibilities. The bank's outsourcing assessment considers the applicable requirements of CBK/PG/16, including retained institutional responsibility for core management and lending decisions [6]. Each approval therefore names the responsible function and the scope of authority being exercised. A shared assistant can organise the evidence across these functions while the institution preserves explicit ownership of each financial decision.

Follow one hypothetical case from evidence to settlement

Consider an illustrative food processor with a KES 500,000 revolving limit, KES 300,000 drawn and a request for an additional KES 100,000 of headroom. A missed instalment and an unpaid customer invoice trigger review. The evidence service links the contractual schedule, verified sales receipts, supplier payments, owner transfers and invoice status as known that morning. It flags the expected invoice receipt as uncertain and distinguishes the requested limit from an actual drawing. For illustration, suppose the candidate-limit model returns a Beta(4,96) posterior for twelve-month default probability. Its mean is 4%, its central 95% credible interval is approximately 1.11%–8.60%, and the probability of exceeding 6% is approximately 14.84%. These constructed figures illustrate the review mechanics and are not estimates from a borrower study [7].

Suppose the bank's illustrative policy requires the probability of exceeding that 6% threshold to be at most 10% for the proposed approval route. The mean alone would give the reviewer an incomplete picture: the exceedance probability directs the case to additional evidence or a different authority. The officer asks for verified invoice acceptance, updates the short-term affordability view and considers retaining the existing limit while the evidence is resolved. Separately, a model supported by an eligible collection experiment may estimate the incremental ninety-day net cash benefit of a supportive action. To illustrate a second uncertainty display, take a Normal distribution with mean KES 8,000 and standard deviation KES 4,000. Its central 90% credible interval is approximately KES 1,421–14,579, and its probability of positive benefit is approximately 97.72%.

Those summaries enter separate decisions. The first concerns uncertainty about default risk at the proposed limit; the second concerns the expected incremental outcome for the defined intervention population under its identification assumptions. The caseworker examines whether the customer falls within that population and whether the action remains appropriate given the disputed or delayed invoice. The officer can approve a support-oriented contact while deferring the higher limit. Meanwhile, the cash forecast represents the timing and amount of the expected receipt, including a delay scenario. Treasury receives the approved portfolio-level aggregation with those timing assumptions. A promised receipt remains distinct from settled liquidity, and an undrawn approved limit contributes to the relevant drawing scenarios and commitment measures under the bank's methodology.

For a simple predictive-cash illustration, suppose the thirty-day model assigns probabilities of 30%, 50% and 20% to available cash of KES 20,000, KES 100,000 and KES 180,000. The predictive mean is KES 92,000. If obligations due within the same period require KES 70,000, the probability of a shortfall is 30%, even though average cash exceeds the requirement. These deliberately simplified scenarios demonstrate why the reviewer needs the distribution and timing alongside the mean. The bank would replace them with a validated model tied to dated receipts, expenses and obligations. A support plan can then address the

plausible low-cash outcome, and Treasury can aggregate scenarios using justified dependence assumptions across accounts rather than summing isolated adverse cases mechanically.

The approved support action creates an expiring instruction with case, customer, channel, content conditions and authority. A separate execution service validates the instruction, checks current case status and records submission, delivery and response using a unique action identifier. If the invoice later pays, the payment reference is reconciled to the account, facility and available-cash date. A failed delivery or duplicate retry remains visible in the operational record. The experiment record keeps the original assignment, actual receipt, human override and matured outcome, allowing analysts to distinguish what was offered from what was delivered. The credit review can then be reopened with the new evidence, using the appropriate model and policy versions rather than treating the earlier approval packet as permanently current.

Test the whole decision service before expanding use

The release programme begins with synthetic cases that exercise ordinary, boundary and failure conditions. Retrospective replay reconstructs historical evidence without including information learned later, and checks whether outputs reconcile to the source systems. Shadow use then produces prospective recommendations without customer action, measuring coverage, latency, disagreement and emerging performance. A bounded pilot follows under a documented mandate, with defined eligibility, action limits, monitoring, review capacity and stopping rules. Each stage tests the model, policy interface, explanation and execution process together. Results are assessed against the existing service and transparent alternatives. NIST's AI Risk Management Framework offers a voluntary structure for governing, mapping, measuring and managing this evaluation across the system's lifecycle [8].

Operational tests should include an outdated curve, a reversed payment, a false entity link, an unavailable reviewer, contradictory source documents and a failed settlement. Security testing should exercise instructions hidden in documents, unauthorised tool requests, excessive data retrieval and attempts to reuse expired execution permissions. Staff simulations assess whether reviewers recognise an interval crossing a decision boundary, understand the stated causal assumptions and use the correction route. Release evidence records the expected result, actual result, exception owner and remediation. CBK's cybersecurity guidance addresses institutional governance, independent assessment, outsourcing and incident reporting [9]. The proposed test programme applies those concerns to the additional interfaces created by agents, generative models and statistical services, with accountable owners for each material dependency.

Learn from outcomes, corrections and human decisions

After deployment, monitoring follows financial results, customer experience and operational performance together. The record retains recommendations, approved actions, overrides, assignment probabilities where relevant, actual delivery and mature outcomes. Corrections and complaints trigger a governed review of affected evidence and decisions, while historical records preserve the information used at the time. Within an approved adaptive experiment, allocation probabilities can change according to the registered update rule. Changes to eligibility, outcomes, model weights or decision authority follow the relevant validation and release process. A signed version, expiry, fallback and rollback route accompany each production release. This gives the institution a clear account of what learned from new evidence, what changed through approval and which customers or portfolios were affected.

The value of this fifth step is practical continuity. The bank can trace a question from the branch through evidence preparation, uncertainty-aware analysis, human judgement, authorised execution and reconciled cash. Credit experiments contribute evidence about offers; sequential models help compare servicing paths; selection analysis clarifies the population supported by available outcomes; Treasury evaluates the institution's resulting cash and risk positions. Agentic software coordinates these activities within defined permissions, and a generative model can make the evidence easier to use. The research becomes an operating capability

when staff can reproduce the numbers, explain the decision, correct the record and act within their authority. That is the intended progression from the four methodological articles to sustainable, accountable credit and balance-sheet management.

Bank-controlled workflow diagram

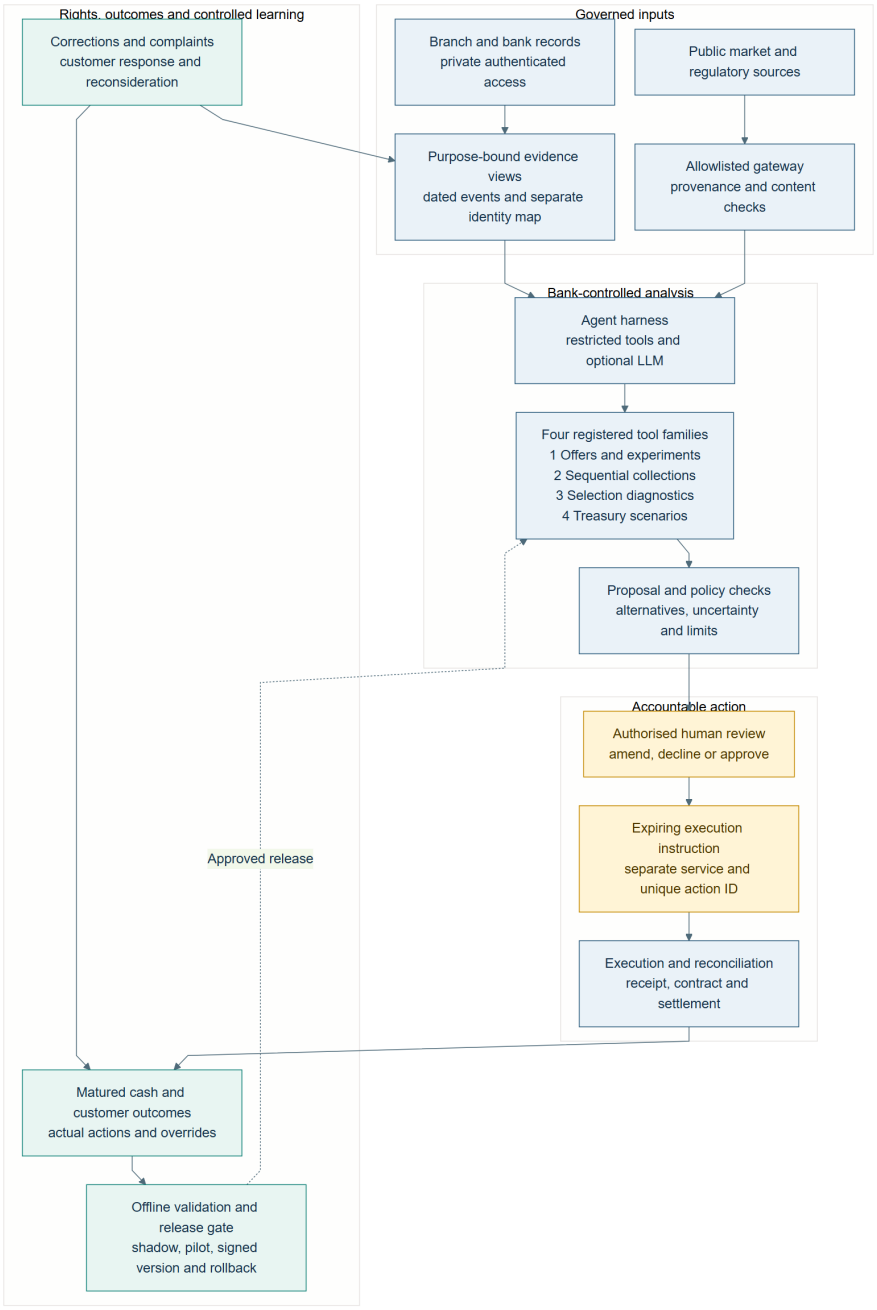


Figure 7: Bank-controlled agentic AI workflow

From governed evidence to accountable action and validated learning. The diagram follows one decision through the bank's evidence, authority and execution records.

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Mathematical appendix to Article 5

A5.1 Three probability statements with separate meanings

Let D denote the registered evidence; a a candidate action; and a_0 a comparison action. The article uses

$$q_p = \Pr\{p_{12}(a) > p_{\max} \mid D\}, \tag{A5.1}$$

$$q_\Delta = \Pr\{\Delta(a, a_0) > \delta \mid D, \mathcal{J}\}, \tag{A5.2}$$

$$q_C = \Pr\{C_{30}(a) < B_{30} \mid D, a\}. \tag{A5.3}$$

Here $p_{12}(a)$ is a modelled twelve-month default probability, $p_{\max}$ its approved threshold, $\Delta(a, a_0)$ an identified incremental outcome for a defined population, δ the minimum worthwhile benefit, and $\mathcal{J}$ the identification assumptions. $C_{30}(a)$ is future available cash over thirty days and B_{30} the required amount on the same basis. Probabilities and $p_{\max}$ are dimensionless; cash, B_{30} and a cash-valued Δ are in KES.

The first probability integrates over uncertainty about a default-risk parameter; the second over uncertainty about a causal estimand under its identification assumptions; the third over a predictive distribution of future cash. Their respective acceptance thresholds have separate purposes and approval routes.

Action notation records the decision being evaluated. A fitted conditional risk $\Pr(D_{12} = 1 \mid X, A = a)$ becomes an intervention risk $\Pr(D_{12}(a) = 1 \mid X)$ only when its data and design justify that interpretation. The same distinction applies to a cash forecast under a changed action. The reviewer receives the prediction target, population and identification assumptions beside the numerical summaries, so an observed association is not silently promoted to an action effect.

A5.2 Derive the Beta posterior illustration

The conjugate beta–binomial calculation follows [7], ch. 2.

For a deliberately simple Bernoulli model, suppose n conditionally independent observations contain d defaults under one specified outcome definition, with default probability p. A Beta prior and binomial likelihood have kernels

$$\pi(p) \propto p^{\alpha_0 - 1}(1-p)^{\beta_0 - 1}, \qquad L(p; D) \propto p^d (1-p)^{n-d}. \tag{A5.4}$$

Multiplying and collecting powers gives

$$p \mid D \sim \text{Beta}(\alpha_0 + d, \beta_0 + n - d). \tag{A5.5}$$

For example, $\alpha_0 = \beta_0 = 1$, $n = 98$ and $d = 3$ produce $\text{Beta}(4, 96)$. This is an algebraic construction of the article's illustrative posterior, not a claim that its hypothetical borrower was estimated from those observations. A fitted hierarchical, action-dependent borrower model would generate its own posterior risk distribution.

With $\alpha = 4$, $\beta = 96$,

$$\mathbb{E}[p \mid D] = \frac{\alpha}{\alpha + \beta} = 0.04, \tag{A5.6}$$

$$\mathrm{Var}(p \mid D) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)} = \frac{384}{1{,}010{,}000} \approx 0.000380198. \tag{A5.7}$$

Let $F(x) = I_x(4, 96)$ be the regularised incomplete-beta cumulative distribution,

$$I_x(\alpha,\beta) = \frac{\int_0^x u^{\alpha-1}(1-u)^{\beta-1}\,du}{\mathrm{B}(\alpha,\beta)}, \qquad \mathrm{B}(\alpha,\beta) = \int_0^1 u^{\alpha-1}(1-u)^{\beta-1}\,du. \tag{A5.8}$$

The central 95% credible interval uses the 2.5% and 97.5% quantiles:

$$[F^{-1}(0.025), F^{-1}(0.975)] = [0.0111167488, 0.0860112643]. \tag{A5.9}$$

Thus the displayed interval is approximately 1.11%–8.60%. The exceedance probability at 6% is

$$q_p = 1 - I_{0.06}(4, 96) = 0.1483731733 \approx 14.84\%. \tag{A5.10}$$

For integer α, β, the result can be reproduced without a special-function library using

$$1 - I_x(\alpha,\beta) = \sum_{k=0}^{\alpha-1} \binom{\alpha+\beta-1}{k} x^k (1-x)^{\alpha+\beta-1-k}. \tag{A5.11}$$

Substituting $(\alpha, \beta) = (4, 96)$ gives a four-term sum with binomial size 99. Quantiles can be computed by bisection on $F(x)$ over $[0, 1]$. The credible interval is a posterior probability statement conditional on the model and evidence; predictive calibration and sensitivity remain separate validation tasks.

A5.3 Derive the incremental-benefit illustration

Define a population-average cash effect

$$\Delta(a, a_0) = \mathbb{E}\left[Y(a) - Y(a_0) \mid \mathcal{P}\right], \tag{A5.12}$$

where $Y(a)$ is the stated ninety-day net-cash outcome under action a and $\mathcal{P}$ the eligible population. Treatment costs are included consistently. Randomisation or a justified observational design, consistency, appropriate interference treatment, overlap and valid outcome measurement support its causal interpretation. Hierarchical priors can stabilise estimation within that design.

The article specifies a constructed posterior

$$\Delta \mid D, \mathcal{I} \sim \mathcal{N}(\mu_\Delta, \sigma_\Delta^2), \qquad \mu_\Delta = 8{,}000, \quad \sigma_\Delta = 4{,}000 \text{ KES}. \tag{A5.13}$$

If $Z = (\Delta - \mu_\Delta)/\sigma_\Delta$, then $Z \sim \mathcal{N}(0, 1)$. Writing Φ for its cumulative distribution and using $\Phi^{-1}(0.95) = 1.6448536269$, the central 90% interval is

$$\begin{aligned} \mu_\Delta \pm 1.6448536269\sigma_\Delta &= 8{,}000 \pm 6{,}579.4145 \\ &= [1{,}420.5855, 14{,}579.4145] \text{ KES}. \end{aligned} \tag{A5.14}$$

For minimum benefit δ,

$$q_{\Delta} = 1 - \Phi\left(\frac{\delta - 8{,}000}{4{,}000}\right). \tag{A5.15}$$

At $\delta = 0$, $q_{\Delta} = \Phi(2) = 0.9772498681$, or approximately 97.72%. This is uncertainty about the defined average effect. An individual customer's realised gain additionally depends on outcome variation and effect heterogeneity, so the same interval is not an individual-outcome forecast.

A5.4 Derive the available-cash mixture

Article 5 constructs the predictive mass function

$$\Pr(C = 20{,}000) = 0.30, \quad \Pr(C = 100{,}000) = 0.50, \quad \Pr(C = 180{,}000) = 0.20, \tag{A5.16}$$

where $C = C_{30}(a)$ is available cash in KES under the specified action. The probabilities sum to one. The predictive mean is

$$\mathbb{E}[C] = 0.30(20{,}000) + 0.50(100{,}000) + 0.20(180{,}000) = 92{,}000 \text{ KES}. \tag{A5.17}$$

The second moment and variance are

$$\mathbb{E}[C^2] = 11{,}600{,}000{,}000 \text{ KES}^2, \tag{A5.18}$$

$$\operatorname{Var}(C) = \mathbb{E}[C^2] - \mathbb{E}[C]^2 = 3{,}136{,}000{,}000 \text{ KES}^2, \tag{A5.19}$$

giving standard deviation KES 56,000. For obligations $B_{30} = 70{,}000$ KES, only the low-cash outcome is below the requirement, hence

$$q_C = \Pr(C < 70{,}000) = 0.30. \tag{A5.20}$$

The expected shortfall amount relative to these obligations is

$$\mathbb{E}[(70{,}000 - C)_+] = 0.30(50{,}000) = 15{,}000 \text{ KES}, \tag{A5.21}$$

where $(x)_+ = \max(x, 0)$. This obligation-shortfall measure has a different definition from a tail-risk measure also called expected shortfall in market-risk applications.

For a full fitted model with uncertain parameters θ, future-shock distribution $p(C \mid \theta, D, a)$ and posterior $p(\theta \mid D)$,

$$p(C \mid D, a) = \int p(C \mid \theta, D, a) p(\theta \mid D)\, d\theta. \tag{A5.22}$$

The law of total variance separates process variation from parameter uncertainty:

$$\operatorname{Var}(C \mid D, a) = \mathbb{E}_{\theta \mid D}[\operatorname{Var}(C \mid \theta, D, a)] + \operatorname{Var}_{\theta \mid D}[\mathbb{E}(C \mid \theta, D, a)]. \tag{A5.23}$$

Thus a credible interval about a risk parameter and a predictive cash interval describe different distributions even when they are generated from the same posterior fit.

A5.5 Limit-review logic and units

The expected-payoff and uncertainty-review distinction follows the decision-analysis framework in [7], ch. 9.

The example starts with limit $L = 500{,}000$ KES and drawn principal $U = 300{,}000$ KES. Existing undrawn headroom is $L - U = 200{,}000$ KES. Increasing the limit by KES 100,000 would make $L' = 600{,}000$ and undrawn headroom KES 300,000, before any new drawing. Cash disbursement changes only when a drawing is executed.

For an illustrative review tolerance $\epsilon_p = 0.10$, the model criterion is

$$\Pr\{p_{12}(a) > 0.06 \mid D\} \leq 0.10. \tag{A5.24}$$

The computed value $0.148373 > 0.10$ does not meet this route's criterion, so the case follows the specified additional-evidence or higher-authority path. This is a rule for handling posterior uncertainty, not a realised-default guarantee. A separate support action can still be considered through its own eligibility and benefit review.

For a continuous strictly increasing posterior cumulative distribution F_p, the gate is equivalent to $F_p(p_{\max}) \geq 1 - \epsilon_p$, or $F_p^{-1}(1 - \epsilon_p) \leq p_{\max}$. It therefore compares an upper posterior quantile with the approved cap. By contrast, the fixed-payoff financial comparison in Article 2 uses $\mathbb{E}[p \mid D]$: with repayment gain KES 1,200 and default loss KES 18,800, its threshold is $1200/(1200 + 18800) = 6\%$ and the expected value at posterior mean 4% is KES 400. A favourable expected value and a failed uncertainty gate can coexist because they answer separate decision questions. Limit changes use incremental action values relative to the existing facility, with uncertain utilisation and recovery retained in joint posterior draws.

A schematic permissible-action set is

$$\mathcal{A}^{\text{review}}(D) = \left\{a : \; G_k(D, a) = 1 \; \forall k, \quad q_p(D, a) \leq \epsilon_p\right\}, \tag{A5.25}$$

where G_k are the applicable contractual, affordability, conduct, mandate and data-quality gates for that action. Additional benefit or cash-risk criteria can be specified separately. Their thresholds are approved policy inputs. Where evidence is insufficient, the permitted response can be information gathering, deferral or another authorised action.

A5.6 Conditional Gaussian computation and Student-t hierarchy

The logistic augmentation follows [3]; the auxiliary half-t representation follows [10], Result 5.

A registered fixed-horizon model can carry the explicit sector, region and vintage hierarchy

$$\text{logit}(p_{i,\tau}) = \alpha^L + x_i^\top \beta^L + u^L_{s(i)} + v^L_{r(i)} + w^L_{c(i)}, \tag{A5.26}$$

where $s(i), r(i), c(i)$ map observation i to sector, region and origination vintage, τ is the default horizon, and L labels the logistic model. A proportional-hazard model has separate parameters labelled H:

$$\eta_i^H(t) = x_i(t)^\top \beta^H + u^H_{s(i)} + v^H_{r(i)} + w^H_{c(i)}, \qquad \lambda_i(t) = \lambda_0(t) e^{\eta_i^H(t)}. \tag{A5.27}$$

The baseline $\lambda_0(t)$ has inverse-time units and supplies the hazard intercept. For each model $m \in \{L, H\}$, the common hierarchy representation is

$$\begin{aligned} u_s^m &= \sigma_u^m z_{us}^m, & z_{us}^m &\sim N(0,1), \\ v_r^m &= \sigma_v^m z_{vr}^m, & z_{vr}^m &\sim N(0,1), \\ w_c^m &= \sigma_w^m z_{wc}^m, & z_{wc}^m &\sim N(0,1), \\ \sigma_k^m &\sim \mathsf{HalfStudentT}(3, s_{0,k}^m), & k &\in \{u, v, w\}, \\ \alpha^L &\sim N(0, s_\alpha^2), & \beta^m &\sim N(0, B^m). \end{aligned} \tag{A5.28}$$

The z variables are independent standard-normal raw effects under this specification; each positive σ is a group-variation scale. The constants $s_{0,k}^m$ and s_α are calibrated prior scales, and B^m is a positive-definite fixed-coefficient covariance. A Bayesian piecewise baseline can use $\lambda_{0\ell} \sim \mathsf{Gamma}(a_\ell, b_\ell)$ for interval ℓ, with positive declared shape and rate. The fixed-horizon outcome and hazard retain their own likelihood and parameter values.

For a mature binary label $y_i = D_i(\tau)$, the logistic likelihood is $\prod_i p_{i,\tau}^{y_i}(1-p_{i,\tau})^{1-y_i}$. For event time t_i and censoring indicator d_i, the corresponding single-event survival likelihood under conditionally independent censoring is

$$\mathcal{L}_H = \prod_i \lambda_i(t_i)^{d_i} \exp\left[-\int_0^{t_i} \lambda_i(u)\, du\right]. \tag{A5.29}$$

The hazard definition gives $d \log S_i(t)/dt = -\lambda_i(t)$, hence conditional survival

$$S_i(t) = \exp\left[-\int_0^{t} \lambda_i(u)\, du\right]. \tag{A5.30}$$

Prediction averages this expression over posterior draws. For competing terminal exits, replace the cumulative hazard by the sum over causes and calculate default incidence by integrating $S_i(t-)\lambda_{iD}(t)$ over the stated horizon. This connects the same named hierarchy to the appropriate probability target while retaining each outcome's likelihood.

For binary observations y_i, logistic predictor $\eta_i = x_i^\mathsf{T}\theta$ and $\kappa_i = y_i - 1/2$, the likelihood has the Pólya–Gamma mixture representation

$$\frac{e^{y_i\eta_i}}{1+e^{\eta_i}} = \frac{1}{2}e^{\kappa_i\eta_i}\int_0^\infty e^{-\omega_i\eta_i^2/2} p_{\mathsf{PG}(1,0)}(\omega_i)\, d\omega_i. \tag{A5.31}$$

Conditional on $\omega_i \sim \mathsf{PG}(1, \eta_i)$ and a Gaussian prior $\theta \sim \mathcal{N}(m_0, Q_0^{-1})$, the quadratic terms give

$$\log p(\theta \mid \omega, \mathcal{H}, D) = -\frac{1}{2}\theta^\mathsf{T}(X^\mathsf{T}\Omega X + Q_0)\theta + \theta^\mathsf{T}(X^\mathsf{T}\kappa + Q_0 m_0) + \text{constant}. \tag{A5.32}$$

Completing the square yields

$$\Sigma_* = (X^\mathsf{T}\Omega X + Q_0)^{-1}, \qquad m_* = \Sigma_*(X^\mathsf{T}\kappa + Q_0 m_0), \tag{A5.33}$$

$$\theta \mid \omega, \mathcal{H}, D \sim \mathcal{N}(m_*, \Sigma_*). \tag{A5.34}$$

X is the expanded design matrix, including intercept, fixed features and sector, region and vintage indicators. Its vector θ stacks their coefficients and centred group effects. The set $\mathcal{H}$ contains the scale and other hyperparameters currently held fixed; Q_0 is the resulting conditional prior precision, m_0 its mean, and $\Omega = \mathrm{diag}(\omega_i)$. In non-centred coordinates the relevant group columns are multiplied by their fixed scales and the raw effects receive standard-normal prior blocks. The same square-completion argument then applies to that transformed design. A separately specified hazard likelihood uses its corresponding inference method; Pólya–Gamma updates apply when its likelihood has the required form.

For the article's Gaussian group effects $u_j \mid \sigma \sim \mathcal{N}(0, \sigma^2)$ with a half-Student-t prior on σ, one useful auxiliary representation is

$$\sigma^2 \mid \xi \sim \mathsf{IG}\left(\frac{\nu}{2}, \frac{\nu}{\xi}\right), \qquad \xi \sim \mathsf{IG}\left(\frac{1}{2}, \frac{1}{A^2}\right). \tag{A5.35}$$

Here $A > 0$ is the half-t scale, $\nu = 3$ its degrees of freedom where selected, and $\mathsf{IG}(a, b)$ has density proportional to $v^{-a-1}e^{-b/v}$ for $v > 0$. Integrating ξ gives

$$p(\sigma) \propto \left(1 + \frac{\sigma^2}{\nu A^2}\right)^{-(\nu+1)/2}, \qquad \sigma > 0. \tag{A5.36}$$

For m independent centred group effects, the associated conditional scale updates are

$$\sigma^2 \mid u, \xi \sim \mathsf{IG}\left(\frac{\nu + m}{2}, \frac{\nu}{\xi} + \frac{1}{2}\sum_{j=1}^{m} u_j^2\right), \tag{A5.37}$$

$$\xi \mid \sigma^2 \sim \mathsf{IG}\left(\frac{\nu + 1}{2}, \frac{\nu}{\sigma^2} + \frac{1}{A^2}\right). \tag{A5.38}$$

These expressions concern the centred scalar-variance hierarchy. Correlation structures or a non-centred parameterisation require their corresponding conditionals. The non-centred identity $u_j = \sigma z_j$, $z_j \sim \mathcal{N}(0, 1)$ expresses the same hierarchy but changes sampling geometry.

To make that distinction computationally explicit, write a non-centred group contribution as σv, where $v = Z_g z$ and the remaining predictor is the fixed offset c. With $\eta = c + \sigma v$, its conditional scale kernel is

$$p(\sigma \mid z, \omega, \ldots) \propto \mathbf{1}\{\sigma > 0\} \exp\left[-\tfrac{1}{2}\sigma^2 v^{\mathsf{T}}\Omega v + \sigma v^{\mathsf{T}}(\kappa - \Omega c)\right] p(\sigma). \tag{A5.39}$$

The centred inverse-Gamma update conditions on u rather than z, so it is used only in the coordinates for which it was derived. Gaussian coefficient blocks, latent-variable updates and scale updates are assessed together through convergence, effective sample size and computation time; the deployed service can then use validated posterior artefacts for the approved refresh period.

An ordinary Student-t coefficient prior has another familiar representation: $\lambda \sim \mathrm{Gamma}(\nu/2, \nu/2)$ in shape–rate convention and $u \mid \lambda \sim \mathcal{N}(0, s^2/\lambda)$. At $\nu = 3$, both Gamma parameters are $3/2$. This coefficient mixture and a half-t prior on a group standard deviation play different roles; the registered model states which is used.

A5.7 Posterior summaries, numerical error and release checks

For retained posterior draws $\theta^{(1)}, \dots, \theta^{(M)}$, calculate each decision-specific risk $p^{(m)} = p_{12}(a; \theta^{(m)})$. The exceedance estimate is

$$\widehat{q}_p = \frac{1}{M} \sum_{m=1}^{M} \mathbf{1}\{p^{(m)} > p_{\mathsf{max}}\}. \tag{A5.40}$$

Empirical quantiles of the same draws estimate credible bounds. Causal-effect draws produce $\widehat{q}_\Delta$ analogously. Predictive cash additionally samples future outcomes conditional on each parameter draw. For independent draws, the standard error of a probability estimate is approximately $\sqrt{\widehat{q}(1-\widehat{q})/M}$; correlated MCMC output requires a variance estimate or effective sample size appropriate to that indicator. A probability near the review threshold merits enough computation that simulation error is visible and acceptably small.

An approved approximation can be compared against reference summaries on a registered test set $\mathcal{T}$:

$$\max_{i \in \mathcal{T}} |\widehat{q}_i^{\,\mathsf{service}} - \widehat{q}_i^{\,\mathsf{reference}}| \le \varepsilon_q. \tag{A5.41}$$

Analogous tolerances cover credible bounds, predictive cash and resulting review routes. Passing these fixtures verifies the tested implementation agreement; it does not establish the model's empirical accuracy outside the tested population.

A5.8 Workflow invariants and what a replay test verifies

Let an action record contain identifier e, case identifier, approved payload digest h, permission scope, expiry t^{exp} and execution state. Proposed engineering invariants include

$$\begin{aligned} \mathsf{Execute}(e,t) \Rightarrow\ & \mathsf{Approved}(e) \wedge \mathsf{Authorised}(e,t) \\ & \wedge\ (t \le t^{\mathsf{exp}}) \\ & \wedge\ \big(\mathsf{Digest}(\mathsf{payload}) = h\big). \end{aligned} \tag{A5.42}$$

A repeated identifier with the same payload returns the existing result; a conflicting payload is rejected for review. Cash availability requires a reconciled settlement record with the required references and availability date. The accounting identity from Article 4 then updates the relevant ledger once for the economic event.

These are design specifications tested through replay, access-control and reconciliation fixtures. They are not statistical theorems about actual bank permissions, source authenticity or external settlement. Deployment evidence must show that the running services enforce them, that reviewers can exercise their authority, and that operational owners resolve exceptions. The resulting audit record connects the mathematical summary to the approved action and realised outcome.

Sources supporting this appendix

Pólya–Gamma inference follows [3], and the half-t auxiliary-variable representation follows [10], Result 5. Numerical summaries are derived from the specified illustrative distributions using the inference and decision framework in [7]. The workflow applies the bank-controlled architecture [1] and the legal sources [4], [5].

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